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**Internet X.509 Public Key Infrastructure Certificate
and Certificate Revocation List (CRL) Profile**

Status of This Memo

This document specifies an Internet standards track protocol for the Internet community, and requests discussion and suggestions for improvements. Please refer to the current edition of the "Internet Official Protocol Standards" (STD 1) for the standardization state and status of this protocol. Distribution of this memo is unlimited.

Abstract

This memo profiles the X.509 v3 certificate and X.509 v2 certificate revocation list (CRL) for use in the Internet. An overview of this approach and model is provided as an introduction. The X.509 v3 certificate format is described in detail, with additional information regarding the format and semantics of Internet name forms. Standard certificate extensions are described and two Internet-specific extensions are defined. A set of required certificate extensions is specified. The X.509 v2 CRL format is described in detail along with standard and Internet-specific extensions. An algorithm for X.509 certification path validation is described. An ASN.1 module and examples are provided in the appendices.

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1. Introduction

This specification is one part of a family of standards for the X.509 Public Key Infrastructure (PKI) for the Internet.

This specification profiles the format and semantics of certificates and certificate revocation lists (CRLs) for the Internet PKI. Procedures are described for processing of certification paths in the Internet environment. Finally, ASN.1 modules are provided in the appendices for all data structures defined or referenced.

Section 2 describes Internet PKI requirements and the assumptions that affect the scope of this document. Section 3 presents an architectural model and describes its relationship to previous IETF and ISO/IEC/ITU-T standards. In particular, this document's relationship with the IETF PEM specifications and the ISO/IEC/ITU-T X.509 documents is described.

Section 4 profiles the X.509 version 3 certificate, and Section 5 profiles the X.509 version 2 CRL. The profiles include the identification of ISO/IEC/ITU-T and ANSI extensions that may be useful in the Internet PKI. The profiles are presented in the 1988 Abstract Syntax Notation One (ASN.1) rather than the 1997 ASN.1 syntax used in the most recent ISO/IEC/ITU-T standards.

Section 6 includes certification path validation procedures. These procedures are based upon the ISO/IEC/ITU-T definition. Implementations are REQUIRED to derive the same results but are not required to use the specified procedures.

Procedures for identification and encoding of public key materials and digital signatures are defined in [RFC3279], [RFC4055], and [RFC4491]. Implementations of this specification are not required to use any particular cryptographic algorithms. However, conforming implementations that use the algorithms identified in [RFC3279], [RFC4055], and [RFC4491] MUST identify and encode the public key materials and digital signatures as described in those specifications.

Finally, three appendices are provided to aid implementers. Appendix A contains all ASN.1 structures defined or referenced within this specification. As above, the material is presented in the 1988 ASN.1. Appendix B contains notes on less familiar features of the ASN.1 notation used within this specification. Appendix C contains examples of conforming certificates and a conforming CRL.

This specification obsoletes [RFC3280]. Differences from RFC 3280 are summarized below:

- * Enhanced support for internationalized names is specified in Section 7, with rules for encoding and comparing Internationalized Domain Names, Internationalized Resource Identifiers (IRIs), and distinguished names. These rules are aligned with comparison rules established in current RFCs, including [RFC3490], [RFC3987], and [RFC4518].
- * Sections 4.1.2.4 and 4.1.2.6 incorporate the conditions for continued use of legacy text encoding schemes that were specified in [RFC4630]. Where in use by an established PKI, transition to UTF8String could cause denial of service based on name chaining failures or incorrect processing of name constraints.
- * Section 4.2.1.4 in RFC 3280, which specified the privateKeyUsagePeriod certificate extension but deprecated its use, was removed. Use of this ISO standard extension is neither deprecated nor recommended for use in the Internet PKI.
- * Section 4.2.1.5 recommends marking the policy mappings extension as critical. RFC 3280 required that the policy mappings extension be marked as non-critical.
- * Section 4.2.1.11 requires marking the policy constraints extension as critical. RFC 3280 permitted the policy constraints extension to be marked as critical or non-critical.
- * The Authority Information Access (AIA) CRL extension, as specified in [RFC4325], was added as Section 5.2.7.

- * Sections 5.2 and 5.3 clarify the rules for handling unrecognized CRL extensions and CRL entry extensions, respectively.
- * Section 5.3.2 in RFC 3280, which specified the holdInstructionCode CRL entry extension, was removed.
- * The path validation algorithm specified in Section 6 no longer tracks the criticality of the certificate policies extensions in a chain of certificates. In RFC 3280, this information was returned to a relying party.
- * The Security Considerations section addresses the risk of circular dependencies arising from the use of https or similar schemes in the CRL distribution points, authority information access, or subject information access extensions.
- * The Security Considerations section addresses risks associated with name ambiguity.
- * The Security Considerations section references RFC 4210 for procedures to signal changes in CA operations.

The ASN.1 modules in Appendix A are unchanged from RFC 3280, except that ub-emailaddress-length was changed from 128 to 255 in order to align with PKCS #9 [RFC2985].

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

2. Requirements and Assumptions

The goal of this specification is to develop a profile to facilitate the use of X.509 certificates within Internet applications for those communities wishing to make use of X.509 technology. Such applications may include WWW, electronic mail, user authentication, and IPsec. In order to relieve some of the obstacles to using X.509 certificates, this document defines a profile to promote the development of certificate management systems, development of application tools, and interoperability determined by policy.

Some communities will need to supplement, or possibly replace, this profile in order to meet the requirements of specialized application domains or environments with additional authorization, assurance, or operational requirements. However, for basic applications, common representations of frequently used attributes are defined so that

application developers can obtain necessary information without regard to the issuer of a particular certificate or certificate revocation list (CRL).

A certificate user should review the certificate policy generated by the certification authority (CA) before relying on the authentication or non-repudiation services associated with the public key in a particular certificate. To this end, this standard does not prescribe legally binding rules or duties.

As supplemental authorization and attribute management tools emerge, such as attribute certificates, it may be appropriate to limit the authenticated attributes that are included in a certificate. These other management tools may provide more appropriate methods of conveying many authenticated attributes.

2.1. Communication and Topology

The users of certificates will operate in a wide range of environments with respect to their communication topology, especially users of secure electronic mail. This profile supports users without high bandwidth, real-time IP connectivity, or high connection availability. In addition, the profile allows for the presence of firewall or other filtered communication.

This profile does not assume the deployment of an X.500 directory system [X.500] or a Lightweight Directory Access Protocol (LDAP) directory system [RFC4510]. The profile does not prohibit the use of an X.500 directory or an LDAP directory; however, any means of distributing certificates and certificate revocation lists (CRLs) may be used.

2.2. Acceptability Criteria

The goal of the Internet Public Key Infrastructure (PKI) is to meet the needs of deterministic, automated identification, authentication, access control, and authorization functions. Support for these services determines the attributes contained in the certificate as well as the ancillary control information in the certificate such as policy data and certification path constraints.

2.3. User Expectations

Users of the Internet PKI are people and processes who use client software and are the subjects named in certificates. These uses include readers and writers of electronic mail, the clients for WWW browsers, WWW servers, and the key manager for IPsec within a router. This profile recognizes the limitations of the platforms these users

employ and the limitations in sophistication and attentiveness of the users themselves. This manifests itself in minimal user configuration responsibility (e.g., trusted CA keys, rules), explicit platform usage constraints within the certificate, certification path constraints that shield the user from many malicious actions, and applications that sensibly automate validation functions.

2.4. Administrator Expectations

As with user expectations, the Internet PKI profile is structured to support the individuals who generally operate CAs. Providing administrators with unbounded choices increases the chances that a subtle CA administrator mistake will result in broad compromise. Also, unbounded choices greatly complicate the software that process and validate the certificates created by the CA.

3. Overview of Approach

Following is a simplified view of the architectural model assumed by the Public-Key Infrastructure using X.509 (PKIX) specifications.

The components in this model are:

end entity: user of PKI certificates and/or end user system that is the subject of a certificate;

CA: certification authority;

RA: registration authority, i.e., an optional system to which a CA delegates certain management functions;

CRL issuer: a system that generates and signs CRLs; and

repository: a system or collection of distributed systems that stores certificates and CRLs and serves as a means of distributing these certificates and CRLs to end entities.

CAs are responsible for indicating the revocation status of the certificates that they issue. Revocation status information may be provided using the Online Certificate Status Protocol (OCSP) [RFC2560], certificate revocation lists (CRLs), or some other mechanism. In general, when revocation status information is provided using CRLs, the CA is also the CRL issuer. However, a CA may delegate the responsibility for issuing CRLs to a different entity.

Note that an Attribute Authority (AA) might also choose to delegate the publication of CRLs to a CRL issuer.

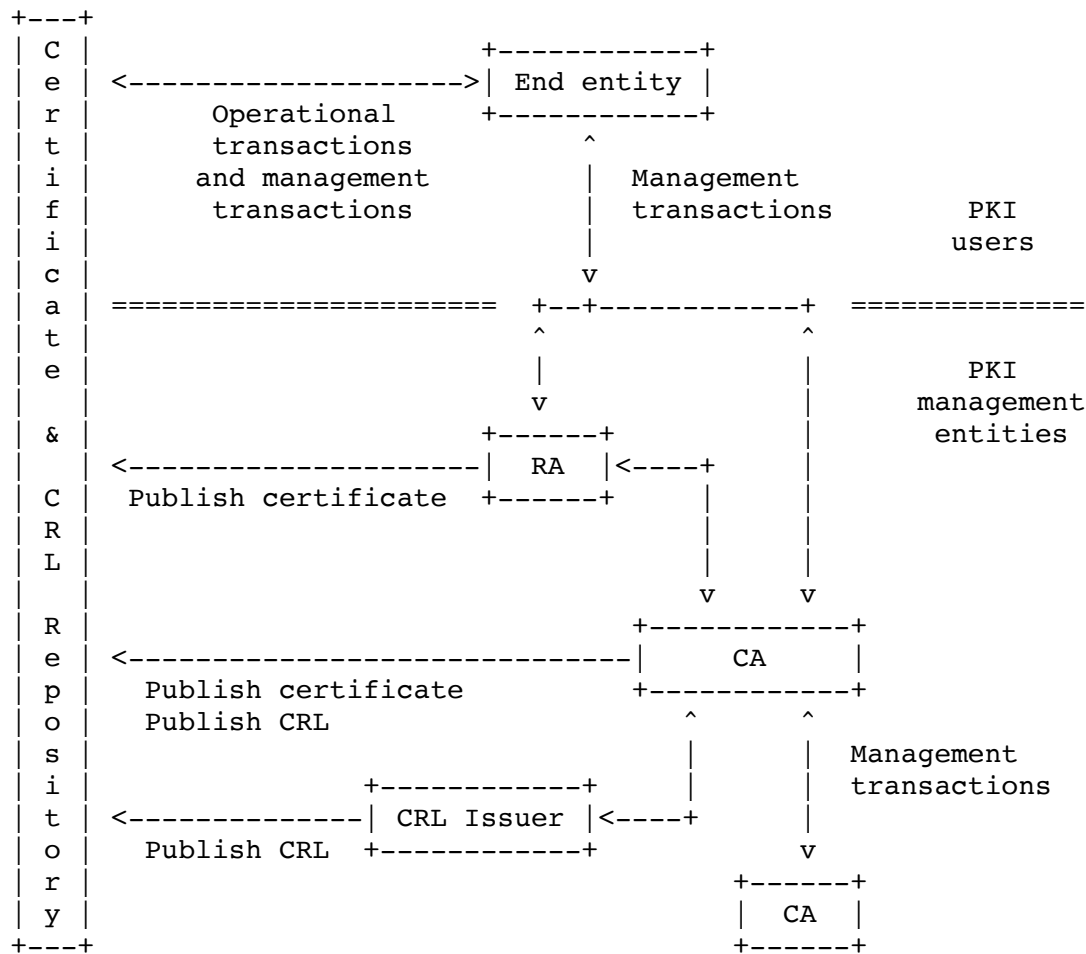


Figure 1. PKI Entities

3.1. X.509 Version 3 Certificate

Users of a public key require confidence that the associated private key is owned by the correct remote subject (person or system) with which an encryption or digital signature mechanism will be used. This confidence is obtained through the use of public key certificates, which are data structures that bind public key values to subjects. The binding is asserted by having a trusted CA digitally sign each certificate. The CA may base this assertion upon technical means (a.k.a., proof of possession through a challenge-response protocol), presentation of the private key, or on an assertion by the subject. A certificate has a limited valid lifetime, which is indicated in its signed contents. Because a certificate's signature and timeliness can be independently checked by a certificate-using client, certificates can be distributed via

untrusted communications and server systems, and can be cached in unsecured storage in certificate-using systems.

ITU-T X.509 (formerly CCITT X.509) or ISO/IEC 9594-8, which was first published in 1988 as part of the X.500 directory recommendations, defines a standard certificate format [X.509]. The certificate format in the 1988 standard is called the version 1 (v1) format. When X.500 was revised in 1993, two more fields were added, resulting in the version 2 (v2) format.

The Internet Privacy Enhanced Mail (PEM) RFCs, published in 1993, include specifications for a public key infrastructure based on X.509 v1 certificates [RFC1422]. The experience gained in attempts to deploy RFC 1422 made it clear that the v1 and v2 certificate formats were deficient in several respects. Most importantly, more fields were needed to carry information that PEM design and implementation experience had proven necessary. In response to these new requirements, the ISO/IEC, ITU-T, and ANSI X9 developed the X.509 version 3 (v3) certificate format. The v3 format extends the v2 format by adding provision for additional extension fields. Particular extension field types may be specified in standards or may be defined and registered by any organization or community. In June 1996, standardization of the basic v3 format was completed [X.509].

ISO/IEC, ITU-T, and ANSI X9 have also developed standard extensions for use in the v3 extensions field [X.509][X9.55]. These extensions can convey such data as additional subject identification information, key attribute information, policy information, and certification path constraints.

However, the ISO/IEC, ITU-T, and ANSI X9 standard extensions are very broad in their applicability. In order to develop interoperable implementations of X.509 v3 systems for Internet use, it is necessary to specify a profile for use of the X.509 v3 extensions tailored for the Internet. It is one goal of this document to specify a profile for Internet WWW, electronic mail, and IPsec applications. Environments with additional requirements may build on this profile or may replace it.

3.2. Certification Paths and Trust

A user of a security service requiring knowledge of a public key generally needs to obtain and validate a certificate containing the required public key. If the public key user does not already hold an assured copy of the public key of the CA that signed the certificate, the CA's name, and related information (such as the validity period or name constraints), then it might need an additional certificate to obtain that public key. In general, a chain of multiple certificates

may be needed, comprising a certificate of the public key owner (the end entity) signed by one CA, and zero or more additional certificates of CAs signed by other CAs. Such chains, called certification paths, are required because a public key user is only initialized with a limited number of assured CA public keys.

There are different ways in which CAs might be configured in order for public key users to be able to find certification paths. For PEM, RFC 1422 defined a rigid hierarchical structure of CAs. There are three types of PEM certification authority:

- (a) Internet Policy Registration Authority (IPRA): This authority, operated under the auspices of the Internet Society, acts as the root of the PEM certification hierarchy at level 1. It issues certificates only for the next level of authorities, PCAs. All certification paths start with the IPRA.
- (b) Policy Certification Authorities (PCAs): PCAs are at level 2 of the hierarchy, each PCA being certified by the IPRA. A PCA shall establish and publish a statement of its policy with respect to certifying users or subordinate certification authorities. Distinct PCAs aim to satisfy different user needs. For example, one PCA (an organizational PCA) might support the general electronic mail needs of commercial organizations, and another PCA (a high-assurance PCA) might have a more stringent policy designed for satisfying legally binding digital signature requirements.
- (c) Certification Authorities (CAs): CAs are at level 3 of the hierarchy and can also be at lower levels. Those at level 3 are certified by PCAs. CAs represent, for example, particular organizations, particular organizational units (e.g., departments, groups, sections), or particular geographical areas.

RFC 1422 furthermore has a name subordination rule, which requires that a CA can only issue certificates for entities whose names are subordinate (in the X.500 naming tree) to the name of the CA itself. The trust associated with a PEM certification path is implied by the PCA name. The name subordination rule ensures that CAs below the PCA are sensibly constrained as to the set of subordinate entities they can certify (e.g., a CA for an organization can only certify entities in that organization's name tree). Certificate user systems are able to mechanically check that the name subordination rule has been followed.

RFC 1422 uses the X.509 v1 certificate format. The limitations of X.509 v1 required imposition of several structural restrictions to clearly associate policy information or restrict the utility of certificates. These restrictions included:

- (a) a pure top-down hierarchy, with all certification paths starting from IPRA;
- (b) a naming subordination rule restricting the names of a CA's subjects; and
- (c) use of the PCA concept, which requires knowledge of individual PCAs to be built into certificate chain verification logic. Knowledge of individual PCAs was required to determine if a chain could be accepted.

With X.509 v3, most of the requirements addressed by RFC 1422 can be addressed using certificate extensions, without a need to restrict the CA structures used. In particular, the certificate extensions relating to certificate policies obviate the need for PCAs and the constraint extensions obviate the need for the name subordination rule. As a result, this document supports a more flexible architecture, including:

- (a) Certification paths start with a public key of a CA in a user's own domain, or with the public key of the top of a hierarchy. Starting with the public key of a CA in a user's own domain has certain advantages. In some environments, the local domain is the most trusted.
- (b) Name constraints may be imposed through explicit inclusion of a name constraints extension in a certificate, but are not required.
- (c) Policy extensions and policy mappings replace the PCA concept, which permits a greater degree of automation. The application can determine if the certification path is acceptable based on the contents of the certificates instead of a priori knowledge of PCAs. This permits automation of certification path processing.

X.509 v3 also includes an extension that identifies the subject of a certificate as being either a CA or an end entity, reducing the reliance on out-of-band information demanded in PEM.

This specification covers two classes of certificates: CA certificates and end entity certificates. CA certificates may be further divided into three classes: cross-certificates, self-issued

certificates, and self-signed certificates. Cross-certificates are CA certificates in which the issuer and subject are different entities. Cross-certificates describe a trust relationship between the two CAs. Self-issued certificates are CA certificates in which the issuer and subject are the same entity. Self-issued certificates are generated to support changes in policy or operations. Self-signed certificates are self-issued certificates where the digital signature may be verified by the public key bound into the certificate. Self-signed certificates are used to convey a public key for use to begin certification paths. End entity certificates are issued to subjects that are not authorized to issue certificates.

3.3. Revocation

When a certificate is issued, it is expected to be in use for its entire validity period. However, various circumstances may cause a certificate to become invalid prior to the expiration of the validity period. Such circumstances include change of name, change of association between subject and CA (e.g., an employee terminates employment with an organization), and compromise or suspected compromise of the corresponding private key. Under such circumstances, the CA needs to revoke the certificate.

X.509 defines one method of certificate revocation. This method involves each CA periodically issuing a signed data structure called a certificate revocation list (CRL). A CRL is a time-stamped list identifying revoked certificates that is signed by a CA or CRL issuer and made freely available in a public repository. Each revoked certificate is identified in a CRL by its certificate serial number. When a certificate-using system uses a certificate (e.g., for verifying a remote user's digital signature), that system not only checks the certificate signature and validity but also acquires a suitably recent CRL and checks that the certificate serial number is not on that CRL. The meaning of "suitably recent" may vary with local policy, but it usually means the most recently issued CRL. A new CRL is issued on a regular periodic basis (e.g., hourly, daily, or weekly). An entry is added to the CRL as part of the next update following notification of revocation. An entry **MUST NOT** be removed from the CRL until it appears on one regularly scheduled CRL issued beyond the revoked certificate's validity period.

An advantage of this revocation method is that CRLs may be distributed by exactly the same means as certificates themselves, namely, via untrusted servers and untrusted communications.

One limitation of the CRL revocation method, using untrusted communications and servers, is that the time granularity of revocation is limited to the CRL issue period. For example, if a

revocation is reported now, that revocation will not be reliably notified to certificate-using systems until all currently issued CRLs are scheduled to be updated -- this may be up to one hour, one day, or one week depending on the frequency that CRLs are issued.

As with the X.509 v3 certificate format, in order to facilitate interoperable implementations from multiple vendors, the X.509 v2 CRL format needs to be profiled for Internet use. It is one goal of this document to specify that profile. However, this profile does not require the issuance of CRLs. Message formats and protocols supporting on-line revocation notification are defined in other PKIX specifications. On-line methods of revocation notification may be applicable in some environments as an alternative to the X.509 CRL. On-line revocation checking may significantly reduce the latency between a revocation report and the distribution of the information to relying parties. Once the CA accepts a revocation report as authentic and valid, any query to the on-line service will correctly reflect the certificate validation impacts of the revocation. However, these methods impose new security requirements: the certificate validator needs to trust the on-line validation service while the repository does not need to be trusted.

3.4. Operational Protocols

Operational protocols are required to deliver certificates and CRLs (or status information) to certificate-using client systems. Provisions are needed for a variety of different means of certificate and CRL delivery, including distribution procedures based on LDAP, HTTP, FTP, and X.500. Operational protocols supporting these functions are defined in other PKIX specifications. These specifications may include definitions of message formats and procedures for supporting all of the above operational environments, including definitions of or references to appropriate MIME content types.

3.5. Management Protocols

Management protocols are required to support on-line interactions between PKI user and management entities. For example, a management protocol might be used between a CA and a client system with which a key pair is associated, or between two CAs that cross-certify each other. The set of functions that potentially need to be supported by management protocols include:

- (a) registration: This is the process whereby a user first makes itself known to a CA (directly, or through an RA), prior to that CA issuing a certificate or certificates for that user.

- (b) initialization: Before a client system can operate securely, it is necessary to install key materials that have the appropriate relationship with keys stored elsewhere in the infrastructure. For example, the client needs to be securely initialized with the public key and other assured information of the trusted CA(s), to be used in validating certificate paths.

Furthermore, a client typically needs to be initialized with its own key pair(s).

- (c) certification: This is the process in which a CA issues a certificate for a user's public key, and returns that certificate to the user's client system and/or posts that certificate in a repository.
- (d) key pair recovery: As an option, user client key materials (e.g., a user's private key used for encryption purposes) may be backed up by a CA or a key backup system. If a user needs to recover these backed-up key materials (e.g., as a result of a forgotten password or a lost key chain file), an on-line protocol exchange may be needed to support such recovery.
- (e) key pair update: All key pairs need to be updated regularly, i.e., replaced with a new key pair, and new certificates issued.
- (f) revocation request: An authorized person advises a CA of an abnormal situation requiring certificate revocation.
- (g) cross-certification: Two CAs exchange information used in establishing a cross-certificate. A cross-certificate is a certificate issued by one CA to another CA that contains a CA signature key used for issuing certificates.

Note that on-line protocols are not the only way of implementing the above functions. For all functions, there are off-line methods of achieving the same result, and this specification does not mandate use of on-line protocols. For example, when hardware tokens are used, many of the functions may be achieved as part of the physical token delivery. Furthermore, some of the above functions may be combined into one protocol exchange. In particular, two or more of the registration, initialization, and certification functions can be combined into one protocol exchange.

The PKIX series of specifications defines a set of standard message formats supporting the above functions. The protocols for conveying these messages in different environments (e.g., email, file transfer, and WWW) are described in those specifications.

4. Certificate and Certificate Extensions Profile

This section presents a profile for public key certificates that will foster interoperability and a reusable PKI. This section is based upon the X.509 v3 certificate format and the standard certificate extensions defined in [X.509]. The ISO/IEC and ITU-T documents use the 1997 version of ASN.1; while this document uses the 1988 ASN.1 syntax, the encoded certificate and standard extensions are equivalent. This section also defines private extensions required to support a PKI for the Internet community.

Certificates may be used in a wide range of applications and environments covering a broad spectrum of interoperability goals and a broader spectrum of operational and assurance requirements. The goal of this document is to establish a common baseline for generic applications requiring broad interoperability and limited special purpose requirements. In particular, the emphasis will be on supporting the use of X.509 v3 certificates for informal Internet electronic mail, IPsec, and WWW applications.

4.1. Basic Certificate Fields

The X.509 v3 certificate basic syntax is as follows. For signature calculation, the data that is to be signed is encoded using the ASN.1 distinguished encoding rules (DER) [X.690]. ASN.1 DER encoding is a tag, length, value encoding system for each element.

```
Certificate ::= SEQUENCE {
    tbsCertificate      TBSCertificate,
    signatureAlgorithm  AlgorithmIdentifier,
    signatureValue      BIT STRING }

TBSCertificate ::= SEQUENCE {
    version             [0] EXPLICIT Version DEFAULT v1,
    serialNumber        CertificateSerialNumber,
    signature            AlgorithmIdentifier,
    issuer              Name,
    validity            Validity,
    subject             Name,
    subjectPublicKeyInfo SubjectPublicKeyInfo,
    issuerUniqueID      [1] IMPLICIT UniqueIdentifier OPTIONAL,
                        -- If present, version MUST be v2 or v3
```



```
    subjectUniqueID [2]  IMPLICIT UniqueIdentifier OPTIONAL,
                        -- If present, version MUST be v2 or v3
    extensions        [3] EXPLICIT Extensions OPTIONAL
                        -- If present, version MUST be v3
  }

Version ::= INTEGER { v1(0), v2(1), v3(2) }

CertificateSerialNumber ::= INTEGER

Validity ::= SEQUENCE {
    notBefore      Time,
    notAfter       Time }

Time ::= CHOICE {
    utcTime        UTCTime,
    generalTime    GeneralizedTime }

UniqueIdentifier ::= BIT STRING

SubjectPublicKeyInfo ::= SEQUENCE {
    algorithm        AlgorithmIdentifier,
    subjectPublicKey  BIT STRING }

Extensions ::= SEQUENCE SIZE (1..MAX) OF Extension

Extension ::= SEQUENCE {
    extnID          OBJECT IDENTIFIER,
    critical         BOOLEAN DEFAULT FALSE,
    extnValue        OCTET STRING
                    -- contains the DER encoding of an ASN.1 value
                    -- corresponding to the extension type identified
                    -- by extnID
  }
```

The following items describe the X.509 v3 certificate for use in the Internet.

4.1.1.1. Certificate Fields

The Certificate is a SEQUENCE of three required fields. The fields are described in detail in the following subsections.

4.1.1.1. **tbsCertificate**

The field contains the names of the subject and issuer, a public key associated with the subject, a validity period, and other associated information. The fields are described in detail in Section 4.1.2; the tbsCertificate usually includes extensions, which are described in Section 4.2.

4.1.1.2. **signatureAlgorithm**

The signatureAlgorithm field contains the identifier for the cryptographic algorithm used by the CA to sign this certificate. [RFC3279], [RFC4055], and [RFC4491] list supported signature algorithms, but other signature algorithms MAY also be supported.

An algorithm identifier is defined by the following ASN.1 structure:

```
AlgorithmIdentifier ::= SEQUENCE {  
    algorithm          OBJECT IDENTIFIER,  
    parameters        ANY DEFINED BY algorithm OPTIONAL }
```

The algorithm identifier is used to identify a cryptographic algorithm. The OBJECT IDENTIFIER component identifies the algorithm (such as DSA with SHA-1). The contents of the optional parameters field will vary according to the algorithm identified.

This field MUST contain the same algorithm identifier as the signature field in the sequence tbsCertificate (Section 4.1.2.3).

4.1.1.3. **signatureValue**

The signatureValue field contains a digital signature computed upon the ASN.1 DER encoded tbsCertificate. The ASN.1 DER encoded tbsCertificate is used as the input to the signature function. This signature value is encoded as a BIT STRING and included in the signature field. The details of this process are specified for each of the algorithms listed in [RFC3279], [RFC4055], and [RFC4491].

By generating this signature, a CA certifies the validity of the information in the tbsCertificate field. In particular, the CA certifies the binding between the public key material and the subject of the certificate.

4.1.2. **TBSCertificate**

The sequence TBSCertificate contains information associated with the subject of the certificate and the CA that issued it. Every TBSCertificate contains the names of the subject and issuer, a public

key associated with the subject, a validity period, a version number, and a serial number; some MAY contain optional unique identifier fields. The remainder of this section describes the syntax and semantics of these fields. A TBSCertificate usually includes extensions. Extensions for the Internet PKI are described in Section 4.2.

4.1.2.1. Version

This field describes the version of the encoded certificate. When extensions are used, as expected in this profile, version MUST be 3 (value is 2). If no extensions are present, but a UniqueIdentifier is present, the version SHOULD be 2 (value is 1); however, the version MAY be 3. If only basic fields are present, the version SHOULD be 1 (the value is omitted from the certificate as the default value); however, the version MAY be 2 or 3.

Implementations SHOULD be prepared to accept any version certificate. At a minimum, conforming implementations MUST recognize version 3 certificates.

Generation of version 2 certificates is not expected by implementations based on this profile.

4.1.2.2. Serial Number

The serial number MUST be a positive integer assigned by the CA to each certificate. It MUST be unique for each certificate issued by a given CA (i.e., the issuer name and serial number identify a unique certificate). CAs MUST force the serialNumber to be a non-negative integer.

Given the uniqueness requirements above, serial numbers can be expected to contain long integers. Certificate users MUST be able to handle serialNumber values up to 20 octets. Conforming CAs MUST NOT use serialNumber values longer than 20 octets.

Note: Non-conforming CAs may issue certificates with serial numbers that are negative or zero. Certificate users SHOULD be prepared to gracefully handle such certificates.

4.1.2.3. Signature

This field contains the algorithm identifier for the algorithm used by the CA to sign the certificate.

This field MUST contain the same algorithm identifier as the signatureAlgorithm field in the sequence Certificate (Section

4.1.1.2). The contents of the optional parameters field will vary according to the algorithm identified. [RFC3279], [RFC4055], and [RFC4491] list supported signature algorithms, but other signature algorithms MAY also be supported.

4.1.2.4. Issuer

The issuer field identifies the entity that has signed and issued the certificate. The issuer field MUST contain a non-empty distinguished name (DN). The issuer field is defined as the X.501 type Name [X.501]. Name is defined by the following ASN.1 structures:

Name ::= CHOICE { -- only one possibility for now --
 rdnSequence RDNSequence }

RDNSequence ::= SEQUENCE OF RelativeDistinguishedName

RelativeDistinguishedName ::= SET SIZE (1..MAX) OF AttributeTypeAndValue

AttributeTypeAndValue ::= SEQUENCE {
 type AttributeType,
 value AttributeValue }

AttributeType ::= OBJECT IDENTIFIER

AttributeValue ::= ANY -- DEFINED BY AttributeType

DirectoryString ::= CHOICE {
 teletexString TeletexString (SIZE (1..MAX)),
 printableString PrintableString (SIZE (1..MAX)),
 universalString UniversalString (SIZE (1..MAX)),
 utf8String UTF8String (SIZE (1..MAX)),
 bmpString BMPString (SIZE (1..MAX)) }

The Name describes a hierarchical name composed of attributes, such as country name, and corresponding values, such as US. The type of the component AttributeValue is determined by the AttributeType; in general it will be a DirectoryString.

The DirectoryString type is defined as a choice of PrintableString, TeletexString, BMPString, UTF8String, and UniversalString. CAs conforming to this profile MUST use either the PrintableString or UTF8String encoding of DirectoryString, with two exceptions. When CAs have previously issued certificates with issuer fields with attributes encoded using TeletexString, BMPString, or UniversalString, then the CA MAY continue to use these encodings of the DirectoryString to preserve backward compatibility. Also, new

CAs that are added to a domain where existing CAs issue certificates with issuer fields with attributes encoded using TeletexString, BMPString, or UniversalString MAY encode attributes that they share with the existing CAs using the same encodings as the existing CAs use.

As noted above, distinguished names are composed of attributes. This specification does not restrict the set of attribute types that may appear in names. However, conforming implementations MUST be prepared to receive certificates with issuer names containing the set of attribute types defined below. This specification RECOMMENDS support for additional attribute types.

Standard sets of attributes have been defined in the X.500 series of specifications [X.520]. Implementations of this specification MUST be prepared to receive the following standard attribute types in issuer and subject (Section 4.1.2.6) names:

- * country,
- * organization,
- * organizational unit,
- * distinguished name qualifier,
- * state or province name,
- * common name (e.g., "Susan Housley"), and
- * serial number.

In addition, implementations of this specification SHOULD be prepared to receive the following standard attribute types in issuer and subject names:

- * locality,
- * title,
- * surname,
- * given name,
- * initials,
- * pseudonym, and
- * generation qualifier (e.g., "Jr.", "3rd", or "IV").

The syntax and associated object identifiers (OIDs) for these attribute types are provided in the ASN.1 modules in Appendix A.

In addition, implementations of this specification MUST be prepared to receive the domainComponent attribute, as defined in [RFC4519]. The Domain Name System (DNS) provides a hierarchical resource labeling system. This attribute provides a convenient mechanism for organizations that wish to use DNS that parallel their DNS names. This is not a replacement for the dNSName component of the alternative name extensions. Implementations are not required to

convert such names into DNS names. The syntax and associated OID for this attribute type are provided in the ASN.1 modules in Appendix A. Rules for encoding internationalized domain names for use with the domainComponent attribute type are specified in Section 7.3.

Certificate users MUST be prepared to process the issuer distinguished name and subject distinguished name (Section 4.1.2.6) fields to perform name chaining for certification path validation (Section 6). Name chaining is performed by matching the issuer distinguished name in one certificate with the subject name in a CA certificate. Rules for comparing distinguished names are specified in Section 7.1. If the names in the issuer and subject field in a certificate match according to the rules specified in Section 7.1, then the certificate is self-issued.

4.1.2.5. Validity

The certificate validity period is the time interval during which the CA warrants that it will maintain information about the status of the certificate. The field is represented as a SEQUENCE of two dates: the date on which the certificate validity period begins (notBefore) and the date on which the certificate validity period ends (notAfter). Both notBefore and notAfter may be encoded as UTCTime or GeneralizedTime.

CAs conforming to this profile MUST always encode certificate validity dates through the year 2049 as UTCTime; certificate validity dates in 2050 or later MUST be encoded as GeneralizedTime. Conforming applications MUST be able to process validity dates that are encoded in either UTCTime or GeneralizedTime.

The validity period for a certificate is the period of time from notBefore through notAfter, inclusive.

In some situations, devices are given certificates for which no good expiration date can be assigned. For example, a device could be issued a certificate that binds its model and serial number to its public key; such a certificate is intended to be used for the entire lifetime of the device.

To indicate that a certificate has no well-defined expiration date, the notAfter SHOULD be assigned the GeneralizedTime value of 99991231235959Z.

When the issuer will not be able to maintain status information until the notAfter date (including when the notAfter date is 99991231235959Z), the issuer MUST ensure that no valid certification path exists for the certificate after maintenance of status

information is terminated. This may be accomplished by expiration or revocation of all CA certificates containing the public key used to verify the signature on the certificate and discontinuing use of the public key used to verify the signature on the certificate as a trust anchor.

4.1.2.5.1. UTCTime

The universal time type, `UTCTime`, is a standard ASN.1 type intended for representation of dates and time. `UTCTime` specifies the year through the two low-order digits and time is specified to the precision of one minute or one second. `UTCTime` includes either `Z` (for Zulu, or Greenwich Mean Time) or a time differential.

For the purposes of this profile, `UTCTime` values **MUST** be expressed in Greenwich Mean Time (Zulu) and **MUST** include seconds (i.e., times are `YYMMDDHHMMSSZ`), even where the number of seconds is zero. Conforming systems **MUST** interpret the year field (`YY`) as follows:

Where `YY` is greater than or equal to 50, the year **SHALL** be interpreted as 19`YY`; and

Where `YY` is less than 50, the year **SHALL** be interpreted as 20`YY`.

4.1.2.5.2. GeneralizedTime

The generalized time type, `GeneralizedTime`, is a standard ASN.1 type for variable precision representation of time. Optionally, the `GeneralizedTime` field can include a representation of the time differential between local and Greenwich Mean Time.

For the purposes of this profile, `GeneralizedTime` values **MUST** be expressed in Greenwich Mean Time (Zulu) and **MUST** include seconds (i.e., times are `YYYYMMDDHHMMSSZ`), even where the number of seconds is zero. `GeneralizedTime` values **MUST NOT** include fractional seconds.

4.1.2.6. Subject

The subject field identifies the entity associated with the public key stored in the subject public key field. The subject name **MAY** be carried in the subject field and/or the `subjectAltName` extension. If the subject is a CA (e.g., the basic constraints extension, as discussed in Section 4.2.1.9, is present and the value of `cA` is `TRUE`), then the subject field **MUST** be populated with a non-empty distinguished name matching the contents of the issuer field (Section 4.1.2.4) in all certificates issued by the subject CA. If the subject is a CRL issuer (e.g., the key usage extension, as discussed in Section 4.2.1.3, is present and the value of `cRLSign` is `TRUE`),

then the subject field **MUST** be populated with a non-empty distinguished name matching the contents of the issuer field (Section 5.1.2.3) in all CRLs issued by the subject CRL issuer. If subject naming information is present only in the subjectAltName extension (e.g., a key bound only to an email address or URI), then the subject name **MUST** be an empty sequence and the subjectAltName extension **MUST** be critical.

Where it is non-empty, the subject field **MUST** contain an X.500 distinguished name (DN). The DN **MUST** be unique for each subject entity certified by the one CA as defined by the issuer field. A CA **MAY** issue more than one certificate with the same DN to the same subject entity.

The subject field is defined as the X.501 type Name. Implementation requirements for this field are those defined for the issuer field (Section 4.1.2.4). Implementations of this specification **MUST** be prepared to receive subject names containing the attribute types required for the issuer field. Implementations of this specification **SHOULD** be prepared to receive subject names containing the recommended attribute types for the issuer field. The syntax and associated object identifiers (OIDs) for these attribute types are provided in the ASN.1 modules in Appendix A. Implementations of this specification **MAY** use the comparison rules in Section 7.1 to process unfamiliar attribute types (i.e., for name chaining) whose attribute values use one of the encoding options from DirectoryString. Binary comparison should be used when unfamiliar attribute types include attribute values with encoding options other than those found in DirectoryString. This allows implementations to process certificates with unfamiliar attributes in the subject name.

When encoding attribute values of type DirectoryString, conforming CAs **MUST** use PrintableString or UTF8String encoding, with the following exceptions:

- (a) When the subject of the certificate is a CA, the subject field **MUST** be encoded in the same way as it is encoded in the issuer field (Section 4.1.2.4) in all certificates issued by the subject CA. Thus, if the subject CA encodes attributes in the issuer fields of certificates that it issues using the TeletexString, BMPString, or UniversalString encodings, then the subject field of certificates issued to that CA **MUST** use the same encoding.
- (b) When the subject of the certificate is a CRL issuer, the subject field **MUST** be encoded in the same way as it is encoded in the issuer field (Section 5.1.2.3) in all CRLs issued by the subject CRL issuer.

- (c) TeletexString, BMPString, and UniversalString are included for backward compatibility, and SHOULD NOT be used for certificates for new subjects. However, these types MAY be used in certificates where the name was previously established, including cases in which a new certificate is being issued to an existing subject or a certificate is being issued to a new subject where the attributes being encoded have been previously established in certificates issued to other subjects. Certificate users SHOULD be prepared to receive certificates with these types.

Legacy implementations exist where an electronic mail address is embedded in the subject distinguished name as an emailAddress attribute [RFC2985]. The attribute value for emailAddress is of type IA5String to permit inclusion of the character '@', which is not part of the PrintableString character set. emailAddress attribute values are not case-sensitive (e.g., "subscriber@example.com" is the same as "SUBSCRIBER@EXAMPLE.COM").

Conforming implementations generating new certificates with electronic mail addresses MUST use the rfc822Name in the subject alternative name extension (Section 4.2.1.6) to describe such identities. Simultaneous inclusion of the emailAddress attribute in the subject distinguished name to support legacy implementations is deprecated but permitted.

4.1.2.7. Subject Public Key Info

This field is used to carry the public key and identify the algorithm with which the key is used (e.g., RSA, DSA, or Diffie-Hellman). The algorithm is identified using the AlgorithmIdentifier structure specified in Section 4.1.1.2. The object identifiers for the supported algorithms and the methods for encoding the public key materials (public key and parameters) are specified in [RFC3279], [RFC4055], and [RFC4491].

4.1.2.8. Unique Identifiers

These fields MUST only appear if the version is 2 or 3 (Section 4.1.2.1). These fields MUST NOT appear if the version is 1. The subject and issuer unique identifiers are present in the certificate to handle the possibility of reuse of subject and/or issuer names over time. This profile RECOMMENDS that names not be reused for different entities and that Internet certificates not make use of unique identifiers. CAs conforming to this profile MUST NOT generate certificates with unique identifiers. Applications conforming to

this profile SHOULD be capable of parsing certificates that include unique identifiers, but there are no processing requirements associated with the unique identifiers.

4.1.2.9. Extensions

This field MUST only appear if the version is 3 (Section 4.1.2.1). If present, this field is a SEQUENCE of one or more certificate extensions. The format and content of certificate extensions in the Internet PKI are defined in Section 4.2.

4.2. Certificate Extensions

The extensions defined for X.509 v3 certificates provide methods for associating additional attributes with users or public keys and for managing relationships between CAs. The X.509 v3 certificate format also allows communities to define private extensions to carry information unique to those communities. Each extension in a certificate is designated as either critical or non-critical. A certificate-using system MUST reject the certificate if it encounters a critical extension it does not recognize or a critical extension that contains information that it cannot process. A non-critical extension MAY be ignored if it is not recognized, but MUST be processed if it is recognized. The following sections present recommended extensions used within Internet certificates and standard locations for information. Communities may elect to use additional extensions; however, caution ought to be exercised in adopting any critical extensions in certificates that might prevent use in a general context.

Each extension includes an OID and an ASN.1 structure. When an extension appears in a certificate, the OID appears as the field extnID and the corresponding ASN.1 DER encoded structure is the value of the octet string extnValue. A certificate MUST NOT include more than one instance of a particular extension. For example, a certificate may contain only one authority key identifier extension (Section 4.2.1.1). An extension includes the boolean critical, with a default value of FALSE. The text for each extension specifies the acceptable values for the critical field for CAs conforming to this profile.

Conforming CAs MUST support key identifiers (Sections 4.2.1.1 and 4.2.1.2), basic constraints (Section 4.2.1.9), key usage (Section 4.2.1.3), and certificate policies (Section 4.2.1.4) extensions. If the CA issues certificates with an empty sequence for the subject field, the CA MUST support the subject alternative name extension (Section 4.2.1.6). Support for the remaining extensions is OPTIONAL. Conforming CAs MAY support extensions that are not identified within

this specification; certificate issuers are cautioned that marking such extensions as critical may inhibit interoperability.

At a minimum, applications conforming to this profile MUST recognize the following extensions: key usage (Section 4.2.1.3), certificate policies (Section 4.2.1.4), subject alternative name (Section 4.2.1.6), basic constraints (Section 4.2.1.9), name constraints (Section 4.2.1.10), policy constraints (Section 4.2.1.11), extended key usage (Section 4.2.1.12), and inhibit anyPolicy (Section 4.2.1.14).

In addition, applications conforming to this profile SHOULD recognize the authority and subject key identifier (Sections 4.2.1.1 and 4.2.1.2) and policy mappings (Section 4.2.1.5) extensions.

4.2.1. Standard Extensions

This section identifies standard certificate extensions defined in [X.509] for use in the Internet PKI. Each extension is associated with an OID defined in [X.509]. These OIDs are members of the id-ce arc, which is defined by the following:

```
id-ce    OBJECT IDENTIFIER ::= { joint-iso-ccitt(2) ds(5) 29 }
```

4.2.1.1. Authority Key Identifier

The authority key identifier extension provides a means of identifying the public key corresponding to the private key used to sign a certificate. This extension is used where an issuer has multiple signing keys (either due to multiple concurrent key pairs or due to changeover). The identification MAY be based on either the key identifier (the subject key identifier in the issuer's certificate) or the issuer name and serial number.

The keyIdentifier field of the authorityKeyIdentifier extension MUST be included in all certificates generated by conforming CAs to facilitate certification path construction. There is one exception; where a CA distributes its public key in the form of a "self-signed" certificate, the authority key identifier MAY be omitted. The signature on a self-signed certificate is generated with the private key associated with the certificate's subject public key. (This proves that the issuer possesses both the public and private keys.) In this case, the subject and authority key identifiers would be identical, but only the subject key identifier is needed for certification path building.

The value of the keyIdentifier field SHOULD be derived from the public key used to verify the certificate's signature or a method

that generates unique values. Two common methods for generating key identifiers from the public key are described in Section 4.2.1.2. Where a key identifier has not been previously established, this specification RECOMMENDS use of one of these methods for generating keyIdentifiers or use of a similar method that uses a different hash algorithm. Where a key identifier has been previously established, the CA SHOULD use the previously established identifier.

This profile RECOMMENDS support for the key identifier method by all certificate users.

Conforming CAs MUST mark this extension as non-critical.

id-ce-authorityKeyIdentifier OBJECT IDENTIFIER ::= { id-ce 35 }

AuthorityKeyIdentifier ::= SEQUENCE {
keyIdentifier [0] KeyIdentifier OPTIONAL,
authorityCertIssuer [1] GeneralNames OPTIONAL,
authorityCertSerialNumber [2] CertificateSerialNumber OPTIONAL }

KeyIdentifier ::= OCTET STRING

4.2.1.2. Subject Key Identifier

The subject key identifier extension provides a means of identifying certificates that contain a particular public key.

To facilitate certification path construction, this extension MUST appear in all conforming CA certificates, that is, all certificates including the basic constraints extension (Section 4.2.1.9) where the value of cA is TRUE. In conforming CA certificates, the value of the subject key identifier MUST be the value placed in the key identifier field of the authority key identifier extension (Section 4.2.1.1) of certificates issued by the subject of this certificate. Applications are not required to verify that key identifiers match when performing certification path validation.

For CA certificates, subject key identifiers SHOULD be derived from the public key or a method that generates unique values. Two common methods for generating key identifiers from the public key are:

- (1) The keyIdentifier is composed of the 160-bit SHA-1 hash of the value of the BIT STRING subjectPublicKey (excluding the tag, length, and number of unused bits).

- (2) The `keyIdentifier` is composed of a four-bit type field with the value 0100 followed by the least significant 60 bits of the SHA-1 hash of the value of the BIT STRING `subjectPublicKey` (excluding the tag, length, and number of unused bits).

Other methods of generating unique numbers are also acceptable.

For end entity certificates, the subject key identifier extension provides a means for identifying certificates containing the particular public key used in an application. Where an end entity has obtained multiple certificates, especially from multiple CAs, the subject key identifier provides a means to quickly identify the set of certificates containing a particular public key. To assist applications in identifying the appropriate end entity certificate, this extension SHOULD be included in all end entity certificates.

For end entity certificates, subject key identifiers SHOULD be derived from the public key. Two common methods for generating key identifiers from the public key are identified above.

Where a key identifier has not been previously established, this specification RECOMMENDS use of one of these methods for generating `keyIdentifiers` or use of a similar method that uses a different hash algorithm. Where a key identifier has been previously established, the CA SHOULD use the previously established identifier.

Conforming CAs MUST mark this extension as non-critical.

`id-ce-subjectKeyIdentifier` OBJECT IDENTIFIER ::= { `id-ce` 14 }

`SubjectKeyIdentifier` ::= `KeyIdentifier`

4.2.1.3. Key Usage

The key usage extension defines the purpose (e.g., encipherment, signature, certificate signing) of the key contained in the certificate. The usage restriction might be employed when a key that could be used for more than one operation is to be restricted. For example, when an RSA key should be used only to verify signatures on objects other than public key certificates and CRLs, the `digitalSignature` and/or `nonRepudiation` bits would be asserted. Likewise, when an RSA key should be used only for key management, the `keyEncipherment` bit would be asserted.

Conforming CAs MUST include this extension in certificates that contain public keys that are used to validate digital signatures on other public key certificates or CRLs. When present, conforming CAs SHOULD mark this extension as critical.

id-ce-keyUsage OBJECT IDENTIFIER ::= { id-ce 15 }

KeyUsage ::= BIT STRING {
 digitalSignature (0),
 nonRepudiation (1), -- recent editions of X.509 have
 -- renamed this bit to contentCommitment
 keyEncipherment (2),
 dataEncipherment (3),
 keyAgreement (4),
 keyCertSign (5),
 cRLSign (6),
 encipherOnly (7),
 decipherOnly (8) }

Bits in the KeyUsage type are used as follows:

The digitalSignature bit is asserted when the subject public key is used for verifying digital signatures, other than signatures on certificates (bit 5) and CRLs (bit 6), such as those used in an entity authentication service, a data origin authentication service, and/or an integrity service.

The nonRepudiation bit is asserted when the subject public key is used to verify digital signatures, other than signatures on certificates (bit 5) and CRLs (bit 6), used to provide a non-repudiation service that protects against the signing entity falsely denying some action. In the case of later conflict, a reliable third party may determine the authenticity of the signed data. (Note that recent editions of X.509 have renamed the nonRepudiation bit to contentCommitment.)

The keyEncipherment bit is asserted when the subject public key is used for enciphering private or secret keys, i.e., for key transport. For example, this bit shall be set when an RSA public key is to be used for encrypting a symmetric content-decryption key or an asymmetric private key.

The dataEncipherment bit is asserted when the subject public key is used for directly enciphering raw user data without the use of an intermediate symmetric cipher. Note that the use of this bit is extremely uncommon; almost all applications use key transport or key agreement to establish a symmetric key.

The `keyAgreement` bit is asserted when the subject public key is used for key agreement. For example, when a Diffie-Hellman key is to be used for key management, then this bit is set.

The `keyCertSign` bit is asserted when the subject public key is used for verifying signatures on public key certificates. If the `keyCertSign` bit is asserted, then the `CA` bit in the basic constraints extension (Section 4.2.1.9) MUST also be asserted.

The `cRLSign` bit is asserted when the subject public key is used for verifying signatures on certificate revocation lists (e.g., CRLs, delta CRLs, or ARLs).

The meaning of the `encipherOnly` bit is undefined in the absence of the `keyAgreement` bit. When the `encipherOnly` bit is asserted and the `keyAgreement` bit is also set, the subject public key may be used only for enciphering data while performing key agreement.

The meaning of the `decipherOnly` bit is undefined in the absence of the `keyAgreement` bit. When the `decipherOnly` bit is asserted and the `keyAgreement` bit is also set, the subject public key may be used only for deciphering data while performing key agreement.

If the `keyUsage` extension is present, then the subject public key MUST NOT be used to verify signatures on certificates or CRLs unless the corresponding `keyCertSign` or `cRLSign` bit is set. If the subject public key is only to be used for verifying signatures on certificates and/or CRLs, then the `digitalSignature` and `nonRepudiation` bits SHOULD NOT be set. However, the `digitalSignature` and/or `nonRepudiation` bits MAY be set in addition to the `keyCertSign` and/or `cRLSign` bits if the subject public key is to be used to verify signatures on certificates and/or CRLs as well as other objects.

Combining the `nonRepudiation` bit in the `keyUsage` certificate extension with other `keyUsage` bits may have security implications depending on the context in which the certificate is to be used. Further distinctions between the `digitalSignature` and `nonRepudiation` bits may be provided in specific certificate policies.

This profile does not restrict the combinations of bits that may be set in an instantiation of the `keyUsage` extension. However, appropriate values for `keyUsage` extensions for particular algorithms are specified in [RFC3279], [RFC4055], and [RFC4491]. When the `keyUsage` extension appears in a certificate, at least one of the bits MUST be set to 1.

4.2.1.4. Certificate Policies

The certificate policies extension contains a sequence of one or more policy information terms, each of which consists of an object identifier (OID) and optional qualifiers. Optional qualifiers, which MAY be present, are not expected to change the definition of the policy. A certificate policy OID MUST NOT appear more than once in a certificate policies extension.

In an end entity certificate, these policy information terms indicate the policy under which the certificate has been issued and the purposes for which the certificate may be used. In a CA certificate, these policy information terms limit the set of policies for certification paths that include this certificate. When a CA does not wish to limit the set of policies for certification paths that include this certificate, it MAY assert the special policy anyPolicy, with a value of { 2 5 29 32 0 }.

Applications with specific policy requirements are expected to have a list of those policies that they will accept and to compare the policy OIDs in the certificate to that list. If this extension is critical, the path validation software MUST be able to interpret this extension (including the optional qualifier), or MUST reject the certificate.

To promote interoperability, this profile RECOMMENDS that policy information terms consist of only an OID. Where an OID alone is insufficient, this profile strongly recommends that the use of qualifiers be limited to those identified in this section. When qualifiers are used with the special policy anyPolicy, they MUST be limited to the qualifiers identified in this section. Only those qualifiers returned as a result of path validation are considered.

This specification defines two policy qualifier types for use by certificate policy writers and certificate issuers. The qualifier types are the CPS Pointer and User Notice qualifiers.

The CPS Pointer qualifier contains a pointer to a Certification Practice Statement (CPS) published by the CA. The pointer is in the form of a URI. Processing requirements for this qualifier are a local matter. No action is mandated by this specification regardless of the criticality value asserted for the extension.

User notice is intended for display to a relying party when a certificate is used. Only user notices returned as a result of path validation are intended for display to the user. If a notice is

duplicated, only one copy need be displayed. To prevent such duplication, this qualifier SHOULD only be present in end entity certificates and CA certificates issued to other organizations.

The user notice has two optional fields: the noticeRef field and the explicitText field. Conforming CAs SHOULD NOT use the noticeRef option.

The noticeRef field, if used, names an organization and identifies, by number, a particular textual statement prepared by that organization. For example, it might identify the organization "CertsRUs" and notice number 1. In a typical implementation, the application software will have a notice file containing the current set of notices for CertsRUs; the application will extract the notice text from the file and display it. Messages MAY be multilingual, allowing the software to select the particular language message for its own environment.

An explicitText field includes the textual statement directly in the certificate. The explicitText field is a string with a maximum size of 200 characters. Conforming CAs SHOULD use the UTF8String encoding for explicitText, but MAY use IA5String. Conforming CAs MUST NOT encode explicitText as VisibleString or BMPString. The explicitText string SHOULD NOT include any control characters (e.g., U+0000 to U+001F and U+007F to U+009F). When the UTF8String encoding is used, all character sequences SHOULD be normalized according to Unicode normalization form C (NFC) [NFC].

If both the noticeRef and explicitText options are included in the one qualifier and if the application software can locate the notice text indicated by the noticeRef option, then that text SHOULD be displayed; otherwise, the explicitText string SHOULD be displayed.

Note: While the explicitText has a maximum size of 200 characters, some non-conforming CAs exceed this limit. Therefore, certificate users SHOULD gracefully handle explicitText with more than 200 characters.

```
id-ce-certificatePolicies OBJECT IDENTIFIER ::= { id-ce 32 }

anyPolicy OBJECT IDENTIFIER ::= { id-ce-certificatePolicies 0 }

certificatePolicies ::= SEQUENCE SIZE (1..MAX) OF PolicyInformation

PolicyInformation ::= SEQUENCE {
    policyIdentifier      CertPolicyId,
    policyQualifiers      SEQUENCE SIZE (1..MAX) OF
                          PolicyQualifierInfo OPTIONAL }

CertPolicyId ::= OBJECT IDENTIFIER

PolicyQualifierInfo ::= SEQUENCE {
    policyQualifierId      PolicyQualifierId,
    qualifier              ANY DEFINED BY policyQualifierId }

-- policyQualifierIds for Internet policy qualifiers

id-qt          OBJECT IDENTIFIER ::= { id-pkix 2 }
id-qt-cps      OBJECT IDENTIFIER ::= { id-qt 1 }
id-qt-unotice  OBJECT IDENTIFIER ::= { id-qt 2 }

PolicyQualifierId ::= OBJECT IDENTIFIER ( id-qt-cps | id-qt-unotice )

Qualifier ::= CHOICE {
    cPSuri          CPSuri,
    userNotice      UserNotice }

CPSuri ::= IA5String

UserNotice ::= SEQUENCE {
    noticeRef        NoticeReference OPTIONAL,
    explicitText      DisplayText OPTIONAL }

NoticeReference ::= SEQUENCE {
    organization      DisplayText,
    noticeNumbers      SEQUENCE OF INTEGER }

DisplayText ::= CHOICE {
    ia5String          IA5String      (SIZE (1..200)),
    visibleString      VisibleString  (SIZE (1..200)),
    bmpString          BMPString      (SIZE (1..200)),
    utf8String         UTF8String     (SIZE (1..200)) }
```

4.2.1.5. Policy Mappings

This extension is used in CA certificates. It lists one or more pairs of OIDs; each pair includes an issuerDomainPolicy and a subjectDomainPolicy. The pairing indicates the issuing CA considers its issuerDomainPolicy equivalent to the subject CA's subjectDomainPolicy.

The issuing CA's users might accept an issuerDomainPolicy for certain applications. The policy mapping defines the list of policies associated with the subject CA that may be accepted as comparable to the issuerDomainPolicy.

Each issuerDomainPolicy named in the policy mappings extension SHOULD also be asserted in a certificate policies extension in the same certificate. Policies MUST NOT be mapped either to or from the special value anyPolicy (Section 4.2.1.4).

In general, certificate policies that appear in the issuerDomainPolicy field of the policy mappings extension are not considered acceptable policies for inclusion in subsequent certificates in the certification path. In some circumstances, a CA may wish to map from one policy (p1) to another (p2), but still wants the issuerDomainPolicy (p1) to be considered acceptable for inclusion in subsequent certificates. This may occur, for example, if the CA is in the process of transitioning from the use of policy p1 to the use of policy p2 and has valid certificates that were issued under each of the policies. A CA may indicate this by including two policy mappings in the CA certificates that it issues. Each policy mapping would have an issuerDomainPolicy of p1; one policy mapping would have a subjectDomainPolicy of p1 and the other would have a subjectDomainPolicy of p2.

This extension MAY be supported by CAs and/or applications. Conforming CAs SHOULD mark this extension as critical.

id-ce-policyMappings OBJECT IDENTIFIER ::= { id-ce 33 }

```
PolicyMappings ::= SEQUENCE SIZE (1..MAX) OF SEQUENCE {
    issuerDomainPolicy      CertPolicyId,
    subjectDomainPolicy     CertPolicyId }
```

4.2.1.6. Subject Alternative Name

The subject alternative name extension allows identities to be bound to the subject of the certificate. These identities may be included in addition to or in place of the identity in the subject field of the certificate. Defined options include an Internet electronic mail

address, a DNS name, an IP address, and a Uniform Resource Identifier (URI). Other options exist, including completely local definitions. Multiple name forms, and multiple instances of each name form, MAY be included. Whenever such identities are to be bound into a certificate, the subject alternative name (or issuer alternative name) extension MUST be used; however, a DNS name MAY also be represented in the subject field using the domainComponent attribute as described in Section 4.1.2.4. Note that where such names are represented in the subject field implementations are not required to convert them into DNS names.

Because the subject alternative name is considered to be definitively bound to the public key, all parts of the subject alternative name MUST be verified by the CA.

Further, if the only subject identity included in the certificate is an alternative name form (e.g., an electronic mail address), then the subject distinguished name MUST be empty (an empty sequence), and the subjectAltName extension MUST be present. If the subject field contains an empty sequence, then the issuing CA MUST include a subjectAltName extension that is marked as critical. When including the subjectAltName extension in a certificate that has a non-empty subject distinguished name, conforming CAs SHOULD mark the subjectAltName extension as non-critical.

When the subjectAltName extension contains an Internet mail address, the address MUST be stored in the rfc822Name. The format of an rfc822Name is a "Mailbox" as defined in Section 4.1.2 of [RFC2821]. A Mailbox has the form "Local-part@Domain". Note that a Mailbox has no phrase (such as a common name) before it, has no comment (text surrounded in parentheses) after it, and is not surrounded by "<" and ">". Rules for encoding Internet mail addresses that include internationalized domain names are specified in Section 7.5.

When the subjectAltName extension contains an ipAddress, the address MUST be stored in the octet string in "network byte order", as specified in [RFC791]. The least significant bit (LSB) of each octet is the LSB of the corresponding byte in the network address. For IP version 4, as specified in [RFC791], the octet string MUST contain exactly four octets. For IP version 6, as specified in [RFC2460], the octet string MUST contain exactly sixteen octets.

When the subjectAltName extension contains a domain name system label, the domain name MUST be stored in the dNSName (an IA5String). The name MUST be in the "preferred name syntax", as specified by Section 3.5 of [RFC1034] and as modified by Section 2.1 of [RFC1123]. Note that while uppercase and lowercase letters are allowed in domain names, no significance is attached to the case. In

addition, while the string " " is a legal domain name, subjectAltName extensions with a dNSName of " " MUST NOT be used. Finally, the use of the DNS representation for Internet mail addresses (subscriber.example.com instead of subscriber@example.com) MUST NOT be used; such identities are to be encoded as rfc822Name. Rules for encoding internationalized domain names are specified in Section 7.2.

When the subjectAltName extension contains a URI, the name MUST be stored in the uniformResourceIdentifier (an IA5String). The name MUST NOT be a relative URI, and it MUST follow the URI syntax and encoding rules specified in [RFC3986]. The name MUST include both a scheme (e.g., "http" or "ftp") and a scheme-specific-part. URIs that include an authority ([RFC3986], Section 3.2) MUST include a fully qualified domain name or IP address as the host. Rules for encoding Internationalized Resource Identifiers (IRIs) are specified in Section 7.4.

As specified in [RFC3986], the scheme name is not case-sensitive (e.g., "http" is equivalent to "HTTP"). The host part, if present, is also not case-sensitive, but other components of the scheme-specific-part may be case-sensitive. Rules for comparing URIs are specified in Section 7.4.

When the subjectAltName extension contains a DN in the directoryName, the encoding rules are the same as those specified for the issuer field in Section 4.1.2.4. The DN MUST be unique for each subject entity certified by the one CA as defined by the issuer field. A CA MAY issue more than one certificate with the same DN to the same subject entity.

The subjectAltName MAY carry additional name types through the use of the otherName field. The format and semantics of the name are indicated through the OBJECT IDENTIFIER in the type-id field. The name itself is conveyed as value field in otherName. For example, Kerberos [RFC4120] format names can be encoded into the otherName, using a Kerberos 5 principal name OID and a SEQUENCE of the Realm and the PrincipalName.

Subject alternative names MAY be constrained in the same manner as subject distinguished names using the name constraints extension as described in Section 4.2.1.10.

If the subjectAltName extension is present, the sequence MUST contain at least one entry. Unlike the subject field, conforming CAs MUST NOT issue certificates with subjectAltNames containing empty GeneralName fields. For example, an rfc822Name is represented as an IA5String. While an empty string is a valid IA5String, such an rfc822Name is not permitted by this profile. The behavior of clients

that encounter such a certificate when processing a certification path is not defined by this profile.

Finally, the semantics of subject alternative names that include wildcard characters (e.g., as a placeholder for a set of names) are not addressed by this specification. Applications with specific requirements MAY use such names, but they must define the semantics.

id-ce-subjectAltName OBJECT IDENTIFIER ::= { id-ce 17 }

SubjectAltName ::= GeneralNames

GeneralNames ::= SEQUENCE SIZE (1..MAX) OF GeneralName

```
GeneralName ::= CHOICE {
    otherName                [0]      OtherName,
    rfc822Name                [1]      IA5String,
    dNSName                   [2]      IA5String,
    x400Address               [3]      ORAddress,
    directoryName             [4]      Name,
    ediPartyName              [5]      EDIPartyName,
    uniformResourceIdentifier [6]      IA5String,
    iPAddress                 [7]      OCTET STRING,
    registeredID              [8]      OBJECT IDENTIFIER }
```

```
OtherName ::= SEQUENCE {
    type-id    OBJECT IDENTIFIER,
    value      [0] EXPLICIT ANY DEFINED BY type-id }
```

```
EDIPartyName ::= SEQUENCE {
    nameAssigner [0]      DirectoryString OPTIONAL,
    partyName    [1]      DirectoryString }
```

4.2.1.7. Issuer Alternative Name

As with Section 4.2.1.6, this extension is used to associate Internet style identities with the certificate issuer. Issuer alternative name MUST be encoded as in 4.2.1.6. Issuer alternative names are not processed as part of the certification path validation algorithm in Section 6. (That is, issuer alternative names are not used in name chaining and name constraints are not enforced.)

Where present, conforming CAs SHOULD mark this extension as non-critical.

id-ce-issuerAltName OBJECT IDENTIFIER ::= { id-ce 18 }

IssuerAltName ::= GeneralNames

4.2.1.8. Subject Directory Attributes

The subject directory attributes extension is used to convey identification attributes (e.g., nationality) of the subject. The extension is defined as a sequence of one or more attributes. Conforming CAs MUST mark this extension as non-critical.

id-ce-subjectDirectoryAttributes OBJECT IDENTIFIER ::= { id-ce 9 }

SubjectDirectoryAttributes ::= SEQUENCE SIZE (1..MAX) OF Attribute

4.2.1.9. Basic Constraints

The basic constraints extension identifies whether the subject of the certificate is a CA and the maximum depth of valid certification paths that include this certificate.

The cA boolean indicates whether the certified public key may be used to verify certificate signatures. If the cA boolean is not asserted, then the keyCertSign bit in the key usage extension MUST NOT be asserted. If the basic constraints extension is not present in a version 3 certificate, or the extension is present but the cA boolean is not asserted, then the certified public key MUST NOT be used to verify certificate signatures.

The pathLenConstraint field is meaningful only if the cA boolean is asserted and the key usage extension, if present, asserts the keyCertSign bit (Section 4.2.1.3). In this case, it gives the maximum number of non-self-issued intermediate certificates that may follow this certificate in a valid certification path. (Note: The last certificate in the certification path is not an intermediate certificate, and is not included in this limit. Usually, the last certificate is an end entity certificate, but it can be a CA certificate.) A pathLenConstraint of zero indicates that no non-self-issued intermediate CA certificates may follow in a valid certification path. Where it appears, the pathLenConstraint field MUST be greater than or equal to zero. Where pathLenConstraint does not appear, no limit is imposed.

Conforming CAs MUST include this extension in all CA certificates that contain public keys used to validate digital signatures on certificates and MUST mark the extension as critical in such certificates. This extension MAY appear as a critical or non-critical extension in CA certificates that contain public keys used exclusively for purposes other than validating digital signatures on certificates. Such CA certificates include ones that contain public keys used exclusively for validating digital signatures on CRLs and ones that contain key management public keys used with certificate

enrollment protocols. This extension MAY appear as a critical or non-critical extension in end entity certificates.

CAs MUST NOT include the pathLenConstraint field unless the cA boolean is asserted and the key usage extension asserts the keyCertSign bit.

id-ce-basicConstraints OBJECT IDENTIFIER ::= { id-ce 19 }

```
BasicConstraints ::= SEQUENCE {  
    cA                      BOOLEAN DEFAULT FALSE,  
    pathLenConstraint       INTEGER (0..MAX) OPTIONAL }
```

4.2.1.10. Name Constraints

The name constraints extension, which MUST be used only in a CA certificate, indicates a name space within which all subject names in subsequent certificates in a certification path MUST be located. Restrictions apply to the subject distinguished name and apply to subject alternative names. Restrictions apply only when the specified name form is present. If no name of the type is in the certificate, the certificate is acceptable.

Name constraints are not applied to self-issued certificates (unless the certificate is the final certificate in the path). (This could prevent CAs that use name constraints from employing self-issued certificates to implement key rollover.)

Restrictions are defined in terms of permitted or excluded name subtrees. Any name matching a restriction in the excludedSubtrees field is invalid regardless of information appearing in the permittedSubtrees. Conforming CAs MUST mark this extension as critical and SHOULD NOT impose name constraints on the x400Address, ediPartyName, or registeredID name forms. Conforming CAs MUST NOT issue certificates where name constraints is an empty sequence. That is, either the permittedSubtrees field or the excludedSubtrees MUST be present.

Applications conforming to this profile MUST be able to process name constraints that are imposed on the directoryName name form and SHOULD be able to process name constraints that are imposed on the rfc822Name, uniformResourceIdentifier, dNSName, and iPAddress name forms. If a name constraints extension that is marked as critical imposes constraints on a particular name form, and an instance of that name form appears in the subject field or subjectAltName extension of a subsequent certificate, then the application MUST either process the constraint or reject the certificate.

Within this profile, the minimum and maximum fields are not used with any name forms, thus, the minimum MUST be zero, and maximum MUST be absent. However, if an application encounters a critical name constraints extension that specifies other values for minimum or maximum for a name form that appears in a subsequent certificate, the application MUST either process these fields or reject the certificate.

For URIs, the constraint applies to the host part of the name. The constraint MUST be specified as a fully qualified domain name and MAY specify a host or a domain. Examples would be "host.example.com" and ".example.com". When the constraint begins with a period, it MAY be expanded with one or more labels. That is, the constraint ".example.com" is satisfied by both host.example.com and my.host.example.com. However, the constraint ".example.com" is not satisfied by "example.com". When the constraint does not begin with a period, it specifies a host. If a constraint is applied to the uniformResourceIdentifier name form and a subsequent certificate includes a subjectAltName extension with a uniformResourceIdentifier that does not include an authority component with a host name specified as a fully qualified domain name (e.g., if the URI either does not include an authority component or includes an authority component in which the host name is specified as an IP address), then the application MUST reject the certificate.

A name constraint for Internet mail addresses MAY specify a particular mailbox, all addresses at a particular host, or all mailboxes in a domain. To indicate a particular mailbox, the constraint is the complete mail address. For example, "root@example.com" indicates the root mailbox on the host "example.com". To indicate all Internet mail addresses on a particular host, the constraint is specified as the host name. For example, the constraint "example.com" is satisfied by any mail address at the host "example.com". To specify any address within a domain, the constraint is specified with a leading period (as with URIs). For example, ".example.com" indicates all the Internet mail addresses in the domain "example.com", but not Internet mail addresses on the host "example.com".

DNS name restrictions are expressed as host.example.com. Any DNS name that can be constructed by simply adding zero or more labels to the left-hand side of the name satisfies the name constraint. For example, www.host.example.com would satisfy the constraint but host1.example.com would not.

Legacy implementations exist where an electronic mail address is embedded in the subject distinguished name in an attribute of type emailAddress (Section 4.1.2.6). When constraints are imposed on the

rfc822Name name form, but the certificate does not include a subject alternative name, the rfc822Name constraint MUST be applied to the attribute of type emailAddress in the subject distinguished name. The ASN.1 syntax for emailAddress and the corresponding OID are supplied in Appendix A.

Restrictions of the form directoryName MUST be applied to the subject field in the certificate (when the certificate includes a non-empty subject field) and to any names of type directoryName in the subjectAltName extension. Restrictions of the form x400Address MUST be applied to any names of type x400Address in the subjectAltName extension.

When applying restrictions of the form directoryName, an implementation MUST compare DN attributes. At a minimum, implementations MUST perform the DN comparison rules specified in Section 7.1. CAs issuing certificates with a restriction of the form directoryName SHOULD NOT rely on implementation of the full ISO DN name comparison algorithm. This implies name restrictions MUST be stated identically to the encoding used in the subject field or subjectAltName extension.

The syntax of ipAddress MUST be as described in Section 4.2.1.6 with the following additions specifically for name constraints. For IPv4 addresses, the ipAddress field of GeneralName MUST contain eight (8) octets, encoded in the style of RFC 4632 (CIDR) to represent an address range [RFC4632]. For IPv6 addresses, the ipAddress field MUST contain 32 octets similarly encoded. For example, a name constraint for "class C" subnet 192.0.2.0 is represented as the octets C0 00 02 00 FF FF FF 00, representing the CIDR notation 192.0.2.0/24 (mask 255.255.255.0).

Additional rules for encoding and processing name constraints are specified in Section 7.

The syntax and semantics for name constraints for otherName, ediPartyName, and registeredID are not defined by this specification, however, syntax and semantics for name constraints for other name forms may be specified in other documents.

```
id-ce-nameConstraints OBJECT IDENTIFIER ::= { id-ce 30 }
```

```
NameConstraints ::= SEQUENCE {  
    permittedSubtrees      [0]      GeneralSubtrees OPTIONAL,  
    excludedSubtrees       [1]      GeneralSubtrees OPTIONAL }
```

```
GeneralSubtrees ::= SEQUENCE SIZE (1..MAX) OF GeneralSubtree
```

```
GeneralSubtree ::= SEQUENCE {
    base                GeneralName,
    minimum              [0]    BaseDistance DEFAULT 0,
    maximum              [1]    BaseDistance OPTIONAL }

BaseDistance ::= INTEGER (0..MAX)
```

4.2.1.11. Policy Constraints

The policy constraints extension can be used in certificates issued to CAs. The policy constraints extension constrains path validation in two ways. It can be used to prohibit policy mapping or require that each certificate in a path contain an acceptable policy identifier.

If the inhibitPolicyMapping field is present, the value indicates the number of additional certificates that may appear in the path before policy mapping is no longer permitted. For example, a value of one indicates that policy mapping may be processed in certificates issued by the subject of this certificate, but not in additional certificates in the path.

If the requireExplicitPolicy field is present, the value of requireExplicitPolicy indicates the number of additional certificates that may appear in the path before an explicit policy is required for the entire path. When an explicit policy is required, it is necessary for all certificates in the path to contain an acceptable policy identifier in the certificate policies extension. An acceptable policy identifier is the identifier of a policy required by the user of the certification path or the identifier of a policy that has been declared equivalent through policy mapping.

Conforming applications MUST be able to process the requireExplicitPolicy field and SHOULD be able to process the inhibitPolicyMapping field. Applications that support the inhibitPolicyMapping field MUST also implement support for the policyMappings extension. If the policyConstraints extension is marked as critical and the inhibitPolicyMapping field is present, applications that do not implement support for the inhibitPolicyMapping field MUST reject the certificate.

Conforming CAs MUST NOT issue certificates where policy constraints is an empty sequence. That is, either the inhibitPolicyMapping field or the requireExplicitPolicy field MUST be present. The behavior of clients that encounter an empty policy constraints field is not addressed in this profile.

Conforming CAs MUST mark this extension as critical.

id-ce-policyConstraints OBJECT IDENTIFIER ::= { id-ce 36 }

PolicyConstraints ::= SEQUENCE {
 requireExplicitPolicy [0] SkipCerts OPTIONAL,
 inhibitPolicyMapping [1] SkipCerts OPTIONAL }

SkipCerts ::= INTEGER (0..MAX)

4.2.1.12. Extended Key Usage

This extension indicates one or more purposes for which the certified public key may be used, in addition to or in place of the basic purposes indicated in the key usage extension. In general, this extension will appear only in end entity certificates. This extension is defined as follows:

id-ce-extKeyUsage OBJECT IDENTIFIER ::= { id-ce 37 }

ExtKeyUsageSyntax ::= SEQUENCE SIZE (1..MAX) OF KeyPurposeId

KeyPurposeId ::= OBJECT IDENTIFIER

Key purposes may be defined by any organization with a need. Object identifiers used to identify key purposes MUST be assigned in accordance with IANA or ITU-T Recommendation X.660 [X.660].

This extension MAY, at the option of the certificate issuer, be either critical or non-critical.

If the extension is present, then the certificate MUST only be used for one of the purposes indicated. If multiple purposes are indicated the application need not recognize all purposes indicated, as long as the intended purpose is present. Certificate using applications MAY require that the extended key usage extension be present and that a particular purpose be indicated in order for the certificate to be acceptable to that application.

If a CA includes extended key usages to satisfy such applications, but does not wish to restrict usages of the key, the CA can include the special KeyPurposeId anyExtendedKeyUsage in addition to the particular key purposes required by the applications. Conforming CAs SHOULD NOT mark this extension as critical if the anyExtendedKeyUsage KeyPurposeId is present. Applications that require the presence of a particular purpose MAY reject certificates that include the anyExtendedKeyUsage OID but not the particular OID expected for the application.

If a certificate contains both a key usage extension and an extended key usage extension, then both extensions MUST be processed independently and the certificate MUST only be used for a purpose consistent with both extensions. If there is no purpose consistent with both extensions, then the certificate MUST NOT be used for any purpose.

The following key usage purposes are defined:

anyExtendedKeyUsage OBJECT IDENTIFIER ::= { id-ce-extKeyUsage 0 }

id-kp OBJECT IDENTIFIER ::= { id-pkix 3 }

id-kp-serverAuth OBJECT IDENTIFIER ::= { id-kp 1 }

-- TLS WWW server authentication

-- Key usage bits that may be consistent: digitalSignature,

-- keyEncipherment or keyAgreement

id-kp-clientAuth OBJECT IDENTIFIER ::= { id-kp 2 }

-- TLS WWW client authentication

-- Key usage bits that may be consistent: digitalSignature

-- and/or keyAgreement

id-kp-codeSigning OBJECT IDENTIFIER ::= { id-kp 3 }

-- Signing of downloadable executable code

-- Key usage bits that may be consistent: digitalSignature

id-kp-emailProtection OBJECT IDENTIFIER ::= { id-kp 4 }

-- Email protection

-- Key usage bits that may be consistent: digitalSignature,

-- nonRepudiation, and/or (keyEncipherment or keyAgreement)

id-kp-timeStamping OBJECT IDENTIFIER ::= { id-kp 8 }

-- Binding the hash of an object to a time

-- Key usage bits that may be consistent: digitalSignature

-- and/or nonRepudiation

id-kp-OCSPSigning OBJECT IDENTIFIER ::= { id-kp 9 }

-- Signing OCSP responses

-- Key usage bits that may be consistent: digitalSignature

-- and/or nonRepudiation

4.2.1.13. CRL Distribution Points

The CRL distribution points extension identifies how CRL information is obtained. The extension SHOULD be non-critical, but this profile RECOMMENDS support for this extension by CAs and applications.

Further discussion of CRL management is contained in Section 5.

The `cRLDistributionPoints` extension is a SEQUENCE of `DistributionPoint`. A `DistributionPoint` consists of three fields, each of which is optional: `distributionPoint`, `reasons`, and `cRLIssuer`. While each of these fields is optional, a `DistributionPoint` MUST NOT consist of only the `reasons` field; either `distributionPoint` or `cRLIssuer` MUST be present. If the certificate issuer is not the CRL issuer, then the `cRLIssuer` field MUST be present and contain the Name of the CRL issuer. If the certificate issuer is also the CRL issuer, then conforming CAs MUST omit the `cRLIssuer` field and MUST include the `distributionPoint` field.

When the `distributionPoint` field is present, it contains either a SEQUENCE of general names or a single value, `nameRelativeToCRLIssuer`. If the `DistributionPointName` contains multiple values, each name describes a different mechanism to obtain the same CRL. For example, the same CRL could be available for retrieval through both LDAP and HTTP.

If the `distributionPoint` field contains a `directoryName`, the entry for that `directoryName` contains the current CRL for the associated reasons and the CRL is issued by the associated `cRLIssuer`. The CRL may be stored in either the `certificateRevocationList` or `authorityRevocationList` attribute. The CRL is to be obtained by the application from whatever directory server is locally configured. The protocol the application uses to access the directory (e.g., DAP or LDAP) is a local matter.

If the `DistributionPointName` contains a general name of type URI, the following semantics MUST be assumed: the URI is a pointer to the current CRL for the associated reasons and will be issued by the associated `cRLIssuer`. When the HTTP or FTP URI scheme is used, the URI MUST point to a single DER encoded CRL as specified in [RFC2585]. HTTP server implementations accessed via the URI SHOULD specify the media type `application/pkix-crl` in the `content-type` header field of the response. When the LDAP URI scheme [RFC4516] is used, the URI MUST include a `<dn>` field containing the distinguished name of the entry holding the CRL, MUST include a single `<attrdesc>` that contains an appropriate attribute description for the attribute that holds the CRL [RFC4523], and SHOULD include a `<host>` (e.g., `<ldap://ldap.example.com/cn=example%20CA,dc=example,dc=com?certificateRevocationList;binary>`). Omitting the `<host>` (e.g., `<ldap:///cn=CA,dc=example,dc=com?authorityRevocationList;binary>`) has the effect of relying on whatever a priori knowledge the client might have to contact an appropriate server. When present, `DistributionPointName` SHOULD include at least one LDAP or HTTP URI.

If the `DistributionPointName` contains the single value `nameRelativeToCRLIssuer`, the value provides a distinguished name

fragment. The fragment is appended to the X.500 distinguished name of the CRL issuer to obtain the distribution point name. If the `cRLIssuer` field in the `DistributionPoint` is present, then the name fragment is appended to the distinguished name that it contains; otherwise, the name fragment is appended to the certificate issuer distinguished name. Conforming CAs SHOULD NOT use `nameRelativeToCRLIssuer` to specify distribution point names. The `DistributionPointName` MUST NOT use the `nameRelativeToCRLIssuer` alternative when `cRLIssuer` contains more than one distinguished name.

If the `DistributionPoint` omits the `reasons` field, the CRL MUST include revocation information for all reasons. This profile RECOMMENDS against segmenting CRLs by reason code. When a conforming CA includes a `cRLDistributionPoints` extension in a certificate, it MUST include at least one `DistributionPoint` that points to a CRL that covers the certificate for all reasons.

The `cRLIssuer` identifies the entity that signs and issues the CRL. If present, the `cRLIssuer` MUST only contain the distinguished name (DN) from the issuer field of the CRL to which the `DistributionPoint` is pointing. The encoding of the name in the `cRLIssuer` field MUST be exactly the same as the encoding in issuer field of the CRL. If the `cRLIssuer` field is included and the DN in that field does not correspond to an X.500 or LDAP directory entry where CRL is located, then conforming CAs MUST include the `distributionPoint` field.

`id-ce-cRLDistributionPoints` OBJECT IDENTIFIER ::= { id-ce 31 }

`cRLDistributionPoints` ::= SEQUENCE SIZE (1..MAX) OF `DistributionPoint`

```
DistributionPoint ::= SEQUENCE {
    distributionPoint      [0]      DistributionPointName OPTIONAL,
    reasons                [1]      ReasonFlags OPTIONAL,
    cRLIssuer              [2]      GeneralNames OPTIONAL }
```

```
DistributionPointName ::= CHOICE {
    fullName               [0]      GeneralNames,
    nameRelativeToCRLIssuer [1]      RelativeDistinguishedName }
```

```
ReasonFlags ::= BIT STRING {  
    unused                (0),  
    keyCompromise         (1),  
    cACompromise          (2),  
    affiliationChanged     (3),  
    superseded            (4),  
    cessationOfOperation  (5),  
    certificateHold        (6),  
    privilegeWithdrawn    (7),  
    aACompromise          (8) }
```

4.2.1.14. Inhibit anyPolicy

The inhibit anyPolicy extension can be used in certificates issued to CAs. The inhibit anyPolicy extension indicates that the special anyPolicy OID, with the value { 2 5 29 32 0 }, is not considered an explicit match for other certificate policies except when it appears in an intermediate self-issued CA certificate. The value indicates the number of additional non-self-issued certificates that may appear in the path before anyPolicy is no longer permitted. For example, a value of one indicates that anyPolicy may be processed in certificates issued by the subject of this certificate, but not in additional certificates in the path.

Conforming CAs MUST mark this extension as critical.

```
id-ce-inhibitAnyPolicy OBJECT IDENTIFIER ::= { id-ce 54 }
```

```
InhibitAnyPolicy ::= SkipCerts
```

```
SkipCerts ::= INTEGER (0..MAX)
```

4.2.1.15. Freshest CRL (a.k.a. Delta CRL Distribution Point)

The freshest CRL extension identifies how delta CRL information is obtained. The extension MUST be marked as non-critical by conforming CAs. Further discussion of CRL management is contained in Section 5.

The same syntax is used for this extension and the cRLDistributionPoints extension, and is described in Section 4.2.1.13. The same conventions apply to both extensions.

```
id-ce-freshestCRL OBJECT IDENTIFIER ::= { id-ce 46 }
```

```
FreshestCRL ::= CRLDistributionPoints
```


4.2.2. Private Internet Extensions

This section defines two extensions for use in the Internet Public Key Infrastructure. These extensions may be used to direct applications to on-line information about the issuer or the subject. Each extension contains a sequence of access methods and access locations. The access method is an object identifier that indicates the type of information that is available. The access location is a GeneralName that implicitly specifies the location and format of the information and the method for obtaining the information.

Object identifiers are defined for the private extensions. The object identifiers associated with the private extensions are defined under the arc id-pe within the arc id-pkix. Any future extensions defined for the Internet PKI are also expected to be defined under the arc id-pe.

```
id-pkix OBJECT IDENTIFIER ::=
    { iso(1) identified-organization(3) dod(6) internet(1)
      security(5) mechanisms(5) pkix(7) }
```

```
id-pe OBJECT IDENTIFIER ::= { id-pkix 1 }
```

4.2.2.1. Authority Information Access

The authority information access extension indicates how to access information and services for the issuer of the certificate in which the extension appears. Information and services may include on-line validation services and CA policy data. (The location of CRLs is not specified in this extension; that information is provided by the CRLDistributionPoints extension.) This extension may be included in end entity or CA certificates. Conforming CAs MUST mark this extension as non-critical.

```
id-pe-authorityInfoAccess OBJECT IDENTIFIER ::= { id-pe 1 }
```

```
AuthorityInfoAccessSyntax ::=
    SEQUENCE SIZE (1..MAX) OF AccessDescription
```

```
AccessDescription ::= SEQUENCE {
    accessMethod      OBJECT IDENTIFIER,
    accessLocation    GeneralName }
```

```
id-ad OBJECT IDENTIFIER ::= { id-pkix 48 }
```

```
id-ad-caIssuers OBJECT IDENTIFIER ::= { id-ad 2 }
```

```
id-ad-ocsp OBJECT IDENTIFIER ::= { id-ad 1 }
```

Each entry in the sequence `AuthorityInfoAccessSyntax` describes the format and location of additional information provided by the issuer of the certificate in which this extension appears. The type and format of the information are specified by the `accessMethod` field; the `accessLocation` field specifies the location of the information. The retrieval mechanism may be implied by the `accessMethod` or specified by `accessLocation`.

This profile defines two `accessMethod` OIDs: `id-ad-caIssuers` and `id-ad-ocsp`.

In a public key certificate, the `id-ad-caIssuers` OID is used when the additional information lists certificates that were issued to the CA that issued the certificate containing this extension. The referenced CA issuers description is intended to aid certificate users in the selection of a certification path that terminates at a point trusted by the certificate user.

When `id-ad-caIssuers` appears as `accessMethod`, the `accessLocation` field describes the referenced description server and the access protocol to obtain the referenced description. The `accessLocation` field is defined as a `GeneralName`, which can take several forms.

When the `accessLocation` is a `directoryName`, the information is to be obtained by the application from whatever directory server is locally configured. The entry for the `directoryName` contains CA certificates in the `crossCertificatePair` and/or `cACertificate` attributes as specified in [RFC4523]. The protocol that application uses to access the directory (e.g., DAP or LDAP) is a local matter.

Where the information is available via LDAP, the `accessLocation` SHOULD be a `uniformResourceIdentifier`. The LDAP URI [RFC4516] MUST include a `<dn>` field containing the distinguished name of the entry holding the certificates, MUST include an `<attributes>` field that lists appropriate attribute descriptions for the attributes that hold the DER encoded certificates or cross-certificate pairs [RFC4523], and SHOULD include a `<host>` (e.g., `<ldap://ldap.example.com/cn=CA,dc=example,dc=com?cACertificate;binary,crossCertificatePair;binary>`). Omitting the `<host>` (e.g., `<ldap:///cn=exampleCA,dc=example,dc=com?cACertificate;binary>`) has the effect of relying on whatever a priori knowledge the client might have to contact an appropriate server.

Where the information is available via HTTP or FTP, `accessLocation` MUST be a `uniformResourceIdentifier` and the URI MUST point to either a single DER encoded certificate as specified in [RFC2585] or a collection of certificates in a BER or DER encoded "certs-only" CMS message as specified in [RFC2797].

Conforming applications that support HTTP or FTP for accessing certificates MUST be able to accept individual DER encoded certificates and SHOULD be able to accept "certs-only" CMS messages.

HTTP server implementations accessed via the URI SHOULD specify the media type application/pkix-cert [RFC2585] in the content-type header field of the response for a single DER encoded certificate and SHOULD specify the media type application/pkcs7-mime [RFC2797] in the content-type header field of the response for "certs-only" CMS messages. For FTP, the name of a file that contains a single DER encoded certificate SHOULD have a suffix of ".cer" [RFC2585] and the name of a file that contains a "certs-only" CMS message SHOULD have a suffix of ".p7c" [RFC2797]. Consuming clients may use the media type or file extension as a hint to the content, but should not depend solely on the presence of the correct media type or file extension in the server response.

The semantics of other id-ad-caIssuers accessLocation name forms are not defined.

An authorityInfoAccess extension may include multiple instances of the id-ad-caIssuers accessMethod. The different instances may specify different methods for accessing the same information or may point to different information. When the id-ad-caIssuers accessMethod is used, at least one instance SHOULD specify an accessLocation that is an HTTP [RFC2616] or LDAP [RFC4516] URI.

The id-ad-ocsp OID is used when revocation information for the certificate containing this extension is available using the Online Certificate Status Protocol (OCSP) [RFC2560].

When id-ad-ocsp appears as accessMethod, the accessLocation field is the location of the OCSP responder, using the conventions defined in [RFC2560].

Additional access descriptors may be defined in other PKIX specifications.

4.2.2.2. Subject Information Access

The subject information access extension indicates how to access information and services for the subject of the certificate in which the extension appears. When the subject is a CA, information and services may include certificate validation services and CA policy data. When the subject is an end entity, the information describes the type of services offered and how to access them. In this case, the contents of this extension are defined in the protocol

specifications for the supported services. This extension may be included in end entity or CA certificates. Conforming CAs MUST mark this extension as non-critical.

id-pe-subjectInfoAccess OBJECT IDENTIFIER ::= { id-pe 11 }

SubjectInfoAccessSyntax ::=
SEQUENCE SIZE (1..MAX) OF AccessDescription

AccessDescription ::= SEQUENCE {
accessMethod OBJECT IDENTIFIER,
accessLocation GeneralName }

Each entry in the sequence SubjectInfoAccessSyntax describes the format and location of additional information provided by the subject of the certificate in which this extension appears. The type and format of the information are specified by the accessMethod field; the accessLocation field specifies the location of the information. The retrieval mechanism may be implied by the accessMethod or specified by accessLocation.

This profile defines one access method to be used when the subject is a CA and one access method to be used when the subject is an end entity. Additional access methods may be defined in the future in the protocol specifications for other services.

The id-ad-caRepository OID is used when the subject is a CA that publishes certificates it issues in a repository. The accessLocation field is defined as a GeneralName, which can take several forms.

When the accessLocation is a directoryName, the information is to be obtained by the application from whatever directory server is locally configured. When the extension is used to point to CA certificates, the entry for the directoryName contains CA certificates in the crossCertificatePair and/or cACertificate attributes as specified in [RFC4523]. The protocol the application uses to access the directory (e.g., DAP or LDAP) is a local matter.

Where the information is available via LDAP, the accessLocation SHOULD be a uniformResourceIdentifier. The LDAP URI [RFC4516] MUST include a <dn> field containing the distinguished name of the entry holding the certificates, MUST include an <attributes> field that lists appropriate attribute descriptions for the attributes that hold the DER encoded certificates or cross-certificate pairs [RFC4523], and SHOULD include a <host> (e.g., <ldap://ldap.example.com/cn=CA, dc=example,dc=com?cACertificate;binary,crossCertificatePair;binary>).

Omitting the <host> (e.g., <ldap:///cn=exampleCA,dc=example,dc=com?cACertificate;binary>) has the effect of relying on whatever a priori knowledge the client might have to contact an appropriate server.

Where the information is available via HTTP or FTP, accessLocation MUST be a uniformResourceIdentifier and the URI MUST point to either a single DER encoded certificate as specified in [RFC2585] or a collection of certificates in a BER or DER encoded "certs-only" CMS message as specified in [RFC2797].

Conforming applications that support HTTP or FTP for accessing certificates MUST be able to accept individual DER encoded certificates and SHOULD be able to accept "certs-only" CMS messages.

HTTP server implementations accessed via the URI SHOULD specify the media type application/pkix-cert [RFC2585] in the content-type header field of the response for a single DER encoded certificate and SHOULD specify the media type application/pkcs7-mime [RFC2797] in the content-type header field of the response for "certs-only" CMS messages. For FTP, the name of a file that contains a single DER encoded certificate SHOULD have a suffix of ".cer" [RFC2585] and the name of a file that contains a "certs-only" CMS message SHOULD have a suffix of ".p7c" [RFC2797]. Consuming clients may use the media type or file extension as a hint to the content, but should not depend solely on the presence of the correct media type or file extension in the server response.

The semantics of other id-ad-caRepository accessLocation name forms are not defined.

A subjectInfoAccess extension may include multiple instances of the id-ad-caRepository accessMethod. The different instances may specify different methods for accessing the same information or may point to different information. When the id-ad-caRepository accessMethod is used, at least one instance SHOULD specify an accessLocation that is an HTTP [RFC2616] or LDAP [RFC4516] URI.

The id-ad-timeStamping OID is used when the subject offers timestamping services using the Time Stamp Protocol defined in [RFC3161]. Where the timestamping services are available via HTTP or FTP, accessLocation MUST be a uniformResourceIdentifier. Where the timestamping services are available via electronic mail, accessLocation MUST be an rfc822Name. Where timestamping services are available using TCP/IP, the dNSName or iPAddress name forms may be used. The semantics of other name forms of accessLocation (when accessMethod is id-ad-timeStamping) are not defined by this specification.

Additional access descriptors may be defined in other PKIX specifications.

id-ad OBJECT IDENTIFIER ::= { id-pkix 48 }

id-ad-caRepository OBJECT IDENTIFIER ::= { id-ad 5 }

id-ad-timeStamping OBJECT IDENTIFIER ::= { id-ad 3 }

5. CRL and CRL Extensions Profile

As discussed above, one goal of this X.509 v2 CRL profile is to foster the creation of an interoperable and reusable Internet PKI. To achieve this goal, guidelines for the use of extensions are specified, and some assumptions are made about the nature of information included in the CRL.

CRLs may be used in a wide range of applications and environments covering a broad spectrum of interoperability goals and an even broader spectrum of operational and assurance requirements. This profile establishes a common baseline for generic applications requiring broad interoperability. The profile defines a set of information that can be expected in every CRL. Also, the profile defines common locations within the CRL for frequently used attributes as well as common representations for these attributes.

CRL issuers issue CRLs. The CRL issuer is either the CA or an entity that has been authorized by the CA to issue CRLs. CAs publish CRLs to provide status information about the certificates they issued. However, a CA may delegate this responsibility to another trusted authority.

Each CRL has a particular scope. The CRL scope is the set of certificates that could appear on a given CRL. For example, the scope could be "all certificates issued by CA X", "all CA certificates issued by CA X", "all certificates issued by CA X that have been revoked for reasons of key compromise and CA compromise", or a set of certificates based on arbitrary local information, such as "all certificates issued to the NIST employees located in Boulder".

A complete CRL lists all unexpired certificates, within its scope, that have been revoked for one of the revocation reasons covered by the CRL scope. A full and complete CRL lists all unexpired certificates issued by a CA that have been revoked for any reason. (Note that since CAs and CRL issuers are identified by name, the scope of a CRL is not affected by the key used to sign the CRL or the key(s) used to sign certificates.)

If the scope of the CRL includes one or more certificates issued by an entity other than the CRL issuer, then it is an indirect CRL. The scope of an indirect CRL may be limited to certificates issued by a single CA or may include certificates issued by multiple CAs. If the issuer of the indirect CRL is a CA, then the scope of the indirect CRL MAY also include certificates issued by the issuer of the CRL.

The CRL issuer MAY also generate delta CRLs. A delta CRL only lists those certificates, within its scope, whose revocation status has changed since the issuance of a referenced complete CRL. The referenced complete CRL is referred to as a base CRL. The scope of a delta CRL MUST be the same as the base CRL that it references.

This profile defines one private Internet CRL extension but does not define any private CRL entry extensions.

Environments with additional or special purpose requirements may build on this profile or may replace it.

Conforming CAs are not required to issue CRLs if other revocation or certificate status mechanisms are provided. When CRLs are issued, the CRLs MUST be version 2 CRLs, include the date by which the next CRL will be issued in the nextUpdate field (Section 5.1.2.5), include the CRL number extension (Section 5.2.3), and include the authority key identifier extension (Section 5.2.1). Conforming applications that support CRLs are REQUIRED to process both version 1 and version 2 complete CRLs that provide revocation information for all certificates issued by one CA. Conforming applications are not required to support processing of delta CRLs, indirect CRLs, or CRLs with a scope other than all certificates issued by one CA.

5.1. CRL Fields

The X.509 v2 CRL syntax is as follows. For signature calculation, the data that is to be signed is ASN.1 DER encoded. ASN.1 DER encoding is a tag, length, value encoding system for each element.

```

CertificateList ::= SEQUENCE {
    tbsCertList      TBSCertList,
    signatureAlgorithm AlgorithmIdentifier,
    signatureValue    BIT STRING }

TBSCertList ::= SEQUENCE {
    version          Version OPTIONAL,
                        -- if present, MUST be v2
    signature        AlgorithmIdentifier,
    issuer           Name,
    thisUpdate       Time,
    nextUpdate       Time OPTIONAL,
    revokedCertificates SEQUENCE OF SEQUENCE {
        userCertificate      CertificateSerialNumber,
        revocationDate       Time,
        crlEntryExtensions   Extensions OPTIONAL
                                -- if present, version MUST be v2
    } OPTIONAL,
    crlExtensions          [0] EXPLICIT Extensions OPTIONAL
                                -- if present, version MUST be v2
}

-- Version, Time, CertificateSerialNumber, and Extensions
-- are all defined in the ASN.1 in Section 4.1

-- AlgorithmIdentifier is defined in Section 4.1.1.2

```

The following items describe the use of the X.509 v2 CRL in the Internet PKI.

5.1.1. CertificateList Fields

The CertificateList is a SEQUENCE of three required fields. The fields are described in detail in the following subsections.

5.1.1.1. tbsCertList

The first field in the sequence is the tbsCertList. This field is itself a sequence containing the name of the issuer, issue date, issue date of the next list, the optional list of revoked certificates, and optional CRL extensions. When there are no revoked certificates, the revoked certificates list is absent. When one or more certificates are revoked, each entry on the revoked certificate list is defined by a sequence of user certificate serial number, revocation date, and optional CRL entry extensions.

5.1.1.2. signatureAlgorithm

The signatureAlgorithm field contains the algorithm identifier for the algorithm used by the CRL issuer to sign the CertificateList. The field is of type AlgorithmIdentifier, which is defined in Section 4.1.1.2. [RFC3279], [RFC4055], and [RFC4491] list supported algorithms for this specification, but other signature algorithms MAY also be supported.

This field MUST contain the same algorithm identifier as the signature field in the sequence tbsCertList (Section 5.1.2.2).

5.1.1.3. signatureValue

The signatureValue field contains a digital signature computed upon the ASN.1 DER encoded tbsCertList. The ASN.1 DER encoded tbsCertList is used as the input to the signature function. This signature value is encoded as a BIT STRING and included in the CRL signatureValue field. The details of this process are specified for each of the supported algorithms in [RFC3279], [RFC4055], and [RFC4491].

CAs that are also CRL issuers MAY use one private key to digitally sign certificates and CRLs, or MAY use separate private keys to digitally sign certificates and CRLs. When separate private keys are employed, each of the public keys associated with these private keys is placed in a separate certificate, one with the keyCertSign bit set in the key usage extension, and one with the cRLSign bit set in the key usage extension (Section 4.2.1.3). When separate private keys are employed, certificates issued by the CA contain one authority key identifier, and the corresponding CRLs contain a different authority key identifier. The use of separate CA certificates for validation of certificate signatures and CRL signatures can offer improved security characteristics; however, it imposes a burden on applications, and it might limit interoperability. Many applications construct a certification path, and then validate the certification path (Section 6). CRL checking in turn requires a separate certification path to be constructed and validated for the CA's CRL signature validation certificate. Applications that perform CRL checking MUST support certification path validation when certificates and CRLs are digitally signed with the same CA private key. These applications SHOULD support certification path validation when certificates and CRLs are digitally signed with different CA private keys.

5.1.2. Certificate List "To Be Signed"

The certificate list to be signed, or TBSCertList, is a sequence of required and optional fields. The required fields identify the CRL issuer, the algorithm used to sign the CRL, and the date and time the CRL was issued.

Optional fields include the date and time by which the CRL issuer will issue the next CRL, lists of revoked certificates, and CRL extensions. The revoked certificate list is optional to support the case where a CA has not revoked any unexpired certificates that it has issued. This profile requires conforming CRL issuers to include the nextUpdate field and the CRL number and authority key identifier CRL extensions in all CRLs issued.

5.1.2.1. Version

This optional field describes the version of the encoded CRL. When extensions are used, as required by this profile, this field **MUST** be present and **MUST** specify version 2 (the integer value is 1).

5.1.2.2. Signature

This field contains the algorithm identifier for the algorithm used to sign the CRL. [RFC3279], [RFC4055], and [RFC4491] list OIDs for the most popular signature algorithms used in the Internet PKI.

This field **MUST** contain the same algorithm identifier as the signatureAlgorithm field in the sequence CertificateList (Section 5.1.1.2).

5.1.2.3. Issuer Name

The issuer name identifies the entity that has signed and issued the CRL. The issuer identity is carried in the issuer field. Alternative name forms may also appear in the issuerAltName extension (Section 5.2.2). The issuer field **MUST** contain a non-empty X.500 distinguished name (DN). The issuer field is defined as the X.501 type Name, and **MUST** follow the encoding rules for the issuer name field in the certificate (Section 4.1.2.4).

5.1.2.4. This Update

This field indicates the issue date of this CRL. thisUpdate may be encoded as UTCTime or GeneralizedTime.

CRL issuers conforming to this profile **MUST** encode thisUpdate as UTCTime for dates through the year 2049. CRL issuers conforming to

this profile MUST encode thisUpdate as GeneralizedTime for dates in the year 2050 or later. Conforming applications MUST be able to process dates that are encoded in either UTCTime or GeneralizedTime.

Where encoded as UTCTime, thisUpdate MUST be specified and interpreted as defined in Section 4.1.2.5.1. Where encoded as GeneralizedTime, thisUpdate MUST be specified and interpreted as defined in Section 4.1.2.5.2.

5.1.2.5. Next Update

This field indicates the date by which the next CRL will be issued. The next CRL could be issued before the indicated date, but it will not be issued any later than the indicated date. CRL issuers SHOULD issue CRLs with a nextUpdate time equal to or later than all previous CRLs. nextUpdate may be encoded as UTCTime or GeneralizedTime.

Conforming CRL issuers MUST include the nextUpdate field in all CRLs. Note that the ASN.1 syntax of TBSCertList describes this field as OPTIONAL, which is consistent with the ASN.1 structure defined in [X.509]. The behavior of clients processing CRLs that omit nextUpdate is not specified by this profile.

CRL issuers conforming to this profile MUST encode nextUpdate as UTCTime for dates through the year 2049. CRL issuers conforming to this profile MUST encode nextUpdate as GeneralizedTime for dates in the year 2050 or later. Conforming applications MUST be able to process dates that are encoded in either UTCTime or GeneralizedTime.

Where encoded as UTCTime, nextUpdate MUST be specified and interpreted as defined in Section 4.1.2.5.1. Where encoded as GeneralizedTime, nextUpdate MUST be specified and interpreted as defined in Section 4.1.2.5.2.

5.1.2.6. Revoked Certificates

When there are no revoked certificates, the revoked certificates list MUST be absent. Otherwise, revoked certificates are listed by their serial numbers. Certificates revoked by the CA are uniquely identified by the certificate serial number. The date on which the revocation occurred is specified. The time for revocationDate MUST be expressed as described in Section 5.1.2.4. Additional information may be supplied in CRL entry extensions; CRL entry extensions are discussed in Section 5.3.

5.1.2.7. Extensions

This field may only appear if the version is 2 (Section 5.1.2.1). If present, this field is a sequence of one or more CRL extensions. CRL extensions are discussed in Section 5.2.

5.2. CRL Extensions

The extensions defined by ANSI X9, ISO/IEC, and ITU-T for X.509 v2 CRLs [X.509] [X9.55] provide methods for associating additional attributes with CRLs. The X.509 v2 CRL format also allows communities to define private extensions to carry information unique to those communities. Each extension in a CRL may be designated as critical or non-critical. If a CRL contains a critical extension that the application cannot process, then the application MUST NOT use that CRL to determine the status of certificates. However, applications may ignore unrecognized non-critical extensions. The following subsections present those extensions used within Internet CRLs. Communities may elect to include extensions in CRLs that are not defined in this specification. However, caution should be exercised in adopting any critical extensions in CRLs that might be used in a general context.

Conforming CRL issuers are REQUIRED to include the authority key identifier (Section 5.2.1) and the CRL number (Section 5.2.3) extensions in all CRLs issued.

5.2.1. Authority Key Identifier

The authority key identifier extension provides a means of identifying the public key corresponding to the private key used to sign a CRL. The identification can be based on either the key identifier (the subject key identifier in the CRL signer's certificate) or the issuer name and serial number. This extension is especially useful where an issuer has more than one signing key, either due to multiple concurrent key pairs or due to changeover.

Conforming CRL issuers MUST use the key identifier method, and MUST include this extension in all CRLs issued.

The syntax for this CRL extension is defined in Section 4.2.1.1.

5.2.2. Issuer Alternative Name

The issuer alternative name extension allows additional identities to be associated with the issuer of the CRL. Defined options include an electronic mail address (rfc822Name), a DNS name, an IP address, and a URI. Multiple instances of a name form and multiple name forms may

be included. Whenever such identities are used, the issuer alternative name extension **MUST** be used; however, a DNS name **MAY** be represented in the issuer field using the domainComponent attribute as described in Section 4.1.2.4.

Conforming CRL issuers **SHOULD** mark the issuerAltName extension as non-critical.

The OID and syntax for this CRL extension are defined in Section 4.2.1.7.

5.2.3. CRL Number

The CRL number is a non-critical CRL extension that conveys a monotonically increasing sequence number for a given CRL scope and CRL issuer. This extension allows users to easily determine when a particular CRL supersedes another CRL. CRL numbers also support the identification of complementary complete CRLs and delta CRLs. CRL issuers conforming to this profile **MUST** include this extension in all CRLs and **MUST** mark this extension as non-critical.

If a CRL issuer generates delta CRLs in addition to complete CRLs for a given scope, the complete CRLs and delta CRLs **MUST** share one numbering sequence. If a delta CRL and a complete CRL that cover the same scope are issued at the same time, they **MUST** have the same CRL number and provide the same revocation information. That is, the combination of the delta CRL and an acceptable complete CRL **MUST** provide the same revocation information as the simultaneously issued complete CRL.

If a CRL issuer generates two CRLs (two complete CRLs, two delta CRLs, or a complete CRL and a delta CRL) for the same scope at different times, the two CRLs **MUST NOT** have the same CRL number. That is, if the this update field (Section 5.1.2.4) in the two CRLs are not identical, the CRL numbers **MUST** be different.

Given the requirements above, CRL numbers can be expected to contain long integers. CRL verifiers **MUST** be able to handle CRLNumber values up to 20 octets. Conforming CRL issuers **MUST NOT** use CRLNumber values longer than 20 octets.

id-ce-CRLNumber OBJECT IDENTIFIER ::= { id-ce 20 }

CRLNumber ::= INTEGER (0..MAX)

5.2.4. Delta CRL Indicator

The delta CRL indicator is a critical CRL extension that identifies a CRL as being a delta CRL. Delta CRLs contain updates to revocation information previously distributed, rather than all the information that would appear in a complete CRL. The use of delta CRLs can significantly reduce network load and processing time in some environments. Delta CRLs are generally smaller than the CRLs they update, so applications that obtain delta CRLs consume less network bandwidth than applications that obtain the corresponding complete CRLs. Applications that store revocation information in a format other than the CRL structure can add new revocation information to the local database without reprocessing information.

The delta CRL indicator extension contains the single value of type BaseCRLNumber. The CRL number identifies the CRL, complete for a given scope, that was used as the starting point in the generation of this delta CRL. A conforming CRL issuer **MUST** publish the referenced base CRL as a complete CRL. The delta CRL contains all updates to the revocation status for that same scope. The combination of a delta CRL plus the referenced base CRL is equivalent to a complete CRL, for the applicable scope, at the time of publication of the delta CRL.

When a conforming CRL issuer generates a delta CRL, the delta CRL **MUST** include a critical delta CRL indicator extension.

When a delta CRL is issued, it **MUST** cover the same set of reasons and the same set of certificates that were covered by the base CRL it references. That is, the scope of the delta CRL **MUST** be the same as the scope of the complete CRL referenced as the base. The referenced base CRL and the delta CRL **MUST** omit the issuing distribution point extension or contain identical issuing distribution point extensions. Further, the CRL issuer **MUST** use the same private key to sign the delta CRL and any complete CRL that it can be used to update.

An application that supports delta CRLs can construct a CRL that is complete for a given scope by combining a delta CRL for that scope with either an issued CRL that is complete for that scope or a locally constructed CRL that is complete for that scope.

When a delta CRL is combined with a complete CRL or a locally constructed CRL, the resulting locally constructed CRL has the CRL number specified in the CRL number extension found in the delta CRL used in its construction. In addition, the resulting locally constructed CRL has the thisUpdate and nextUpdate times specified in

the corresponding fields of the delta CRL used in its construction. In addition, the locally constructed CRL inherits the issuing distribution point from the delta CRL.

A complete CRL and a delta CRL MAY be combined if the following four conditions are satisfied:

- (a) The complete CRL and delta CRL have the same issuer.
- (b) The complete CRL and delta CRL have the same scope. The two CRLs have the same scope if either of the following conditions are met:
 - (1) The issuingDistributionPoint extension is omitted from both the complete CRL and the delta CRL.
 - (2) The issuingDistributionPoint extension is present in both the complete CRL and the delta CRL, and the values for each of the fields in the extensions are the same in both CRLs.
- (c) The CRL number of the complete CRL is equal to or greater than the BaseCRLNumber specified in the delta CRL. That is, the complete CRL contains (at a minimum) all the revocation information held by the referenced base CRL.
- (d) The CRL number of the complete CRL is less than the CRL number of the delta CRL. That is, the delta CRL follows the complete CRL in the numbering sequence.

CRL issuers MUST ensure that the combination of a delta CRL and any appropriate complete CRL accurately reflects the current revocation status. The CRL issuer MUST include an entry in the delta CRL for each certificate within the scope of the delta CRL whose status has changed since the generation of the referenced base CRL:

- (a) If the certificate is revoked for a reason included in the scope of the CRL, list the certificate as revoked.
- (b) If the certificate is valid and was listed on the referenced base CRL or any subsequent CRL with reason code `certificateHold`, and the reason code `certificateHold` is included in the scope of the CRL, list the certificate with the reason code `removeFromCRL`.

- (c) If the certificate is revoked for a reason outside the scope of the CRL, but the certificate was listed on the referenced base CRL or any subsequent CRL with a reason code included in the scope of this CRL, list the certificate as revoked but omit the reason code.
- (d) If the certificate is revoked for a reason outside the scope of the CRL and the certificate was neither listed on the referenced base CRL nor any subsequent CRL with a reason code included in the scope of this CRL, do not list the certificate on this CRL.

The status of a certificate is considered to have changed if it is revoked (for any revocation reason, including `certificateHold`), if it is released from hold, or if its revocation reason changes.

It is appropriate to list a certificate with reason code `removeFromCRL` on a delta CRL even if the certificate was not on hold in the referenced base CRL. If the certificate was placed on hold in any CRL issued after the base but before this delta CRL and then released from hold, it **MUST** be listed on the delta CRL with revocation reason `removeFromCRL`.

A CRL issuer **MAY** optionally list a certificate on a delta CRL with reason code `removeFromCRL` if the `notAfter` time specified in the certificate precedes the `thisUpdate` time specified in the delta CRL and the certificate was listed on the referenced base CRL or in any CRL issued after the base but before this delta CRL.

If a certificate revocation notice first appears on a delta CRL, then it is possible for the certificate validity period to expire before the next complete CRL for the same scope is issued. In this case, the revocation notice **MUST** be included in all subsequent delta CRLs until the revocation notice is included on at least one explicitly issued complete CRL for this scope.

An application that supports delta CRLs **MUST** be able to construct a current complete CRL by combining a previously issued complete CRL and the most current delta CRL. An application that supports delta CRLs **MAY** also be able to construct a current complete CRL by combining a previously locally constructed complete CRL and the current delta CRL. A delta CRL is considered to be the current one if the current time is between the times contained in the `thisUpdate` and `nextUpdate` fields. Under some circumstances, the CRL issuer may publish one or more delta CRLs before the time indicated by the `nextUpdate` field. If more than one current delta CRL for a given scope is encountered, the application **SHOULD** consider the one with the latest value in `thisUpdate` to be the most current one.

id-ce-deltaCRLIndicator OBJECT IDENTIFIER ::= { id-ce 27 }

BaseCRLNumber ::= CRLNumber

5.2.5. Issuing Distribution Point

The issuing distribution point is a critical CRL extension that identifies the CRL distribution point and scope for a particular CRL, and it indicates whether the CRL covers revocation for end entity certificates only, CA certificates only, attribute certificates only, or a limited set of reason codes. Although the extension is critical, conforming implementations are not required to support this extension. However, implementations that do not support this extension MUST either treat the status of any certificate not listed on this CRL as unknown or locate another CRL that does not contain any unrecognized critical extensions.

The CRL is signed using the CRL issuer's private key. CRL distribution points do not have their own key pairs. If the CRL is stored in the X.500 directory, it is stored in the directory entry corresponding to the CRL distribution point, which may be different from the directory entry of the CRL issuer.

The reason codes associated with a distribution point MUST be specified in onlySomeReasons. If onlySomeReasons does not appear, the distribution point MUST contain revocations for all reason codes. CAs may use CRL distribution points to partition the CRL on the basis of compromise and routine revocation. In this case, the revocations with reason code keyCompromise (1), cACompromise (2), and aACompromise (8) appear in one distribution point, and the revocations with other reason codes appear in another distribution point.

If a CRL includes an issuingDistributionPoint extension with onlySomeReasons present, then every certificate in the scope of the CRL that is revoked MUST be assigned a revocation reason other than unspecified. The assigned revocation reason is used to determine on which CRL(s) to list the revoked certificate, however, there is no requirement to include the reasonCode CRL entry extension in the corresponding CRL entry.

The syntax and semantics for the distributionPoint field are the same as for the distributionPoint field in the cRLDistributionPoints extension (Section 4.2.1.13). If the distributionPoint field is present, then it MUST include at least one of names from the corresponding distributionPoint field of the cRLDistributionPoints

extension of every certificate that is within the scope of this CRL. The identical encoding MUST be used in the distributionPoint fields of the certificate and the CRL.

If the distributionPoint field is absent, the CRL MUST contain entries for all revoked unexpired certificates issued by the CRL issuer, if any, within the scope of the CRL.

If the scope of the CRL only includes certificates issued by the CRL issuer, then the indirectCRL boolean MUST be set to FALSE. Otherwise, if the scope of the CRL includes certificates issued by one or more authorities other than the CRL issuer, the indirectCRL boolean MUST be set to TRUE. The authority responsible for each entry is indicated by the certificate issuer CRL entry extension (Section 5.3.3).

If the scope of the CRL only includes end entity public key certificates, then onlyContainsUserCerts MUST be set to TRUE. If the scope of the CRL only includes CA certificates, then onlyContainsCACerts MUST be set to TRUE. If either onlyContainsUserCerts or onlyContainsCACerts is set to TRUE, then the scope of the CRL MUST NOT include any version 1 or version 2 certificates. Conforming CRLs issuers MUST set the onlyContainsAttributeCerts boolean to FALSE.

Conforming CRLs issuers MUST NOT issue CRLs where the DER encoding of the issuing distribution point extension is an empty sequence. That is, if onlyContainsUserCerts, onlyContainsCACerts, indirectCRL, and onlyContainsAttributeCerts are all FALSE, then either the distributionPoint field or the onlySomeReasons field MUST be present.

id-ce-issuingDistributionPoint OBJECT IDENTIFIER ::= { id-ce 28 }

```
IssuingDistributionPoint ::= SEQUENCE {
    distributionPoint          [0] DistributionPointName OPTIONAL,
    onlyContainsUserCerts     [1] BOOLEAN DEFAULT FALSE,
    onlyContainsCACerts       [2] BOOLEAN DEFAULT FALSE,
    onlySomeReasons           [3] ReasonFlags OPTIONAL,
    indirectCRL               [4] BOOLEAN DEFAULT FALSE,
    onlyContainsAttributeCerts [5] BOOLEAN DEFAULT FALSE }
```

-- at most one of onlyContainsUserCerts, onlyContainsCACerts,
-- and onlyContainsAttributeCerts may be set to TRUE.

5.2.6. Freshest CRL (a.k.a. Delta CRL Distribution Point)

The freshest CRL extension identifies how delta CRL information for this complete CRL is obtained. Conforming CRL issuers **MUST** mark this extension as non-critical. This extension **MUST NOT** appear in delta CRLs.

The same syntax is used for this extension as the `cRLDistributionPoints` certificate extension, and is described in Section 4.2.1.13. However, only the distribution point field is meaningful in this context. The reasons and `cRLIssuer` fields **MUST** be omitted from this CRL extension.

Each distribution point name provides the location at which a delta CRL for this complete CRL can be found. The scope of these delta CRLs **MUST** be the same as the scope of this complete CRL. The contents of this CRL extension are only used to locate delta CRLs; the contents are not used to validate the CRL or the referenced delta CRLs. The encoding conventions defined for distribution points in Section 4.2.1.13 apply to this extension.

`id-ce-freshestCRL OBJECT IDENTIFIER ::= { id-ce 46 }`

`FreshestCRL ::= CRLDistributionPoints`

5.2.7. Authority Information Access

This section defines the use of the Authority Information Access extension in a CRL. The syntax and semantics defined in Section 4.2.2.1 for the certificate extension are also used for the CRL extension.

This CRL extension **MUST** be marked as non-critical.

When present in a CRL, this extension **MUST** include at least one `AccessDescription` specifying `id-ad-caIssuers` as the `accessMethod`. The `id-ad-caIssuers` OID is used when the information available lists certificates that can be used to verify the signature on the CRL (i.e., certificates that have a subject name that matches the issuer name on the CRL and that have a subject public key that corresponds to the private key used to sign the CRL). Access method types other than `id-ad-caIssuers` **MUST NOT** be included. At least one instance of `AccessDescription` **SHOULD** specify an `accessLocation` that is an HTTP [RFC2616] or LDAP [RFC4516] URI.

Where the information is available via HTTP or FTP, `accessLocation` MUST be a `uniformResourceIdentifier` and the URI MUST point to either a single DER encoded certificate as specified in [RFC2585] or a collection of certificates in a BER or DER encoded "certs-only" CMS message as specified in [RFC2797].

Conforming applications that support HTTP or FTP for accessing certificates MUST be able to accept individual DER encoded certificates and SHOULD be able to accept "certs-only" CMS messages.

HTTP server implementations accessed via the URI SHOULD specify the media type `application/pkix-cert` [RFC2585] in the `content-type` header field of the response for a single DER encoded certificate and SHOULD specify the media type `application/pkcs7-mime` [RFC2797] in the `content-type` header field of the response for "certs-only" CMS messages. For FTP, the name of a file that contains a single DER encoded certificate SHOULD have a suffix of ".cer" [RFC2585] and the name of a file that contains a "certs-only" CMS message SHOULD have a suffix of ".p7c" [RFC2797]. Consuming clients may use the media type or file extension as a hint to the content, but should not depend solely on the presence of the correct media type or file extension in the server response.

When the `accessLocation` is a `directoryName`, the information is to be obtained by the application from whatever directory server is locally configured. When one CA public key is used to validate signatures on certificates and CRLs, the desired CA certificate is stored in the `crossCertificatePair` and/or `cACertificate` attributes as specified in [RFC4523]. When different public keys are used to validate signatures on certificates and CRLs, the desired certificate is stored in the `userCertificate` attribute as specified in [RFC4523]. Thus, implementations that support the `directoryName` form of `accessLocation` MUST be prepared to find the needed certificate in any of these three attributes. The protocol that an application uses to access the directory (e.g., DAP or LDAP) is a local matter.

Where the information is available via LDAP, the `accessLocation` SHOULD be a `uniformResourceIdentifier`. The LDAP URI [RFC4516] MUST include a `<dn>` field containing the distinguished name of the entry holding the certificates, MUST include an `<attributes>` field that lists appropriate attribute descriptions for the attributes that hold the DER encoded certificates or cross-certificate pairs [RFC4523], and SHOULD include a `<host>` (e.g., `<ldap://ldap.example.com/cn=CA,dc=example,dc=com?cACertificate;binary,crossCertificatePair;binary>`). Omitting the `<host>` (e.g., `<ldap:///cn=exampleCA,dc=example,dc=com?cACertificate;binary>`) has the effect of relying on whatever a priori knowledge the client might have to contact an appropriate server.

5.3. CRL Entry Extensions

The CRL entry extensions defined by ISO/IEC, ITU-T, and ANSI X9 for X.509 v2 CRLs provide methods for associating additional attributes with CRL entries [X.509] [X9.55]. The X.509 v2 CRL format also allows communities to define private CRL entry extensions to carry information unique to those communities. Each extension in a CRL entry may be designated as critical or non-critical. If a CRL contains a critical CRL entry extension that the application cannot process, then the application MUST NOT use that CRL to determine the status of any certificates. However, applications may ignore unrecognized non-critical CRL entry extensions.

The following subsections present recommended extensions used within Internet CRL entries and standard locations for information. Communities may elect to use additional CRL entry extensions; however, caution should be exercised in adopting any critical CRL entry extensions in CRLs that might be used in a general context.

Support for the CRL entry extensions defined in this specification is optional for conforming CRL issuers and applications. However, CRL issuers SHOULD include reason codes (Section 5.3.1) and invalidity dates (Section 5.3.2) whenever this information is available.

5.3.1. Reason Code

The reasonCode is a non-critical CRL entry extension that identifies the reason for the certificate revocation. CRL issuers are strongly encouraged to include meaningful reason codes in CRL entries; however, the reason code CRL entry extension SHOULD be absent instead of using the unspecified (0) reasonCode value.

The removeFromCRL (8) reasonCode value may only appear in delta CRLs and indicates that a certificate is to be removed from a CRL because either the certificate expired or was removed from hold. All other reason codes may appear in any CRL and indicate that the specified certificate should be considered revoked.

```
id-ce-cRLReasons OBJECT IDENTIFIER ::= { id-ce 21 }
```

```
-- reasonCode ::= { CRLReason }
```

```
CRLReason ::= ENUMERATED {  
    unspecified             (0),  
    keyCompromise           (1),  
    cACompromise            (2),  
    affiliationChanged      (3),  
    superseded              (4),  
    cessationOfOperation    (5),  
    certificateHold         (6),  
    -- value 7 is not used  
    removeFromCRL           (8),  
    privilegeWithdrawn      (9),  
    aACompromise            (10) }
```

5.3.2. Invalidity Date

The invalidity date is a non-critical CRL entry extension that provides the date on which it is known or suspected that the private key was compromised or that the certificate otherwise became invalid. This date may be earlier than the revocation date in the CRL entry, which is the date at which the CA processed the revocation. When a revocation is first posted by a CRL issuer in a CRL, the invalidity date may precede the date of issue of earlier CRLs, but the revocation date SHOULD NOT precede the date of issue of earlier CRLs. Whenever this information is available, CRL issuers are strongly encouraged to share it with CRL users.

The GeneralizedTime values included in this field MUST be expressed in Greenwich Mean Time (Zulu), and MUST be specified and interpreted as defined in Section 4.1.2.5.2.

```
id-ce-invalidityDate OBJECT IDENTIFIER ::= { id-ce 24 }
```

```
InvalidityDate ::= GeneralizedTime
```

5.3.3. Certificate Issuer

This CRL entry extension identifies the certificate issuer associated with an entry in an indirect CRL, that is, a CRL that has the indirectCRL indicator set in its issuing distribution point extension. When present, the certificate issuer CRL entry extension includes one or more names from the issuer field and/or issuer alternative name extension of the certificate that corresponds to the CRL entry. If this extension is not present on the first entry in an indirect CRL, the certificate issuer defaults to the CRL issuer. On

subsequent entries in an indirect CRL, if this extension is not present, the certificate issuer for the entry is the same as that for the preceding entry. This field is defined as follows:

id-ce-certificateIssuer OBJECT IDENTIFIER ::= { id-ce 29 }

CertificateIssuer ::= GeneralNames

Conforming CRL issuers MUST include in this extension the distinguished name (DN) from the issuer field of the certificate that corresponds to this CRL entry. The encoding of the DN MUST be identical to the encoding used in the certificate.

CRL issuers MUST mark this extension as critical since an implementation that ignored this extension could not correctly attribute CRL entries to certificates. This specification RECOMMENDS that implementations recognize this extension.

6. Certification Path Validation

Certification path validation procedures for the Internet PKI are based on the algorithm supplied in [X.509]. Certification path processing verifies the binding between the subject distinguished name and/or subject alternative name and subject public key. The binding is limited by constraints that are specified in the certificates that comprise the path and inputs that are specified by the relying party. The basic constraints and policy constraints extensions allow the certification path processing logic to automate the decision making process.

This section describes an algorithm for validating certification paths. Conforming implementations of this specification are not required to implement this algorithm, but MUST provide functionality equivalent to the external behavior resulting from this procedure. Any algorithm may be used by a particular implementation so long as it derives the correct result.

In Section 6.1, the text describes basic path validation. Valid paths begin with certificates issued by a trust anchor. The algorithm requires the public key of the CA, the CA's name, and any constraints upon the set of paths that may be validated using this key.

The selection of a trust anchor is a matter of policy: it could be the top CA in a hierarchical PKI, the CA that issued the verifier's own certificate(s), or any other CA in a network PKI. The path

validation procedure is the same regardless of the choice of trust anchor. In addition, different applications may rely on different trust anchors, or may accept paths that begin with any of a set of trust anchors.

Section 6.2 describes methods for using the path validation algorithm in specific implementations.

Section 6.3 describes the steps necessary to determine if a certificate is revoked when CRLs are the revocation mechanism used by the certificate issuer.

6.1. Basic Path Validation

This text describes an algorithm for X.509 path processing. A conforming implementation **MUST** include an X.509 path processing procedure that is functionally equivalent to the external behavior of this algorithm. However, support for some of the certificate extensions processed in this algorithm are **OPTIONAL** for compliant implementations. Clients that do not support these extensions **MAY** omit the corresponding steps in the path validation algorithm.

For example, clients are not required to support the policy mappings extension. Clients that do not support this extension **MAY** omit the path validation steps where policy mappings are processed. Note that clients **MUST** reject the certificate if it contains an unsupported critical extension.

While the certificate and CRL profiles specified in Sections 4 and 5 of this document specify values for certificate and CRL fields and extensions that are considered to be appropriate for the Internet PKI, the algorithm presented in this section is not limited to accepting certificates and CRLs that conform to these profiles. Therefore, the algorithm only includes checks to verify that the certification path is valid according to X.509 and does not include checks to verify that the certificates and CRLs conform to this profile. While the algorithm could be extended to include checks for conformance to the profiles in Sections 4 and 5, this profile **RECOMMENDS** against including such checks.

The algorithm presented in this section validates the certificate with respect to the current date and time. A conforming implementation **MAY** also support validation with respect to some point in the past. Note that mechanisms are not available for validating a certificate with respect to a time outside the certificate validity period.

The trust anchor is an input to the algorithm. There is no requirement that the same trust anchor be used to validate all certification paths. Different trust anchors MAY be used to validate different paths, as discussed further in Section 6.2.

The primary goal of path validation is to verify the binding between a subject distinguished name or a subject alternative name and subject public key, as represented in the target certificate, based on the public key of the trust anchor. In most cases, the target certificate will be an end entity certificate, but the target certificate may be a CA certificate as long as the subject public key is to be used for a purpose other than verifying the signature on a public key certificate. Verifying the binding between the name and subject public key requires obtaining a sequence of certificates that support that binding. The procedure performed to obtain this sequence of certificates is outside the scope of this specification.

To meet this goal, the path validation process verifies, among other things, that a prospective certification path (a sequence of n certificates) satisfies the following conditions:

- (a) for all x in $\{1, \dots, n-1\}$, the subject of certificate x is the issuer of certificate $x+1$;
- (b) certificate 1 is issued by the trust anchor;
- (c) certificate n is the certificate to be validated (i.e., the target certificate); and
- (d) for all x in $\{1, \dots, n\}$, the certificate was valid at the time in question.

A certificate MUST NOT appear more than once in a prospective certification path.

When the trust anchor is provided in the form of a self-signed certificate, this self-signed certificate is not included as part of the prospective certification path. Information about trust anchors is provided as inputs to the certification path validation algorithm (Section 6.1.1).

A particular certification path may not, however, be appropriate for all applications. Therefore, an application MAY augment this algorithm to further limit the set of valid paths. The path validation process also determines the set of certificate policies that are valid for this path, based on the certificate policies extension, policy mappings extension, policy constraints extension, and inhibit anyPolicy extension. To achieve this, the path

validation algorithm constructs a valid policy tree. If the set of certificate policies that are valid for this path is not empty, then the result will be a valid policy tree of depth n , otherwise the result will be a null valid policy tree.

A certificate is self-issued if the same DN appears in the subject and issuer fields (the two DNs are the same if they match according to the rules specified in Section 7.1). In general, the issuer and subject of the certificates that make up a path are different for each certificate. However, a CA may issue a certificate to itself to support key rollover or changes in certificate policies. These self-issued certificates are not counted when evaluating path length or name constraints.

This section presents the algorithm in four basic steps: (1) initialization, (2) basic certificate processing, (3) preparation for the next certificate, and (4) wrap-up. Steps (1) and (4) are performed exactly once. Step (2) is performed for all certificates in the path. Step (3) is performed for all certificates in the path except the final certificate. Figure 2 provides a high-level flowchart of this algorithm.

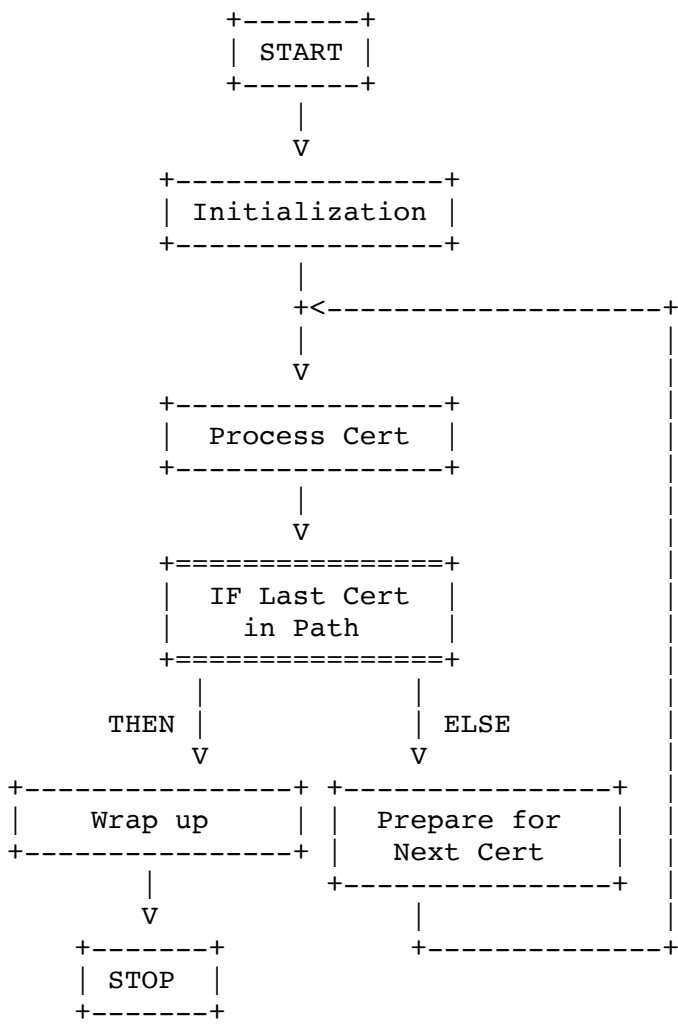


Figure 2. Certification Path Processing Flowchart

6.1.1.1. Inputs

This algorithm assumes that the following nine inputs are provided to the path processing logic:

- (a) a prospective certification path of length n.
- (b) the current date/time.

- (c) `user-initial-policy-set`: A set of certificate policy identifiers naming the policies that are acceptable to the certificate user. The `user-initial-policy-set` contains the special value `any-policy` if the user is not concerned about certificate policy.
- (d) trust anchor information, describing a CA that serves as a trust anchor for the certification path. The trust anchor information includes:
 - (1) the trusted issuer name,
 - (2) the trusted public key algorithm,
 - (3) the trusted public key, and
 - (4) optionally, the trusted public key parameters associated with the public key.

The trust anchor information may be provided to the path processing procedure in the form of a self-signed certificate. When the trust anchor information is provided in the form of a certificate, the name in the subject field is used as the trusted issuer name and the contents of the `subjectPublicKeyInfo` field is used as the source of the trusted public key algorithm and the trusted public key. The trust anchor information is trusted because it was delivered to the path processing procedure by some trustworthy out-of-band procedure. If the trusted public key algorithm requires parameters, then the parameters are provided along with the trusted public key.

- (e) `initial-policy-mapping-inhibit`, which indicates if policy mapping is allowed in the certification path.
- (f) `initial-explicit-policy`, which indicates if the path must be valid for at least one of the certificate policies in the `user-initial-policy-set`.
- (g) `initial-any-policy-inhibit`, which indicates whether the `anyPolicy` OID should be processed if it is included in a certificate.
- (h) `initial-permitted-subtrees`, which indicates for each name type (e.g., X.500 distinguished names, email addresses, or IP addresses) a set of subtrees within which all subject names in every certificate in the certification path MUST fall. The `initial-permitted-subtrees` input includes a set for each name type. For each name type, the set may consist of a

single subtree that includes all names of that name type or one or more subtrees that each specifies a subset of the names of that name type, or the set may be empty. If the set for a name type is empty, then the certification path will be considered invalid if any certificate in the certification path includes a name of that name type.

- (i) `initial-excluded-subtrees`, which indicates for each name type (e.g., X.500 distinguished names, email addresses, or IP addresses) a set of subtrees within which no subject name in any certificate in the certification path may fall. The `initial-excluded-subtrees` input includes a set for each name type. For each name type, the set may be empty or may consist of one or more subtrees that each specifies a subset of the names of that name type. If the set for a name type is empty, then no names of that name type are excluded.

Conforming implementations are not required to support the setting of all of these inputs. For example, a conforming implementation may be designed to validate all certification paths using a value of `FALSE` for `initial-any-policy-inhibit`.

6.1.2. Initialization

This initialization phase establishes eleven state variables based upon the nine inputs:

- (a) `valid_policy_tree`: A tree of certificate policies with their optional qualifiers; each of the leaves of the tree represents a valid policy at this stage in the certification path validation. If valid policies exist at this stage in the certification path validation, the depth of the tree is equal to the number of certificates in the chain that have been processed. If valid policies do not exist at this stage in the certification path validation, the tree is set to `NULL`. Once the tree is set to `NULL`, policy processing ceases.

Each node in the `valid_policy_tree` includes three data objects: the valid policy, a set of associated policy qualifiers, and a set of one or more expected policy values. If the node is at depth `x`, the components of the node have the following semantics:

- (1) The `valid_policy` is a single policy OID representing a valid policy for the path of length `x`.

- (2) The `qualifier_set` is a set of policy qualifiers associated with the valid policy in certificate `x`.
- (3) The `expected_policy_set` contains one or more policy OIDs that would satisfy this policy in the certificate `x+1`.

The initial value of the `valid_policy_tree` is a single node with `valid_policy` anyPolicy, an empty `qualifier_set`, and an `expected_policy_set` with the single value anyPolicy. This node is considered to be at depth zero.

Figure 3 is a graphic representation of the initial state of the `valid_policy_tree`. Additional figures will use this format to describe changes in the `valid_policy_tree` during path processing.

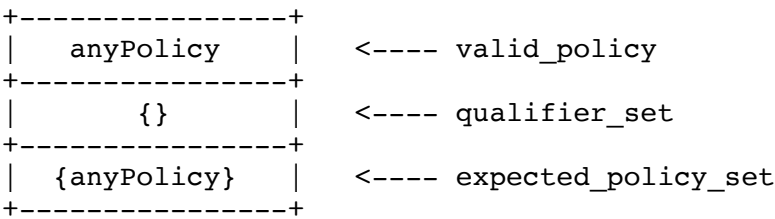


Figure 3. Initial Value of the `valid_policy_tree` State Variable

- (b) `permitted_subtrees`: a set of root names for each name type (e.g., X.500 distinguished names, email addresses, or IP addresses) defining a set of subtrees within which all subject names in subsequent certificates in the certification path **MUST** fall. This variable includes a set for each name type, and the initial value is `initial-permitted-subtrees`.
- (c) `excluded_subtrees`: a set of root names for each name type (e.g., X.500 distinguished names, email addresses, or IP addresses) defining a set of subtrees within which no subject name in subsequent certificates in the certification path may fall. This variable includes a set for each name type, and the initial value is `initial-excluded-subtrees`.
- (d) `explicit_policy`: an integer that indicates if a non-NULL `valid_policy_tree` is required. The integer indicates the number of non-self-issued certificates to be processed before this requirement is imposed. Once set, this variable may be decreased, but may not be increased. That is, if a certificate in the path requires a non-NULL `valid_policy_tree`, a later certificate cannot remove this requirement. If `initial-explicit-policy` is set, then the initial value is 0, otherwise the initial value is `n+1`.

- (e) `inhibit_anyPolicy`: an integer that indicates whether the `anyPolicy` policy identifier is considered a match. The integer indicates the number of non-self-issued certificates to be processed before the `anyPolicy` OID, if asserted in a certificate other than an intermediate self-issued certificate, is ignored. Once set, this variable may be decreased, but may not be increased. That is, if a certificate in the path inhibits processing of `anyPolicy`, a later certificate cannot permit it. If `initial-any-policy-inhibit` is set, then the initial value is 0, otherwise the initial value is `n+1`.
- (f) `policy_mapping`: an integer that indicates if policy mapping is permitted. The integer indicates the number of non-self-issued certificates to be processed before policy mapping is inhibited. Once set, this variable may be decreased, but may not be increased. That is, if a certificate in the path specifies that policy mapping is not permitted, it cannot be overridden by a later certificate. If `initial-policy-mapping-inhibit` is set, then the initial value is 0, otherwise the initial value is `n+1`.
- (g) `working_public_key_algorithm`: the digital signature algorithm used to verify the signature of a certificate. The `working_public_key_algorithm` is initialized from the trusted public key algorithm provided in the trust anchor information.
- (h) `working_public_key`: the public key used to verify the signature of a certificate. The `working_public_key` is initialized from the trusted public key provided in the trust anchor information.
- (i) `working_public_key_parameters`: parameters associated with the current public key that may be required to verify a signature (depending upon the algorithm). The `working_public_key_parameters` variable is initialized from the trusted public key parameters provided in the trust anchor information.
- (j) `working_issuer_name`: the issuer distinguished name expected in the next certificate in the chain. The `working_issuer_name` is initialized to the trusted issuer name provided in the trust anchor information.

- (k) `max_path_length`: this integer is initialized to `n`, is decremented for each non-self-issued certificate in the path, and may be reduced to the value in the path length constraint field within the basic constraints extension of a CA certificate.

Upon completion of the initialization steps, perform the basic certificate processing steps specified in 6.1.3.

6.1.3. Basic Certificate Processing

The basic path processing actions to be performed for certificate `i` (for all `i` in `[1..n]`) are listed below.

- (a) Verify the basic certificate information. The certificate MUST satisfy each of the following:
 - (1) The signature on the certificate can be verified using `working_public_key_algorithm`, the `working_public_key`, and the `working_public_key_parameters`.
 - (2) The certificate validity period includes the current time.
 - (3) At the current time, the certificate is not revoked. This may be determined by obtaining the appropriate CRL (Section 6.3), by status information, or by out-of-band mechanisms.
 - (4) The certificate issuer name is the `working_issuer_name`.
- (b) If certificate `i` is self-issued and it is not the final certificate in the path, skip this step for certificate `i`. Otherwise, verify that the subject name is within one of the `permitted_subtrees` for X.500 distinguished names, and verify that each of the alternative names in the `subjectAltName` extension (critical or non-critical) is within one of the `permitted_subtrees` for that name type.
- (c) If certificate `i` is self-issued and it is not the final certificate in the path, skip this step for certificate `i`. Otherwise, verify that the subject name is not within any of the `excluded_subtrees` for X.500 distinguished names, and verify that each of the alternative names in the `subjectAltName` extension (critical or non-critical) is not within any of the `excluded_subtrees` for that name type.

- (d) If the certificate policies extension is present in the certificate and the `valid_policy_tree` is not NULL, process the policy information by performing the following steps in order:
 - (1) For each policy P not equal to anyPolicy in the certificate policies extension, let P-OID denote the OID for policy P and P-Q denote the qualifier set for policy P. Perform the following steps in order:
 - (i) For each node of depth i-1 in the `valid_policy_tree` where P-OID is in the `expected_policy_set`, create a child node as follows: set the `valid_policy` to P-OID, set the `qualifier_set` to P-Q, and set the `expected_policy_set` to {P-OID}.

For example, consider a `valid_policy_tree` with a node of depth i-1 where the `expected_policy_set` is {Gold, White}. Assume the certificate policies Gold and Silver appear in the certificate policies extension of certificate i. The Gold policy is matched, but the Silver policy is not. This rule will generate a child node of depth i for the Gold policy. The result is shown as Figure 4.

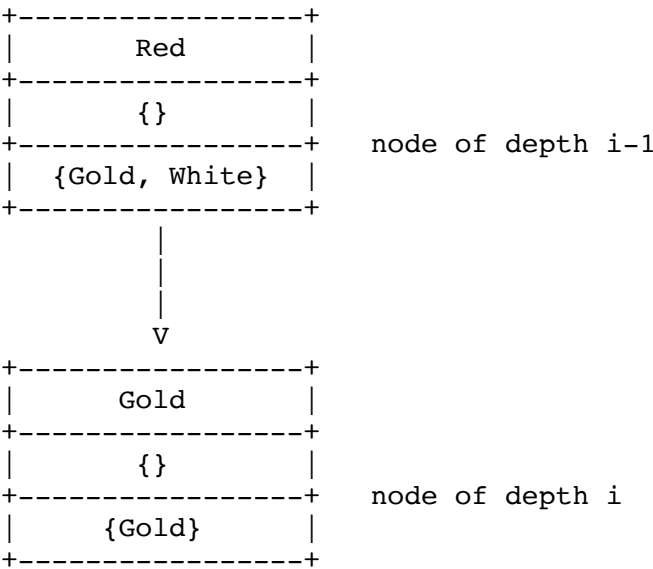


Figure 4. Processing an Exact Match

- (ii) If there was no match in step (i) and the valid_policy_tree includes a node of depth i-1 with the valid_policy anyPolicy, generate a child node with the following values: set the valid_policy to P-OID, set the qualifier_set to P-Q, and set the expected_policy_set to {P-OID}.

For example, consider a valid_policy_tree with a node of depth i-1 where the valid_policy is anyPolicy. Assume the certificate policies Gold and Silver appear in the certificate policies extension of certificate i. The Gold policy does not have a qualifier, but the Silver policy has the qualifier Q-Silver. If Gold and Silver were not matched in (i) above, this rule will generate two child nodes of depth i, one for each policy. The result is shown as Figure 5.

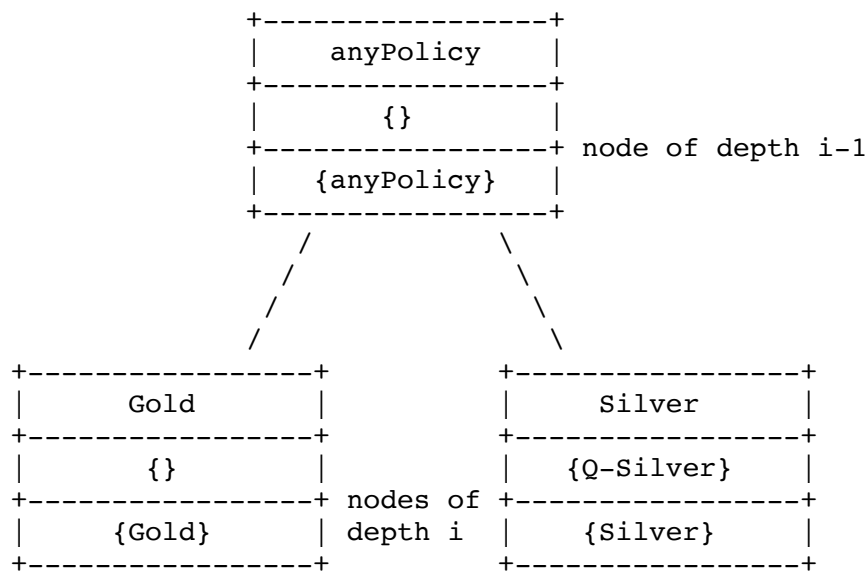


Figure 5. Processing Unmatched Policies when a Leaf Node Specifies anyPolicy

- (2) If the certificate policies extension includes the policy anyPolicy with the qualifier set AP-Q and either (a) inhibit_anyPolicy is greater than 0 or (b) i<n and the certificate is self-issued, then:

For each node in the valid_policy_tree of depth i-1, for each value in the expected_policy_set (including anyPolicy) that does not appear in a child node, create a child node with the following values: set the valid_policy

to the value from the `expected_policy_set` in the parent node, set the `qualifier_set` to `AP-Q`, and set the `expected_policy_set` to the value in the `valid_policy` from this node.

For example, consider a `valid_policy_tree` with a node of depth `i-1` where the `expected_policy_set` is `{Gold, Silver}`. Assume `anyPolicy` appears in the certificate policies extension of certificate `i` with no policy qualifiers, but `Gold` and `Silver` do not appear. This rule will generate two child nodes of depth `i`, one for each policy. The result is shown below as Figure 6.

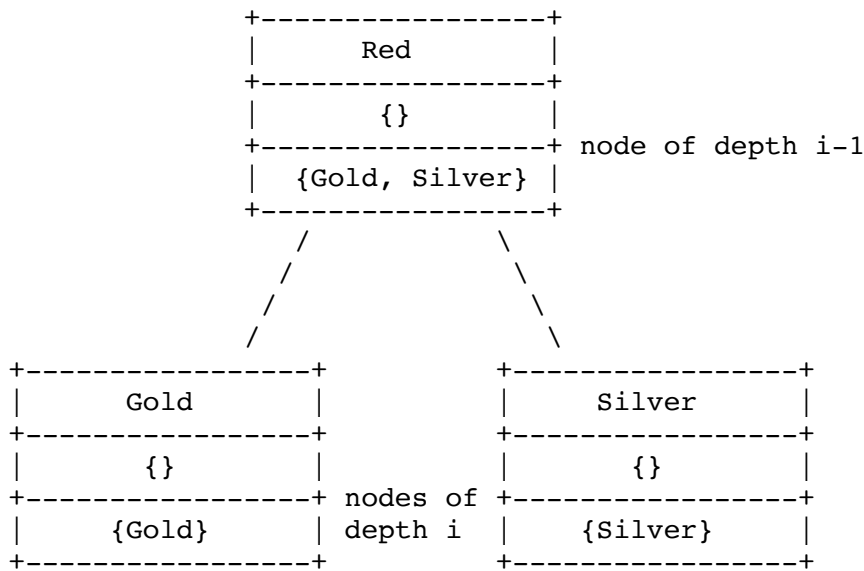


Figure 6. Processing Unmatched Policies When the Certificate Policies Extension Specifies `anyPolicy`

- (3) If there is a node in the `valid_policy_tree` of depth `i-1` or less without any child nodes, delete that node. Repeat this step until there are no nodes of depth `i-1` or less without children.

For example, consider the `valid_policy_tree` shown in Figure 7 below. The two nodes at depth `i-1` that are marked with an 'X' have no children, and they are deleted. Applying this rule to the resulting tree will cause the node at depth `i-2` that is marked with a 'Y' to be deleted. In the resulting tree, there are no nodes of depth `i-1` or less without children, and this step is complete.

- (e) If the certificate policies extension is not present, set the `valid_policy_tree` to `NULL`.
- (f) Verify that either `explicit_policy` is greater than 0 or the `valid_policy_tree` is not equal to `NULL`;

If any of steps (a), (b), (c), or (f) fails, the procedure terminates, returning a failure indication and an appropriate reason.

If `i` is not equal to `n`, continue by performing the preparatory steps listed in Section 6.1.4. If `i` is equal to `n`, perform the wrap-up steps listed in Section 6.1.5.

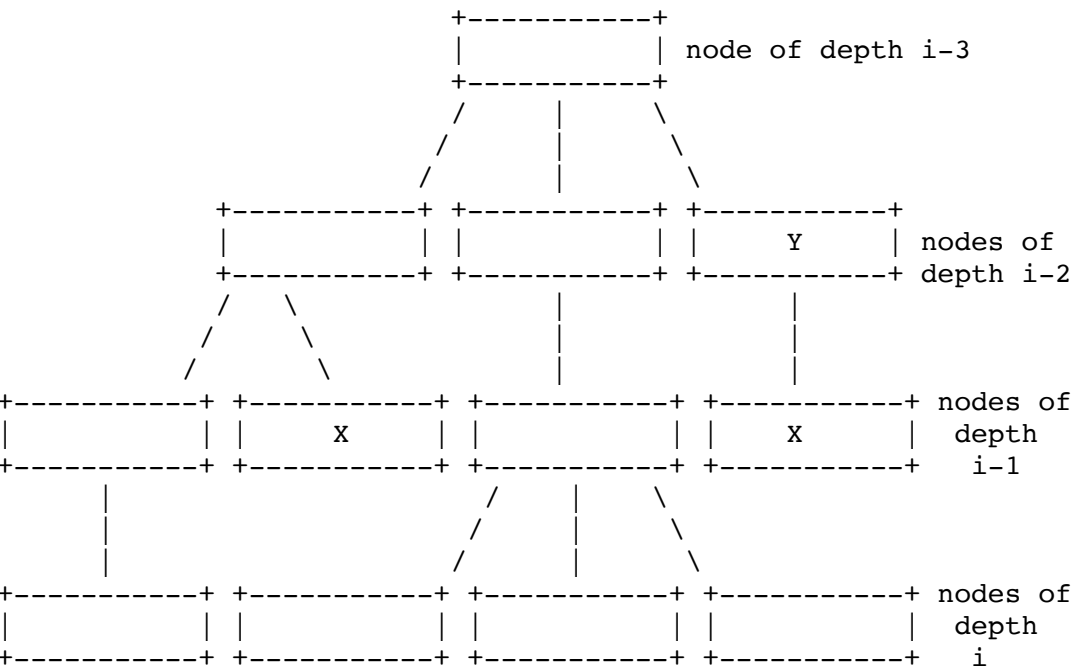


Figure 7. Pruning the `valid_policy_tree`

6.1.4. Preparation for Certificate `i+1`

To prepare for processing of certificate `i+1`, perform the following steps for certificate `i`:

- (a) If a policy mappings extension is present, verify that the special value `anyPolicy` does not appear as an `issuerDomainPolicy` or a `subjectDomainPolicy`.
- (b) If a policy mappings extension is present, then for each `issuerDomainPolicy` `ID-P` in the policy mappings extension:

- (1) If the `policy_mapping` variable is greater than 0, for each node in the `valid_policy_tree` of depth `i` where `ID-P` is the `valid_policy`, set `expected_policy_set` to the set of `subjectDomainPolicy` values that are specified as equivalent to `ID-P` by the policy mappings extension.

If no node of depth `i` in the `valid_policy_tree` has a `valid_policy` of `ID-P` but there is a node of depth `i` with a `valid_policy` of `anyPolicy`, then generate a child node of the node of depth `i-1` that has a `valid_policy` of `anyPolicy` as follows:

- (i) set the `valid_policy` to `ID-P`;
- (ii) set the `qualifier_set` to the `qualifier` set of the policy `anyPolicy` in the certificate policies extension of certificate `i`; and
- (iii) set the `expected_policy_set` to the set of `subjectDomainPolicy` values that are specified as equivalent to `ID-P` by the policy mappings extension.

- (2) If the `policy_mapping` variable is equal to 0:

- (i) delete each node of depth `i` in the `valid_policy_tree` where `ID-P` is the `valid_policy`.
- (ii) If there is a node in the `valid_policy_tree` of depth `i-1` or less without any child nodes, delete that node. Repeat this step until there are no nodes of depth `i-1` or less without children.

- (c) Assign the certificate subject name to `working_issuer_name`.
- (d) Assign the certificate `subjectPublicKey` to `working_public_key`.
- (e) If the `subjectPublicKeyInfo` field of the certificate contains an `algorithm` field with non-null parameters, assign the parameters to the `working_public_key_parameters` variable.

If the `subjectPublicKeyInfo` field of the certificate contains an `algorithm` field with null parameters or parameters are omitted, compare the certificate `subjectPublicKey` algorithm to the `working_public_key_algorithm`. If the certificate `subjectPublicKey` algorithm and the `working_public_key_algorithm` are different, set the `working_public_key_parameters` to null.

- (f) Assign the certificate `subjectPublicKey` algorithm to the `working_public_key_algorithm` variable.
- (g) If a name constraints extension is included in the certificate, modify the `permitted_subtrees` and `excluded_subtrees` state variables as follows:
 - (1) If `permittedSubtrees` is present in the certificate, set the `permitted_subtrees` state variable to the intersection of its previous value and the value indicated in the extension field. If `permittedSubtrees` does not include a particular name type, the `permitted_subtrees` state variable is unchanged for that name type. For example, the intersection of `example.com` and `foo.example.com` is `foo.example.com`. And the intersection of `example.com` and `example.net` is the empty set.
 - (2) If `excludedSubtrees` is present in the certificate, set the `excluded_subtrees` state variable to the union of its previous value and the value indicated in the extension field. If `excludedSubtrees` does not include a particular name type, the `excluded_subtrees` state variable is unchanged for that name type. For example, the union of the name spaces `example.com` and `foo.example.com` is `example.com`. And the union of `example.com` and `example.net` is both name spaces.
- (h) If certificate `i` is not self-issued:
 - (1) If `explicit_policy` is not 0, decrement `explicit_policy` by 1.
 - (2) If `policy_mapping` is not 0, decrement `policy_mapping` by 1.
 - (3) If `inhibit_anyPolicy` is not 0, decrement `inhibit_anyPolicy` by 1.
- (i) If a policy constraints extension is included in the certificate, modify the `explicit_policy` and `policy_mapping` state variables as follows:
 - (1) If `requireExplicitPolicy` is present and is less than `explicit_policy`, set `explicit_policy` to the value of `requireExplicitPolicy`.
 - (2) If `inhibitPolicyMapping` is present and is less than `policy_mapping`, set `policy_mapping` to the value of `inhibitPolicyMapping`.

- (j) If the `inhibitAnyPolicy` extension is included in the certificate and is less than `inhibit_anyPolicy`, set `inhibit_anyPolicy` to the value of `inhibitAnyPolicy`.
- (k) If certificate `i` is a version 3 certificate, verify that the `basicConstraints` extension is present and that `ca` is set to `TRUE`. (If certificate `i` is a version 1 or version 2 certificate, then the application **MUST** either verify that certificate `i` is a CA certificate through out-of-band means or reject the certificate. Conforming implementations may choose to reject all version 1 and version 2 intermediate certificates.)
- (l) If the certificate was not self-issued, verify that `max_path_length` is greater than zero and decrement `max_path_length` by 1.
- (m) If `pathLenConstraint` is present in the certificate and is less than `max_path_length`, set `max_path_length` to the value of `pathLenConstraint`.
- (n) If a key usage extension is present, verify that the `keyCertSign` bit is set.
- (o) Recognize and process any other critical extension present in the certificate. Process any other recognized non-critical extension present in the certificate that is relevant to path processing.

If check (a), (k), (l), (n), or (o) fails, the procedure terminates, returning a failure indication and an appropriate reason.

If (a), (k), (l), (n), and (o) have completed successfully, increment `i` and perform the basic certificate processing specified in Section 6.1.3.

6.1.5. Wrap-Up Procedure

To complete the processing of the target certificate, perform the following steps for certificate `n`:

- (a) If `explicit_policy` is not 0, decrement `explicit_policy` by 1.
- (b) If a policy constraints extension is included in the certificate and `requireExplicitPolicy` is present and has a value of 0, set the `explicit_policy` state variable to 0.

- (c) Assign the certificate `subjectPublicKey` to `working_public_key`.
- (d) If the `subjectPublicKeyInfo` field of the certificate contains an `algorithm` field with non-null parameters, assign the parameters to the `working_public_key_parameters` variable.

If the `subjectPublicKeyInfo` field of the certificate contains an `algorithm` field with null parameters or parameters are omitted, compare the certificate `subjectPublicKey` algorithm to the `working_public_key_algorithm`. If the certificate `subjectPublicKey` algorithm and the `working_public_key_algorithm` are different, set the `working_public_key_parameters` to null.

- (e) Assign the certificate `subjectPublicKey` algorithm to the `working_public_key_algorithm` variable.
- (f) Recognize and process any other critical extension present in the certificate `n`. Process any other recognized non-critical extension present in certificate `n` that is relevant to path processing.
- (g) Calculate the intersection of the `valid_policy_tree` and the `user-initial-policy-set`, as follows:
 - (i) If the `valid_policy_tree` is NULL, the intersection is NULL.
 - (ii) If the `valid_policy_tree` is not NULL and the `user-initial-policy-set` is any-policy, the intersection is the entire `valid_policy_tree`.
 - (iii) If the `valid_policy_tree` is not NULL and the `user-initial-policy-set` is not any-policy, calculate the intersection of the `valid_policy_tree` and the `user-initial-policy-set` as follows:
 1. Determine the set of policy nodes whose parent nodes have a `valid_policy` of anyPolicy. This is the `valid_policy_node_set`.
 2. If the `valid_policy` of any node in the `valid_policy_node_set` is not in the `user-initial-policy-set` and is not anyPolicy, delete this node and all its children.

3. If the `valid_policy_tree` includes a node of depth `n` with the `valid_policy` `anyPolicy` and the `user-initial-policy-set` is not `any-policy`, perform the following steps:
 - a. Set `P-Q` to the `qualifier_set` in the node of depth `n` with `valid_policy` `anyPolicy`.
 - b. For each `P-OID` in the `user-initial-policy-set` that is not the `valid_policy` of a node in the `valid_policy_node_set`, create a child node whose parent is the node of depth `n-1` with the `valid_policy` `anyPolicy`. Set the values in the child node as follows: set the `valid_policy` to `P-OID`, set the `qualifier_set` to `P-Q`, and set the `expected_policy_set` to `{P-OID}`.
 - c. Delete the node of depth `n` with the `valid_policy` `anyPolicy`.
4. If there is a node in the `valid_policy_tree` of depth `n-1` or less without any child nodes, delete that node. Repeat this step until there are no nodes of depth `n-1` or less without children.

If either (1) the value of `explicit_policy` variable is greater than zero or (2) the `valid_policy_tree` is not `NULL`, then path processing has succeeded.

6.1.6. Outputs

If path processing succeeds, the procedure terminates, returning a success indication together with final value of the `valid_policy_tree`, the `working_public_key`, the `working_public_key_algorithm`, and the `working_public_key_parameters`.

6.2. Using the Path Validation Algorithm

The path validation algorithm describes the process of validating a single certification path. While each certification path begins with a specific trust anchor, there is no requirement that all certification paths validated by a particular system share a single trust anchor. The selection of one or more trusted CAs is a local decision. A system may provide any one of its trusted CAs as the trust anchor for a particular path. The inputs to the path validation algorithm may be different for each path. The inputs used to process a path may reflect application-specific requirements or limitations in the trust accorded a particular trust anchor. For

example, a trusted CA may only be trusted for a particular certificate policy. This restriction can be expressed through the inputs to the path validation procedure.

An implementation MAY augment the algorithm presented in Section 6.1 to further limit the set of valid certification paths that begin with a particular trust anchor. For example, an implementation MAY modify the algorithm to apply a path length constraint to a specific trust anchor during the initialization phase, or the application MAY require the presence of a particular alternative name form in the target certificate, or the application MAY impose requirements on application-specific extensions. Thus, the path validation algorithm presented in Section 6.1 defines the minimum conditions for a path to be considered valid.

Where a CA distributes self-signed certificates to specify trust anchor information, certificate extensions can be used to specify recommended inputs to path validation. For example, a policy constraints extension could be included in the self-signed certificate to indicate that paths beginning with this trust anchor should be trusted only for the specified policies. Similarly, a name constraints extension could be included to indicate that paths beginning with this trust anchor should be trusted only for the specified name spaces. The path validation algorithm presented in Section 6.1 does not assume that trust anchor information is provided in self-signed certificates and does not specify processing rules for additional information included in such certificates.

Implementations that use self-signed certificates to specify trust anchor information are free to process or ignore such information.

6.3. CRL Validation

This section describes the steps necessary to determine if a certificate is revoked when CRLs are the revocation mechanism used by the certificate issuer. Conforming implementations that support CRLs are not required to implement this algorithm, but they MUST be functionally equivalent to the external behavior resulting from this procedure when processing CRLs that are issued in conformance with this profile. Any algorithm may be used by a particular implementation so long as it derives the correct result.

This algorithm assumes that all of the needed CRLs are available in a local cache. Further, if the next update time of a CRL has passed, the algorithm assumes a mechanism to fetch a current CRL and place it in the local CRL cache.

This algorithm defines a set of inputs, a set of state variables, and processing steps that are performed for each certificate in the path. The algorithm output is the revocation status of the certificate.

6.3.1. Revocation Inputs

To support revocation processing, the algorithm requires two inputs:

- (a) **certificate:** The algorithm requires the certificate serial number and issuer name to determine whether a certificate is on a particular CRL. The `basicConstraints` extension is used to determine whether the supplied certificate is associated with a CA or an end entity. If present, the algorithm uses the `cRLDistributionPoints` and `freshestCRL` extensions to determine revocation status.
- (b) **use-deltas:** This boolean input determines whether delta CRLs are applied to CRLs.

6.3.2. Initialization and Revocation State Variables

To support CRL processing, the algorithm requires the following state variables:

- (a) **reasons_mask:** This variable contains the set of revocation reasons supported by the CRLs and delta CRLs processed so far. The legal members of the set are the possible revocation reason values minus `unspecified`: `keyCompromise`, `cACompromise`, `affiliationChanged`, `superseded`, `cessationOfOperation`, `certificateHold`, `privilegeWithdrawn`, and `aACompromise`. The special value `all-reasons` is used to denote the set of all legal members. This variable is initialized to the empty set.
- (b) **cert_status:** This variable contains the status of the certificate. This variable may be assigned one of the following values: `unspecified`, `keyCompromise`, `cACompromise`, `affiliationChanged`, `superseded`, `cessationOfOperation`, `certificateHold`, `removeFromCRL`, `privilegeWithdrawn`, `aACompromise`, the special value `UNREVOKED`, or the special value `UNDETERMINED`. This variable is initialized to the special value `UNREVOKED`.
- (c) **interim_reasons_mask:** This contains the set of revocation reasons supported by the CRL or delta CRL currently being processed.

Note: In some environments, it is not necessary to check all reason codes. For example, some environments are only concerned with `cACompromise` and `keyCompromise` for CA certificates. This algorithm checks all reason codes. Additional processing and state variables may be necessary to limit the checking to a subset of the reason codes.

6.3.3. CRL Processing

This algorithm begins by assuming that the certificate is not revoked. The algorithm checks one or more CRLs until either the certificate status is determined to be revoked or sufficient CRLs have been checked to cover all reason codes.

For each distribution point (DP) in the certificate's CRL distribution points extension, for each corresponding CRL in the local CRL cache, while `((reasons_mask is not all-reasons) and (cert_status is UNREVOKED))` perform the following:

- (a) Update the local CRL cache by obtaining a complete CRL, a delta CRL, or both, as required:
 - (1) If the current time is after the value of the CRL next update field, then do one of the following:
 - (i) If `use-deltas` is set and either the certificate or the CRL contains the freshest CRL extension, obtain a delta CRL with a next update value that is after the current time and can be used to update the locally cached CRL as specified in Section 5.2.4.
 - (ii) Update the local CRL cache with a current complete CRL, verify that the current time is before the next update value in the new CRL, and continue processing with the new CRL. If `use-deltas` is set and either the certificate or the CRL contains the freshest CRL extension, then obtain the current delta CRL that can be used to update the new locally cached complete CRL as specified in Section 5.2.4.
 - (2) If the current time is before the value of the next update field, `use-deltas` is set, and either the certificate or the CRL contains the freshest CRL extension, then obtain the current delta CRL that can be used to update the locally cached complete CRL as specified in Section 5.2.4.

- (b) Verify the issuer and scope of the complete CRL as follows:
 - (1) If the DP includes `cRLIssuer`, then verify that the issuer field in the complete CRL matches `cRLIssuer` in the DP and that the complete CRL contains an issuing distribution point extension with the `indirectCRL` boolean asserted. Otherwise, verify that the CRL issuer matches the certificate issuer.
 - (2) If the complete CRL includes an issuing distribution point (IDP) CRL extension, check the following:
 - (i) If the distribution point name is present in the IDP CRL extension and the distribution field is present in the DP, then verify that one of the names in the IDP matches one of the names in the DP. If the distribution point name is present in the IDP CRL extension and the distribution field is omitted from the DP, then verify that one of the names in the IDP matches one of the names in the `cRLIssuer` field of the DP.
 - (ii) If the `onlyContainsUserCerts` boolean is asserted in the IDP CRL extension, verify that the certificate does not include the basic constraints extension with the `ca` boolean asserted.
 - (iii) If the `onlyContainsCACerts` boolean is asserted in the IDP CRL extension, verify that the certificate includes the basic constraints extension with the `ca` boolean asserted.
 - (iv) Verify that the `onlyContainsAttributeCerts` boolean is not asserted.
- (c) If `use-deltas` is set, verify the issuer and scope of the delta CRL as follows:
 - (1) Verify that the delta CRL issuer matches the complete CRL issuer.
 - (2) If the complete CRL includes an issuing distribution point (IDP) CRL extension, verify that the delta CRL contains a matching IDP CRL extension. If the complete CRL omits an IDP CRL extension, verify that the delta CRL also omits an IDP CRL extension.

- (3) Verify that the delta CRL authority key identifier extension matches the complete CRL authority key identifier extension.
- (d) Compute the `interim_reasons_mask` for this CRL as follows:
 - (1) If the issuing distribution point (IDP) CRL extension is present and includes `onlySomeReasons` and the DP includes reasons, then set `interim_reasons_mask` to the intersection of reasons in the DP and `onlySomeReasons` in the IDP CRL extension.
 - (2) If the IDP CRL extension includes `onlySomeReasons` but the DP omits reasons, then set `interim_reasons_mask` to the value of `onlySomeReasons` in the IDP CRL extension.
 - (3) If the IDP CRL extension is not present or omits `onlySomeReasons` but the DP includes reasons, then set `interim_reasons_mask` to the value of DP reasons.
 - (4) If the IDP CRL extension is not present or omits `onlySomeReasons` and the DP omits reasons, then set `interim_reasons_mask` to the special value `all-reasons`.
- (e) Verify that `interim_reasons_mask` includes one or more reasons that are not included in the `reasons_mask`.
- (f) Obtain and validate the certification path for the issuer of the complete CRL. The trust anchor for the certification path MUST be the same as the trust anchor used to validate the target certificate. If a key usage extension is present in the CRL issuer's certificate, verify that the `cRLSign` bit is set.
- (g) Validate the signature on the complete CRL using the public key validated in step (f).
- (h) If `use-deltas` is set, then validate the signature on the delta CRL using the public key validated in step (f).
- (i) If `use-deltas` is set, then search for the certificate on the delta CRL. If an entry is found that matches the certificate issuer and serial number as described in Section 5.3.3, then set the `cert_status` variable to the indicated reason as follows:

- (1) If the reason code CRL entry extension is present, set the `cert_status` variable to the value of the reason code CRL entry extension.
- (2) If the reason code CRL entry extension is not present, set the `cert_status` variable to the value unspecified.
- (j) If (`cert_status` is `UNREVOKED`), then search for the certificate on the complete CRL. If an entry is found that matches the certificate issuer and serial number as described in Section 5.3.3, then set the `cert_status` variable to the indicated reason as described in step (i).
- (k) If (`cert_status` is `removeFromCRL`), then set `cert_status` to `UNREVOKED`.
- (l) Set the `reasons_mask` state variable to the union of its previous value and the value of the `interim_reasons_mask` state variable.

If ((`reasons_mask` is `all-reasons`) OR (`cert_status` is not `UNREVOKED`)), then the revocation status has been determined, so return `cert_status`.

If the revocation status has not been determined, repeat the process above with any available CRLs not specified in a distribution point but issued by the certificate issuer. For the processing of such a CRL, assume a DP with both the `reasons` and the `cRLIssuer` fields omitted and a distribution point name of the certificate issuer. That is, the sequence of names in `fullName` is generated from the certificate issuer field as well as the certificate issuerAltName extension. After processing such CRLs, if the revocation status has still not been determined, then return the `cert_status` `UNDETERMINED`.

7. Processing Rules for Internationalized Names

Internationalized names may be encountered in numerous certificate and CRL fields and extensions, including distinguished names, internationalized domain names, electronic mail addresses, and Internationalized Resource Identifiers (IRIs). Storage, comparison, and presentation of such names require special care. Some characters may be encoded in multiple ways. The same names could be represented in multiple encodings (e.g., ASCII or UTF8). This section establishes conformance requirements for storage or comparison of each of these name forms. Informative guidance on presentation is provided for some of these name forms.

7.1. Internationalized Names in Distinguished Names

Representation of internationalized names in distinguished names is covered in Sections 4.1.2.4, Issuer Name, and 4.1.2.6, Subject Name. Standard naming attributes, such as common name, employ the `DirectoryString` type, which supports internationalized names through a variety of language encodings. Conforming implementations **MUST** support `UTF8String` and `PrintableString`. RFC 3280 required only binary comparison of attribute values encoded in `UTF8String`, however, this specification requires a more comprehensive handling of comparison. Implementations may encounter certificates and CRLs with names encoded using `TeletexString`, `BMPString`, or `UniversalString`, but support for these is **OPTIONAL**.

Conforming implementations **MUST** use the LDAP StringPrep profile (including insignificant space handling), as specified in [RFC4518], as the basis for comparison of distinguished name attributes encoded in either `PrintableString` or `UTF8String`. Conforming implementations **MUST** support name comparisons using `caseIgnoreMatch`. Support for attribute types that use other equality matching rules is optional.

Before comparing names using the `caseIgnoreMatch` matching rule, conforming implementations **MUST** perform the six-step string preparation algorithm described in [RFC4518] for each attribute of type `DirectoryString`, with the following clarifications:

- * In step 2, Map, the mapping shall include case folding as specified in Appendix B.2 of [RFC3454].
- * In step 6, Insignificant Character Removal, perform white space compression as specified in Section 2.6.1, Insignificant Space Handling, of [RFC4518].

When performing the string preparation algorithm, attributes **MUST** be treated as stored values.

Comparisons of `domainComponent` attributes **MUST** be performed as specified in Section 7.3.

Two naming attributes match if the attribute types are the same and the values of the attributes are an exact match after processing with the string preparation algorithm. Two relative distinguished names `RDN1` and `RDN2` match if they have the same number of naming attributes and for each naming attribute in `RDN1` there is a matching naming attribute in `RDN2`. Two distinguished names `DN1` and `DN2` match if they have the same number of `RDNs`, for each `RDN` in `DN1` there is a matching `RDN` in `DN2`, and the matching `RDNs` appear in the same order in both `DNs`. A distinguished name `DN1` is within the subtree defined by the

distinguished name DN2 if DN1 contains at least as many RDNs as DN2, and DN1 and DN2 are a match when trailing RDNs in DN1 are ignored.

7.2. Internationalized Domain Names in GeneralName

Internationalized Domain Names (IDNs) may be included in certificates and CRLs in the subjectAltName and issuerAltName extensions, name constraints extension, authority information access extension, subject information access extension, CRL distribution points extension, and issuing distribution point extension. Each of these extensions uses the GeneralName type; one choice in GeneralName is the dNSName field, which is defined as type IA5String.

IA5String is limited to the set of ASCII characters. To accommodate internationalized domain names in the current structure, conforming implementations MUST convert internationalized domain names to the ASCII Compatible Encoding (ACE) format as specified in Section 4 of RFC 3490 before storage in the dNSName field. Specifically, conforming implementations MUST perform the conversion operation specified in Section 4 of RFC 3490, with the following clarifications:

- * in step 1, the domain name SHALL be considered a "stored string". That is, the AllowUnassigned flag SHALL NOT be set;
- * in step 3, set the flag called "UseSTD3ASCIIRules";
- * in step 4, process each label with the "ToASCII" operation; and
- * in step 5, change all label separators to U+002E (full stop).

When comparing DNS names for equality, conforming implementations MUST perform a case-insensitive exact match on the entire DNS name. When evaluating name constraints, conforming implementations MUST perform a case-insensitive exact match on a label-by-label basis. As noted in Section 4.2.1.10, any DNS name that may be constructed by adding labels to the left-hand side of the domain name given as the constraint is considered to fall within the indicated subtree.

Implementations should convert IDNs to Unicode before display. Specifically, conforming implementations should perform the conversion operation specified in Section 4 of RFC 3490, with the following clarifications:

- * in step 1, the domain name SHALL be considered a "stored string". That is, the AllowUnassigned flag SHALL NOT be set;
- * in step 3, set the flag called "UseSTD3ASCIIRules";

- * in step 4, process each label with the "ToUnicode" operation;
and
- * skip step 5.

Note: Implementations MUST allow for increased space requirements for IDNs. An IDN ACE label will begin with the four additional characters "xn--" and may require as many as five ASCII characters to specify a single international character.

7.3. Internationalized Domain Names in Distinguished Names

Domain Names may also be represented as distinguished names using domain components in the subject field, the issuer field, the subjectAltName extension, or the issuerAltName extension. As with the dNSName in the GeneralName type, the value of this attribute is defined as an IA5String. Each domainComponent attribute represents a single label. To represent a label from an IDN in the distinguished name, the implementation MUST perform the "ToASCII" label conversion specified in Section 4.1 of RFC 3490. The label SHALL be considered a "stored string". That is, the AllowUnassigned flag SHALL NOT be set.

Conforming implementations shall perform a case-insensitive exact match when comparing domainComponent attributes in distinguished names, as described in Section 7.2.

Implementations should convert ACE labels to Unicode before display. Specifically, conforming implementations should perform the "ToUnicode" conversion operation specified, as described in Section 7.2, on each ACE label before displaying the name.

7.4. Internationalized Resource Identifiers

Internationalized Resource Identifiers (IRIs) are the internationalized complement to the Uniform Resource Identifier (URI). IRIs are sequences of characters from Unicode, while URIs are sequences of characters from the ASCII character set. [RFC3987] defines a mapping from IRIs to URIs. While IRIs are not encoded directly in any certificate fields or extensions, their mapped URIs may be included in certificates and CRLs. URIs may appear in the subjectAltName and issuerAltName extensions, name constraints extension, authority information access extension, subject information access extension, issuing distribution point extension, and CRL distribution points extension. Each of these extensions uses the GeneralName type; URIs are encoded in the uniformResourceIdentifier field in GeneralName, which is defined as type IA5String.

To accommodate IRIs in the current structure, conforming implementations MUST map IRIs to URIs as specified in Section 3.1 of [RFC3987], with the following clarifications:

- * in step 1, generate a UCS character sequence from the original IRI format normalizing according to the NFC as specified in Variant b (normalization according to NFC);
- * perform step 2 using the output from step 1.

Implementations MUST NOT convert the ireg-name component before performing step 2.

Before URIs may be compared, conforming implementations MUST perform a combination of the syntax-based and scheme-based normalization techniques described in [RFC3987]. Specifically, conforming implementations MUST prepare URIs for comparison as follows:

- * Step 1: Where IRIs allow the usage of IDNs, those names MUST be converted to ASCII Compatible Encoding as specified in Section 7.2 above.
- * Step 2: The scheme and host are normalized to lowercase, as described in Section 5.3.2.1 of [RFC3987].
- * Step 3: Perform percent-encoding normalization, as specified in Section 5.3.2.3 of [RFC3987].
- * Step 4: Perform path segment normalization, as specified in Section 5.3.2.4 of [RFC3987].
- * Step 5: If recognized, the implementation MUST perform scheme-based normalization as specified in Section 5.3.3 of [RFC3987].

Conforming implementations MUST recognize and perform scheme-based normalization for the following schemes: ldap, http, https, and ftp. If the scheme is not recognized, step 5 is omitted.

When comparing URIs for equivalence, conforming implementations shall perform a case-sensitive exact match.

Implementations should convert URIs to Unicode before display. Specifically, conforming implementations should perform the conversion operation specified in Section 3.2 of [RFC3987].

7.5. Internationalized Electronic Mail Addresses

Electronic Mail addresses may be included in certificates and CRLs in the subjectAltName and issuerAltName extensions, name constraints extension, authority information access extension, subject information access extension, issuing distribution point extension, or CRL distribution points extension. Each of these extensions uses the GeneralName construct; GeneralName includes the rfc822Name choice, which is defined as type IA5String. To accommodate email addresses with internationalized domain names using the current structure, conforming implementations MUST convert the addresses into an ASCII representation.

Where the host-part (the Domain of the Mailbox) contains an internationalized name, the domain name MUST be converted from an IDN to the ASCII Compatible Encoding (ACE) format as specified in Section 7.2.

Two email addresses are considered to match if:

- 1) the local-part of each name is an exact match, AND
- 2) the host-part of each name matches using a case-insensitive ASCII comparison.

Implementations should convert the host-part of internationalized email addresses specified in these extensions to Unicode before display. Specifically, conforming implementations should perform the conversion of the host-part of the Mailbox as described in Section 7.2.

8. Security Considerations

The majority of this specification is devoted to the format and content of certificates and CRLs. Since certificates and CRLs are digitally signed, no additional integrity service is necessary. Neither certificates nor CRLs need be kept secret, and unrestricted and anonymous access to certificates and CRLs has no security implications.

However, security factors outside the scope of this specification will affect the assurance provided to certificate users. This section highlights critical issues to be considered by implementers, administrators, and users.

The procedures performed by CAs and RAs to validate the binding of the subject's identity to their public key greatly affect the assurance that ought to be placed in the certificate. Relying

parties might wish to review the CA's certification practice statement. This is particularly important when issuing certificates to other CAs.

The use of a single key pair for both signature and other purposes is strongly discouraged. Use of separate key pairs for signature and key management provides several benefits to the users. The ramifications associated with loss or disclosure of a signature key are different from loss or disclosure of a key management key. Using separate key pairs permits a balanced and flexible response. Similarly, different validity periods or key lengths for each key pair may be appropriate in some application environments. Unfortunately, some legacy applications (e.g., Secure Sockets Layer (SSL)) use a single key pair for signature and key management.

The protection afforded private keys is a critical security factor. On a small scale, failure of users to protect their private keys will permit an attacker to masquerade as them or decrypt their personal information. On a larger scale, compromise of a CA's private signing key may have a catastrophic effect. If an attacker obtains the private key unnoticed, the attacker may issue bogus certificates and CRLs. Existence of bogus certificates and CRLs will undermine confidence in the system. If such a compromise is detected, all certificates issued to the compromised CA MUST be revoked, preventing services between its users and users of other CAs. Rebuilding after such a compromise will be problematic, so CAs are advised to implement a combination of strong technical measures (e.g., tamper-resistant cryptographic modules) and appropriate management procedures (e.g., separation of duties) to avoid such an incident.

Loss of a CA's private signing key may also be problematic. The CA would not be able to produce CRLs or perform normal key rollover. CAs SHOULD maintain secure backup for signing keys. The security of the key backup procedures is a critical factor in avoiding key compromise.

The availability and freshness of revocation information affects the degree of assurance that ought to be placed in a certificate. While certificates expire naturally, events may occur during its natural lifetime that negate the binding between the subject and public key. If revocation information is untimely or unavailable, the assurance associated with the binding is clearly reduced. Relying parties might not be able to process every critical extension that can appear in a CRL. CAs SHOULD take extra care when making revocation information available only through CRLs that contain critical extensions, particularly if support for those extensions is not mandated by this profile. For example, if revocation information is supplied using a combination of delta CRLs and full CRLs, and the

delta CRLs are issued more frequently than the full CRLs, then relying parties that cannot handle the critical extensions related to delta CRL processing will not be able to obtain the most recent revocation information. Alternatively, if a full CRL is issued whenever a delta CRL is issued, then timely revocation information will be available to all relying parties. Similarly, implementations of the certification path validation mechanism described in Section 6 that omit revocation checking provide less assurance than those that support it.

The certification path validation algorithm depends on the certain knowledge of the public keys (and other information) about one or more trusted CAs. The decision to trust a CA is an important decision as it ultimately determines the trust afforded a certificate. The authenticated distribution of trusted CA public keys (usually in the form of a "self-signed" certificate) is a security critical out-of-band process that is beyond the scope of this specification.

In addition, where a key compromise or CA failure occurs for a trusted CA, the user will need to modify the information provided to the path validation routine. Selection of too many trusted CAs makes the trusted CA information difficult to maintain. On the other hand, selection of only one trusted CA could limit users to a closed community of users.

The quality of implementations that process certificates also affects the degree of assurance provided. The path validation algorithm described in Section 6 relies upon the integrity of the trusted CA information, and especially the integrity of the public keys associated with the trusted CAs. By substituting public keys for which an attacker has the private key, an attacker could trick the user into accepting false certificates.

The binding between a key and certificate subject cannot be stronger than the cryptographic module implementation and algorithms used to generate the signature. Short key lengths or weak hash algorithms will limit the utility of a certificate. CAs are encouraged to note advances in cryptology so they can employ strong cryptographic techniques. In addition, CAs SHOULD decline to issue certificates to CAs or end entities that generate weak signatures.

Inconsistent application of name comparison rules can result in acceptance of invalid X.509 certification paths or rejection of valid ones. The X.500 series of specifications defines rules for comparing distinguished names that require comparison of strings without regard

to case, character set, multi-character white space substring, or leading and trailing white space. This specification relaxes these requirements, requiring support for binary comparison at a minimum.

CAs MUST encode the distinguished name in the subject field of a CA certificate identically to the distinguished name in the issuer field in certificates issued by that CA. If CAs use different encodings, implementations might fail to recognize name chains for paths that include this certificate. As a consequence, valid paths could be rejected.

In addition, name constraints for distinguished names MUST be stated identically to the encoding used in the subject field or subjectAltName extension. If not, then name constraints stated as excludedSubtrees will not match and invalid paths will be accepted and name constraints expressed as permittedSubtrees will not match and valid paths will be rejected. To avoid acceptance of invalid paths, CAs SHOULD state name constraints for distinguished names as permittedSubtrees wherever possible.

In general, using the nameConstraints extension to constrain one name form (e.g., DNS names) offers no protection against use of other name forms (e.g., electronic mail addresses).

While X.509 mandates that names be unambiguous, there is a risk that two unrelated authorities will issue certificates and/or CRLs under the same issuer name. As a means of reducing problems and security issues related to issuer name collisions, CA and CRL issuer names SHOULD be formed in a way that reduces the likelihood of name collisions. Implementers should take into account the possible existence of multiple unrelated CAs and CRL issuers with the same name. At a minimum, implementations validating CRLs MUST ensure that the certification path of a certificate and the CRL issuer certification path used to validate the certificate terminate at the same trust anchor.

While the local-part of an electronic mail address is case sensitive [RFC2821], emailAddress attribute values are not case sensitive [RFC2985]. As a result, there is a risk that two different email addresses will be treated as the same address when the matching rule for the emailAddress attribute is used, if the email server exploits the case sensitivity of mailbox local-parts. Implementers should not include an email address in the emailAddress attribute if the email server that hosts the email address treats the local-part of email addresses as case sensitive.

Implementers should be aware of risks involved if the CRL distribution points or authority information access extensions of

corrupted certificates or CRLs contain links to malicious code. Implementers should always take the steps of validating the retrieved data to ensure that the data is properly formed.

When certificates include a `cRLDistributionPoints` extension with an `https` URI or similar scheme, circular dependencies can be introduced. The relying party is forced to perform an additional path validation in order to obtain the CRL required to complete the initial path validation! Circular conditions can also be created with an `https` URI (or similar scheme) in the `authorityInfoAccess` or `subjectInfoAccess` extensions. At worst, this situation can create unresolvable dependencies.

CAs SHOULD NOT include URIs that specify `https`, `ldaps`, or similar schemes in extensions. CAs that include an `https` URI in one of these extensions MUST ensure that the server's certificate can be validated without using the information that is pointed to by the URI. Relying parties that choose to validate the server's certificate when obtaining information pointed to by an `https` URI in the `cRLDistributionPoints`, `authorityInfoAccess`, or `subjectInfoAccess` extensions MUST be prepared for the possibility that this will result in unbounded recursion.

Self-issued certificates provide CAs with one automated mechanism to indicate changes in the CA's operations. In particular, self-issued certificates may be used to implement a graceful change-over from one non-compromised CA key pair to the next. Detailed procedures for "CA key update" are specified in [RFC4210], where the CA protects its new public key using its previous private key and vice versa using two self-issued certificates. Conforming client implementations will process the self-issued certificate and determine whether certificates issued under the new key may be trusted. Self-issued certificates MAY be used to support other changes in CA operations, such as additions to the CA's policy set, using similar procedures.

Some legacy implementations support names encoded in the ISO 8859-1 character set (`Latin1String`) [ISO8859] but tag them as `TeletexString`. `TeletexString` encodes a larger character set than ISO 8859-1, but it encodes some characters differently. The name comparison rules specified in Section 7.1 assume that `TeletexStrings` are encoded as described in the ASN.1 standard. When comparing names encoded using the `Latin1String` character set, false positives and negatives are possible.

When strings are mapped from internal representations to visual representations, sometimes two different strings will have the same or similar visual representations. This can happen for many different reasons, including use of similar glyphs and use of

composed characters (such as e + ' equaling U+00E9, the Korean composed characters, and vowels above consonant clusters in certain languages). As a result of this situation, people doing visual comparisons between two different names may think they are the same when in fact they are not. Also, people may mistake one string for another. Issuers of certificates and relying parties both need to be aware of this situation.

9. IANA Considerations

Extensions in certificates and CRLs are identified using object identifiers. The objects are defined in an arc delegated by IANA to the PKIX Working Group. No further action by IANA is necessary for this document or any anticipated updates.

10. Acknowledgments

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Appendix A. Pseudo-ASN.1 Structures and OIDs

This appendix describes data objects used by conforming PKI components in an "ASN.1-like" syntax. This syntax is a hybrid of the 1988 and 1993 ASN.1 syntaxes. The 1988 ASN.1 syntax is augmented with 1993 UNIVERSAL Types UniversalString, BMPString, and UTF8String.

The ASN.1 syntax does not permit the inclusion of type statements in the ASN.1 module, and the 1993 ASN.1 standard does not permit use of the new UNIVERSAL types in modules using the 1988 syntax. As a result, this module does not conform to either version of the ASN.1 standard.

This appendix may be converted into 1988 ASN.1 by replacing the definitions for the UNIVERSAL Types with the 1988 catch-all "ANY".

A.1. Explicitly Tagged Module, 1988 Syntax

```
PKIX1Explicit88 { iso(1) identified-organization(3) dod(6) internet(1)
  security(5) mechanisms(5) pkix(7) id-mod(0) id-pkix1-explicit(18) }

DEFINITIONS EXPLICIT TAGS ::=

BEGIN

-- EXPORTS ALL --

-- IMPORTS NONE --

-- UNIVERSAL Types defined in 1993 and 1998 ASN.1
-- and required by this specification

UniversalString ::= [UNIVERSAL 28] IMPLICIT OCTET STRING
  -- UniversalString is defined in ASN.1:1993

BMPString ::= [UNIVERSAL 30] IMPLICIT OCTET STRING
  -- BMPString is the subtype of UniversalString and models
  -- the Basic Multilingual Plane of ISO/IEC 10646

UTF8String ::= [UNIVERSAL 12] IMPLICIT OCTET STRING
  -- The content of this type conforms to RFC 3629.

-- PKIX specific OIDs

id-pkix OBJECT IDENTIFIER ::=
  { iso(1) identified-organization(3) dod(6) internet(1)
    security(5) mechanisms(5) pkix(7) }
```

```
-- PKIX arcs

id-pe OBJECT IDENTIFIER ::= { id-pkix 1 }
    -- arc for private certificate extensions
id-qt OBJECT IDENTIFIER ::= { id-pkix 2 }
    -- arc for policy qualifier types
id-kp OBJECT IDENTIFIER ::= { id-pkix 3 }
    -- arc for extended key purpose OIDS
id-ad OBJECT IDENTIFIER ::= { id-pkix 48 }
    -- arc for access descriptors

-- policyQualifierIds for Internet policy qualifiers

id-qt-cps      OBJECT IDENTIFIER ::= { id-qt 1 }
    -- OID for CPS qualifier
id-qt-unotice  OBJECT IDENTIFIER ::= { id-qt 2 }
    -- OID for user notice qualifier

-- access descriptor definitions

id-ad-ocsp      OBJECT IDENTIFIER ::= { id-ad 1 }
id-ad-caIssuers OBJECT IDENTIFIER ::= { id-ad 2 }
id-ad-timeStamping OBJECT IDENTIFIER ::= { id-ad 3 }
id-ad-caRepository OBJECT IDENTIFIER ::= { id-ad 5 }

-- attribute data types

Attribute      ::= SEQUENCE {
    type      AttributeType,
    values    SET OF AttributeValue }
    -- at least one value is required

AttributeType  ::= OBJECT IDENTIFIER

AttributeValue ::= ANY -- DEFINED BY AttributeType

AttributeTypeAndValue ::= SEQUENCE {
    type      AttributeType,
    value     AttributeValue }

-- suggested naming attributes: Definition of the following
-- information object set may be augmented to meet local
-- requirements. Note that deleting members of the set may
-- prevent interoperability with conforming implementations.
-- presented in pairs: the AttributeType followed by the
-- type definition for the corresponding AttributeValue
```

```
-- Arc for standard naming attributes

id-at OBJECT IDENTIFIER ::= { joint-iso-ccitt(2) ds(5) 4 }

-- Naming attributes of type X520name

id-at-name           AttributeType ::= { id-at 41 }
id-at-surname        AttributeType ::= { id-at  4 }
id-at-givenName       AttributeType ::= { id-at 42 }
id-at-initials        AttributeType ::= { id-at 43 }
id-at-generationQualifier AttributeType ::= { id-at 44 }

-- Naming attributes of type X520Name:
--   X520name ::= DirectoryString (SIZE (1..ub-name))
--
-- Expanded to avoid parameterized type:
X520name ::= CHOICE {
    teletexString      TeletexString      (SIZE (1..ub-name)),
    printableString    PrintableString    (SIZE (1..ub-name)),
    universalString     UniversalString    (SIZE (1..ub-name)),
    utf8String          UTF8String         (SIZE (1..ub-name)),
    bmpString           BMPString          (SIZE (1..ub-name)) }

-- Naming attributes of type X520CommonName

id-at-commonName      AttributeType ::= { id-at 3 }

-- Naming attributes of type X520CommonName:
--   X520CommonName ::= DirectoryName (SIZE (1..ub-common-name))
--
-- Expanded to avoid parameterized type:
X520CommonName ::= CHOICE {
    teletexString      TeletexString      (SIZE (1..ub-common-name)),
    printableString    PrintableString    (SIZE (1..ub-common-name)),
    universalString     UniversalString    (SIZE (1..ub-common-name)),
    utf8String          UTF8String         (SIZE (1..ub-common-name)),
    bmpString           BMPString          (SIZE (1..ub-common-name)) }
```


-- Naming attributes of type X520LocalityName

id-at-localityName AttributeType ::= { id-at 7 }

-- Naming attributes of type X520LocalityName:

-- X520LocalityName ::= DirectoryName (SIZE (1..ub-locality-name))

--

-- Expanded to avoid parameterized type:

X520LocalityName ::= CHOICE {
 teletexString TeletexString (SIZE (1..ub-locality-name)),
 printableString PrintableString (SIZE (1..ub-locality-name)),
 universalString UniversalString (SIZE (1..ub-locality-name)),
 utf8String UTF8String (SIZE (1..ub-locality-name)),
 bmpString BMPString (SIZE (1..ub-locality-name)) }

-- Naming attributes of type X520StateOrProvinceName

id-at-stateOrProvinceName AttributeType ::= { id-at 8 }

-- Naming attributes of type X520StateOrProvinceName:

-- X520StateOrProvinceName ::= DirectoryName (SIZE (1..ub-state-name))

--

-- Expanded to avoid parameterized type:

X520StateOrProvinceName ::= CHOICE {
 teletexString TeletexString (SIZE (1..ub-state-name)),
 printableString PrintableString (SIZE (1..ub-state-name)),
 universalString UniversalString (SIZE (1..ub-state-name)),
 utf8String UTF8String (SIZE (1..ub-state-name)),
 bmpString BMPString (SIZE (1..ub-state-name)) }

```
-- Naming attributes of type X520OrganizationName

id-at-organizationName AttributeType ::= { id-at 10 }

-- Naming attributes of type X520OrganizationName:
--   X520OrganizationName ::=
--       DirectoryName (SIZE (1..ub-organization-name))
--
-- Expanded to avoid parameterized type:
X520OrganizationName ::= CHOICE {
    teletexString      TeletexString
                        (SIZE (1..ub-organization-name)),
    printableString    PrintableString
                        (SIZE (1..ub-organization-name)),
    universalString    UniversalString
                        (SIZE (1..ub-organization-name)),
    utf8String         UTF8String
                        (SIZE (1..ub-organization-name)),
    bmpString          BMPString
                        (SIZE (1..ub-organization-name)) }

-- Naming attributes of type X520OrganizationalUnitName

id-at-organizationalUnitName AttributeType ::= { id-at 11 }

-- Naming attributes of type X520OrganizationalUnitName:
--   X520OrganizationalUnitName ::=
--       DirectoryName (SIZE (1..ub-organizational-unit-name))
--
-- Expanded to avoid parameterized type:
X520OrganizationalUnitName ::= CHOICE {
    teletexString      TeletexString
                        (SIZE (1..ub-organizational-unit-name)),
    printableString    PrintableString
                        (SIZE (1..ub-organizational-unit-name)),
    universalString    UniversalString
                        (SIZE (1..ub-organizational-unit-name)),
    utf8String         UTF8String
                        (SIZE (1..ub-organizational-unit-name)),
    bmpString          BMPString
                        (SIZE (1..ub-organizational-unit-name)) }
```

```
-- Naming attributes of type X520Title

id-at-title                AttributeType ::= { id-at 12 }

-- Naming attributes of type X520Title:
--   X520Title ::= DirectoryName (SIZE (1..ub-title))
--
-- Expanded to avoid parameterized type:
X520Title ::= CHOICE {
    teletexString      TeletexString      (SIZE (1..ub-title)),
    printableString    PrintableString    (SIZE (1..ub-title)),
    universalString    UniversalString    (SIZE (1..ub-title)),
    utf8String         UTF8String         (SIZE (1..ub-title)),
    bmpString          BMPString          (SIZE (1..ub-title)) }

-- Naming attributes of type X520dnQualifier

id-at-dnQualifier          AttributeType ::= { id-at 46 }

X520dnQualifier ::=      PrintableString

-- Naming attributes of type X520countryName (digraph from IS 3166)

id-at-countryName          AttributeType ::= { id-at 6 }

X520countryName ::=      PrintableString (SIZE (2))

-- Naming attributes of type X520SerialNumber

id-at-serialNumber         AttributeType ::= { id-at 5 }

X520SerialNumber ::=      PrintableString (SIZE (1..ub-serial-number))

-- Naming attributes of type X520Pseudonym

id-at-pseudonym            AttributeType ::= { id-at 65 }

-- Naming attributes of type X520Pseudonym:
--   X520Pseudonym ::= DirectoryName (SIZE (1..ub-pseudonym))
--
-- Expanded to avoid parameterized type:
X520Pseudonym ::= CHOICE {
    teletexString      TeletexString      (SIZE (1..ub-pseudonym)),
    printableString    PrintableString    (SIZE (1..ub-pseudonym)),
    universalString    UniversalString    (SIZE (1..ub-pseudonym)),
    utf8String         UTF8String         (SIZE (1..ub-pseudonym)),
    bmpString          BMPString          (SIZE (1..ub-pseudonym)) }
```

```
-- Naming attributes of type DomainComponent (from RFC 4519)

id-domainComponent    AttributeType ::= { 0 9 2342 19200300 100 1 25 }

DomainComponent ::=   IA5String

-- Legacy attributes

pkcs-9 OBJECT IDENTIFIER ::=
    { iso(1) member-body(2) us(840) rsadsi(113549) pkcs(1) 9 }

id-emailAddress       AttributeType ::= { pkcs-9 1 }

EmailAddress ::=      IA5String (SIZE (1..ub-emailaddress-length))

-- naming data types --

Name ::= CHOICE { -- only one possibility for now --
    rdnSequence    RDNSequence }

RDNSequence ::= SEQUENCE OF RelativeDistinguishedName

DistinguishedName ::=   RDNSequence

RelativeDistinguishedName ::= SET SIZE (1..MAX) OF AttributeTypeAndValue

-- Directory string type --

DirectoryString ::= CHOICE {
    teletexString      TeletexString      (SIZE (1..MAX)),
    printableString    PrintableString    (SIZE (1..MAX)),
    universalString    UniversalString    (SIZE (1..MAX)),
    utf8String         UTF8String         (SIZE (1..MAX)),
    bmpString          BMPString          (SIZE (1..MAX)) }

-- certificate and CRL specific structures begin here

Certificate ::= SEQUENCE {
    tbsCertificate      TBSCertificate,
    signatureAlgorithm  AlgorithmIdentifier,
    signature            BIT STRING  }
```

```

TBSCertificate ::= SEQUENCE {
    version          [0] Version DEFAULT v1,
    serialNumber      CertificateSerialNumber,
    signature         AlgorithmIdentifier,
    issuer            Name,
    validity          Validity,
    subject           Name,
    subjectPublicKeyInfo SubjectPublicKeyInfo,
    issuerUniqueID    [1] IMPLICIT UniqueIdentifier OPTIONAL,
                      -- If present, version MUST be v2 or v3
    subjectUniqueID   [2] IMPLICIT UniqueIdentifier OPTIONAL,
                      -- If present, version MUST be v2 or v3
    extensions        [3] Extensions OPTIONAL
                      -- If present, version MUST be v3 -- }

Version ::= INTEGER { v1(0), v2(1), v3(2) }

CertificateSerialNumber ::= INTEGER

Validity ::= SEQUENCE {
    notBefore      Time,
    notAfter       Time }

Time ::= CHOICE {
    utcTime        UTCTime,
    generalTime    GeneralizedTime }

UniqueIdentifier ::= BIT STRING

SubjectPublicKeyInfo ::= SEQUENCE {
    algorithm        AlgorithmIdentifier,
    subjectPublicKey  BIT STRING }

Extensions ::= SEQUENCE SIZE (1..MAX) OF Extension

Extension ::= SEQUENCE {
    extnID          OBJECT IDENTIFIER,
    critical         BOOLEAN DEFAULT FALSE,
    extnValue        OCTET STRING
                    -- contains the DER encoding of an ASN.1 value
                    -- corresponding to the extension type identified
                    -- by extnID
}
    
```

-- CRL structures

```
CertificateList ::= SEQUENCE {
    tbsCertList      TBSCertList,
    signatureAlgorithm AlgorithmIdentifier,
    signature         BIT STRING }
```

```
TBSCertList ::= SEQUENCE {
    version          Version OPTIONAL,
                        -- if present, MUST be v2
    signature         AlgorithmIdentifier,
    issuer            Name,
    thisUpdate        Time,
    nextUpdate        Time OPTIONAL,
    revokedCertificates SEQUENCE OF SEQUENCE {
        userCertificate      CertificateSerialNumber,
        revocationDate       Time,
        crlEntryExtensions   Extensions OPTIONAL
                        -- if present, version MUST be v2
    } OPTIONAL,
    crlExtensions       [0] Extensions OPTIONAL }
                        -- if present, version MUST be v2
```

-- Version, Time, CertificateSerialNumber, and Extensions were
-- defined earlier for use in the certificate structure

```
AlgorithmIdentifier ::= SEQUENCE {
    algorithm         OBJECT IDENTIFIER,
    parameters        ANY DEFINED BY algorithm OPTIONAL }
    -- contains a value of the type
    -- registered for use with the
    -- algorithm object identifier value
```

-- X.400 address syntax starts here

```
ORAddress ::= SEQUENCE {
    built-in-standard-attributes BuiltInStandardAttributes,
    built-in-domain-defined-attributes
        BuiltInDomainDefinedAttributes OPTIONAL,
    -- see also teletex-domain-defined-attributes
    extension-attributes ExtensionAttributes OPTIONAL }
```

-- Built-in Standard Attributes

```
BuiltInStandardAttributes ::= SEQUENCE {
    country-name                CountryName OPTIONAL,
    administration-domain-name  AdministrationDomainName OPTIONAL,
    network-address              [0] IMPLICIT NetworkAddress OPTIONAL,
    -- see also extended-network-address
    terminal-identifier          [1] IMPLICIT TerminalIdentifier OPTIONAL,
    private-domain-name          [2] PrivateDomainName OPTIONAL,
    organization-name            [3] IMPLICIT OrganizationName OPTIONAL,
    -- see also teletex-organization-name
    numeric-user-identifier      [4] IMPLICIT NumericUserIdentifier
                                OPTIONAL,
    personal-name                [5] IMPLICIT PersonalName OPTIONAL,
    -- see also teletex-personal-name
    organizational-unit-names    [6] IMPLICIT OrganizationalUnitNames
                                OPTIONAL }
    -- see also teletex-organizational-unit-names

CountryName ::= [APPLICATION 1] CHOICE {
    x121-dcc-code                NumericString
                                (SIZE (ub-country-name-numeric-length)),
    iso-3166-alpha2-code         PrintableString
                                (SIZE (ub-country-name-alpha-length)) }

AdministrationDomainName ::= [APPLICATION 2] CHOICE {
    numeric      NumericString (SIZE (0..ub-domain-name-length)),
    printable    PrintableString (SIZE (0..ub-domain-name-length)) }

NetworkAddress ::= X121Address -- see also extended-network-address

X121Address ::= NumericString (SIZE (1..ub-x121-address-length))

TerminalIdentifier ::= PrintableString (SIZE (1..ub-terminal-id-length))

PrivateDomainName ::= CHOICE {
    numeric      NumericString (SIZE (1..ub-domain-name-length)),
    printable    PrintableString (SIZE (1..ub-domain-name-length)) }

OrganizationName ::= PrintableString
                    (SIZE (1..ub-organization-name-length))
    -- see also teletex-organization-name

NumericUserIdentifier ::= NumericString
                        (SIZE (1..ub-numeric-user-id-length))
```

```

PersonalName ::= SET {
    surname      [0] IMPLICIT PrintableString
                    (SIZE (1..ub-surname-length)),
    given-name   [1] IMPLICIT PrintableString
                    (SIZE (1..ub-given-name-length)) OPTIONAL,
    initials     [2] IMPLICIT PrintableString
                    (SIZE (1..ub-initials-length)) OPTIONAL,
    generation-qualifier [3] IMPLICIT PrintableString
                    (SIZE (1..ub-generation-qualifier-length))
                    OPTIONAL }
-- see also teletex-personal-name

OrganizationalUnitNames ::= SEQUENCE SIZE (1..ub-organizational-units)
    OF OrganizationalUnitName
-- see also teletex-organizational-unit-names

OrganizationalUnitName ::= PrintableString (SIZE
    (1..ub-organizational-unit-name-length))

-- Built-in Domain-defined Attributes

BuiltInDomainDefinedAttributes ::= SEQUENCE SIZE
    (1..ub-domain-defined-attributes) OF
    BuiltInDomainDefinedAttribute

BuiltInDomainDefinedAttribute ::= SEQUENCE {
    type PrintableString (SIZE
        (1..ub-domain-defined-attribute-type-length)),
    value PrintableString (SIZE
        (1..ub-domain-defined-attribute-value-length)) }

-- Extension Attributes

ExtensionAttributes ::= SET SIZE (1..ub-extension-attributes) OF
    ExtensionAttribute

ExtensionAttribute ::= SEQUENCE {
    extension-attribute-type [0] IMPLICIT INTEGER
        (0..ub-extension-attributes),
    extension-attribute-value [1]
        ANY DEFINED BY extension-attribute-type }

-- Extension types and attribute values

common-name INTEGER ::= 1

CommonName ::= PrintableString (SIZE (1..ub-common-name-length))
    
```


teletex-common-name INTEGER ::= 2

TeletexCommonName ::= TeletexString (SIZE (1..ub-common-name-length))

teletex-organization-name INTEGER ::= 3

TeletexOrganizationName ::=

TeletexString (SIZE (1..ub-organization-name-length))

teletex-personal-name INTEGER ::= 4

TeletexPersonalName ::= SET {

surname [0] IMPLICIT TeletexString

(SIZE (1..ub-surname-length)),

given-name [1] IMPLICIT TeletexString

(SIZE (1..ub-given-name-length)) OPTIONAL,

initials [2] IMPLICIT TeletexString

(SIZE (1..ub-initials-length)) OPTIONAL,

generation-qualifier [3] IMPLICIT TeletexString

(SIZE (1..ub-generation-qualifier-length))

OPTIONAL }

teletex-organizational-unit-names INTEGER ::= 5

TeletexOrganizationalUnitNames ::= SEQUENCE SIZE

(1..ub-organizational-units) OF TeletexOrganizationalUnitName

TeletexOrganizationalUnitName ::= TeletexString

(SIZE (1..ub-organizational-unit-name-length))

pds-name INTEGER ::= 7

PDSName ::= PrintableString (SIZE (1..ub-pds-name-length))

physical-delivery-country-name INTEGER ::= 8

PhysicalDeliveryCountryName ::= CHOICE {

x121-dcc-code NumericString (SIZE (ub-country-name-numeric-length)),

iso-3166-alpha2-code PrintableString

(SIZE (ub-country-name-alpha-length)) }

postal-code INTEGER ::= 9

PostalCode ::= CHOICE {

numeric-code NumericString (SIZE (1..ub-postal-code-length)),

printable-code PrintableString (SIZE (1..ub-postal-code-length)) }

physical-delivery-office-name INTEGER ::= 10

PhysicalDeliveryOfficeName ::= PDSPParameter

physical-delivery-office-number INTEGER ::= 11

PhysicalDeliveryOfficeNumber ::= PDSPParameter

extension-OR-address-components INTEGER ::= 12

ExtensionORAddressComponents ::= PDSPParameter

physical-delivery-personal-name INTEGER ::= 13

PhysicalDeliveryPersonalName ::= PDSPParameter

physical-delivery-organization-name INTEGER ::= 14

PhysicalDeliveryOrganizationName ::= PDSPParameter

extension-physical-delivery-address-components INTEGER ::= 15

ExtensionPhysicalDeliveryAddressComponents ::= PDSPParameter

unformatted-postal-address INTEGER ::= 16

UnformattedPostalAddress ::= SET {
 printable-address SEQUENCE SIZE (1..ub-pds-physical-address-lines)
 OF PrintableString (SIZE (1..ub-pds-parameter-length)) OPTIONAL,
 teletex-string TeletexString
 (SIZE (1..ub-unformatted-address-length)) OPTIONAL }

street-address INTEGER ::= 17

StreetAddress ::= PDSPParameter

post-office-box-address INTEGER ::= 18

PostOfficeBoxAddress ::= PDSPParameter

poste-restante-address INTEGER ::= 19

PosteRestanteAddress ::= PDSPParameter

unique-postal-name INTEGER ::= 20

UniquePostalName ::= PDSPParameter

local-postal-attributes INTEGER ::= 21

```
LocalPostalAttributes ::= PDSParameter

PDSParameter ::= SET {
    printable-string PrintableString
        (SIZE(1..ub-pds-parameter-length)) OPTIONAL,
    teletex-string TeletexString
        (SIZE(1..ub-pds-parameter-length)) OPTIONAL }

extended-network-address INTEGER ::= 22

ExtendedNetworkAddress ::= CHOICE {
    e163-4-address SEQUENCE {
        number [0] IMPLICIT NumericString
            (SIZE (1..ub-e163-4-number-length)),
        sub-address [1] IMPLICIT NumericString
            (SIZE (1..ub-e163-4-sub-address-length))
            OPTIONAL },
    psap-address [0] IMPLICIT PresentationAddress }

PresentationAddress ::= SEQUENCE {
    pSelector [0] EXPLICIT OCTET STRING OPTIONAL,
    sSelector [1] EXPLICIT OCTET STRING OPTIONAL,
    tSelector [2] EXPLICIT OCTET STRING OPTIONAL,
    nAddresses [3] EXPLICIT SET SIZE (1..MAX) OF OCTET STRING }

terminal-type INTEGER ::= 23

TerminalType ::= INTEGER {
    telex (3),
    teletex (4),
    g3-facsimile (5),
    g4-facsimile (6),
    ia5-terminal (7),
    videotex (8) } (0..ub-integer-options)

-- Extension Domain-defined Attributes

teletex-domain-defined-attributes INTEGER ::= 6

TeletexDomainDefinedAttributes ::= SEQUENCE SIZE
    (1..ub-domain-defined-attributes) OF TeletexDomainDefinedAttribute

TeletexDomainDefinedAttribute ::= SEQUENCE {
    type TeletexString
        (SIZE (1..ub-domain-defined-attribute-type-length)),
    value TeletexString
        (SIZE (1..ub-domain-defined-attribute-value-length)) }
```

-- specifications of Upper Bounds MUST be regarded as mandatory
-- from Annex B of ITU-T X.411 Reference Definition of MTS Parameter
-- Upper Bounds

-- Upper Bounds

ub-name INTEGER ::= 32768
ub-common-name INTEGER ::= 64
ub-locality-name INTEGER ::= 128
ub-state-name INTEGER ::= 128
ub-organization-name INTEGER ::= 64
ub-organizational-unit-name INTEGER ::= 64
ub-title INTEGER ::= 64
ub-serial-number INTEGER ::= 64
ub-match INTEGER ::= 128
ub-emailaddress-length INTEGER ::= 255
ub-common-name-length INTEGER ::= 64
ub-country-name-alpha-length INTEGER ::= 2
ub-country-name-numeric-length INTEGER ::= 3
ub-domain-defined-attributes INTEGER ::= 4
ub-domain-defined-attribute-type-length INTEGER ::= 8
ub-domain-defined-attribute-value-length INTEGER ::= 128
ub-domain-name-length INTEGER ::= 16
ub-extension-attributes INTEGER ::= 256
ub-el63-4-number-length INTEGER ::= 15
ub-el63-4-sub-address-length INTEGER ::= 40
ub-generation-qualifier-length INTEGER ::= 3
ub-given-name-length INTEGER ::= 16
ub-initials-length INTEGER ::= 5
ub-integer-options INTEGER ::= 256
ub-numeric-user-id-length INTEGER ::= 32
ub-organization-name-length INTEGER ::= 64
ub-organizational-unit-name-length INTEGER ::= 32
ub-organizational-units INTEGER ::= 4
ub-pds-name-length INTEGER ::= 16
ub-pds-parameter-length INTEGER ::= 30
ub-pds-physical-address-lines INTEGER ::= 6
ub-postal-code-length INTEGER ::= 16
ub-pseudonym INTEGER ::= 128
ub-surname-length INTEGER ::= 40
ub-terminal-id-length INTEGER ::= 24
ub-unformatted-address-length INTEGER ::= 180
ub-x121-address-length INTEGER ::= 16

-- Note - upper bounds on string types, such as TeletexString, are
-- measured in characters. Excepting PrintableString or IA5String, a
-- significantly greater number of octets will be required to hold
-- such a value. As a minimum, 16 octets, or twice the specified
-- upper bound, whichever is the larger, should be allowed for

-- TeletexString. For UTF8String or UniversalString at least four
-- times the upper bound should be allowed.

END

A.2. Implicitly Tagged Module, 1988 Syntax

PKIX1Implicit88 { iso(1) identified-organization(3) dod(6) internet(1)
security(5) mechanisms(5) pkix(7) id-mod(0) id-pkix1-implicit(19) }

DEFINITIONS IMPLICIT TAGS ::=

BEGIN

-- EXPORTS ALL --

IMPORTS

id-pe, id-kp, id-qt-unotice, id-qt-cps,
-- delete following line if "new" types are supported --
BMPString, UTF8String, -- end "new" types --
ORAddress, Name, RelativeDistinguishedName,
CertificateSerialNumber, Attribute, DirectoryString
FROM PKIX1Explicit88 { iso(1) identified-organization(3)
dod(6) internet(1) security(5) mechanisms(5) pkix(7)
id-mod(0) id-pkix1-explicit(18) };

-- ISO arc for standard certificate and CRL extensions

id-ce OBJECT IDENTIFIER ::= {joint-iso-ccitt(2) ds(5) 29}

-- authority key identifier OID and syntax

id-ce-authorityKeyIdentifier OBJECT IDENTIFIER ::= { id-ce 35 }

AuthorityKeyIdentifier ::= SEQUENCE {
keyIdentifier [0] KeyIdentifier OPTIONAL,
authorityCertIssuer [1] GeneralNames OPTIONAL,
authorityCertSerialNumber [2] CertificateSerialNumber OPTIONAL }
-- authorityCertIssuer and authorityCertSerialNumber MUST both
-- be present or both be absent

KeyIdentifier ::= OCTET STRING

```
-- subject key identifier OID and syntax

id-ce-subjectKeyIdentifier OBJECT IDENTIFIER ::= { id-ce 14 }

SubjectKeyIdentifier ::= KeyIdentifier

-- key usage extension OID and syntax

id-ce-keyUsage OBJECT IDENTIFIER ::= { id-ce 15 }

KeyUsage ::= BIT STRING {
    digitalSignature          (0),
    nonRepudiation            (1),  -- recent editions of X.509 have
                                   -- renamed this bit to contentCommitment
    keyEncipherment           (2),
    dataEncipherment          (3),
    keyAgreement              (4),
    keyCertSign               (5),
    cRLSign                   (6),
    encipherOnly              (7),
    decipherOnly              (8) }

-- private key usage period extension OID and syntax

id-ce-privateKeyUsagePeriod OBJECT IDENTIFIER ::= { id-ce 16 }

PrivateKeyUsagePeriod ::= SEQUENCE {
    notBefore      [0]    GeneralizedTime OPTIONAL,
    notAfter       [1]    GeneralizedTime OPTIONAL }
    -- either notBefore or notAfter MUST be present

-- certificate policies extension OID and syntax

id-ce-certificatePolicies OBJECT IDENTIFIER ::= { id-ce 32 }

anyPolicy OBJECT IDENTIFIER ::= { id-ce-certificatePolicies 0 }

CertificatePolicies ::= SEQUENCE SIZE (1..MAX) OF PolicyInformation

PolicyInformation ::= SEQUENCE {
    policyIdentifier    CertPolicyId,
    policyQualifiers    SEQUENCE SIZE (1..MAX) OF
        PolicyQualifierInfo OPTIONAL }

CertPolicyId ::= OBJECT IDENTIFIER
```

```

PolicyQualifierInfo ::= SEQUENCE {
    policyQualifierId  PolicyQualifierId,
    qualifier          ANY DEFINED BY policyQualifierId }

-- Implementations that recognize additional policy qualifiers MUST
-- augment the following definition for PolicyQualifierId

PolicyQualifierId ::= OBJECT IDENTIFIER ( id-qt-cps | id-qt-unotice )

-- CPS pointer qualifier

CPSuri ::= IA5String

-- user notice qualifier

UserNotice ::= SEQUENCE {
    noticeRef          NoticeReference OPTIONAL,
    explicitText       DisplayText OPTIONAL }

NoticeReference ::= SEQUENCE {
    organization       DisplayText,
    noticeNumbers      SEQUENCE OF INTEGER }

DisplayText ::= CHOICE {
    ia5String          IA5String          (SIZE (1..200)),
    visibleString      VisibleString      (SIZE (1..200)),
    bmpString          BMPString          (SIZE (1..200)),
    utf8String         UTF8String         (SIZE (1..200)) }

-- policy mapping extension OID and syntax

id-ce-policyMappings OBJECT IDENTIFIER ::= { id-ce 33 }

PolicyMappings ::= SEQUENCE SIZE (1..MAX) OF SEQUENCE {
    issuerDomainPolicy  CertPolicyId,
    subjectDomainPolicy CertPolicyId }

-- subject alternative name extension OID and syntax

id-ce-subjectAltName OBJECT IDENTIFIER ::= { id-ce 17 }

SubjectAltName ::= GeneralNames

GeneralNames ::= SEQUENCE SIZE (1..MAX) OF GeneralName
    
```

```
GeneralName ::= CHOICE {
    otherName          [0]  AnotherName,
    rfc822Name         [1]  IA5String,
    dNSName            [2]  IA5String,
    x400Address        [3]  ORAddress,
    directoryName      [4]  Name,
    ediPartyName       [5]  EDIPartyName,
    uniformResourceIdentifier [6] IA5String,
    iPAddress          [7]  OCTET STRING,
    registeredID       [8]  OBJECT IDENTIFIER }

-- AnotherName replaces OTHER-NAME ::= TYPE-IDENTIFIER, as
-- TYPE-IDENTIFIER is not supported in the '88 ASN.1 syntax

AnotherName ::= SEQUENCE {
    type-id    OBJECT IDENTIFIER,
    value      [0] EXPLICIT ANY DEFINED BY type-id }

EDIPartyName ::= SEQUENCE {
    nameAssigner      [0]  DirectoryString OPTIONAL,
    partyName         [1]  DirectoryString }

-- issuer alternative name extension OID and syntax

id-ce-issuerAltName OBJECT IDENTIFIER ::=  { id-ce 18 }

IssuerAltName ::= GeneralNames

id-ce-subjectDirectoryAttributes OBJECT IDENTIFIER ::=  { id-ce 9 }

SubjectDirectoryAttributes ::= SEQUENCE SIZE (1..MAX) OF Attribute

-- basic constraints extension OID and syntax

id-ce-basicConstraints OBJECT IDENTIFIER ::=  { id-ce 19 }

BasicConstraints ::= SEQUENCE {
    cA          BOOLEAN DEFAULT FALSE,
    pathLenConstraint INTEGER (0..MAX) OPTIONAL }
```



```
-- name constraints extension OID and syntax

id-ce-nameConstraints OBJECT IDENTIFIER ::=  { id-ce 30 }

NameConstraints ::= SEQUENCE {
    permittedSubtrees      [0]      GeneralSubtrees OPTIONAL,
    excludedSubtrees       [1]      GeneralSubtrees OPTIONAL }

GeneralSubtrees ::= SEQUENCE SIZE (1..MAX) OF GeneralSubtree

GeneralSubtree ::= SEQUENCE {
    base                    GeneralName,
    minimum                 [0]      BaseDistance DEFAULT 0,
    maximum                 [1]      BaseDistance OPTIONAL }

BaseDistance ::= INTEGER (0..MAX)

-- policy constraints extension OID and syntax

id-ce-policyConstraints OBJECT IDENTIFIER ::=  { id-ce 36 }

PolicyConstraints ::= SEQUENCE {
    requireExplicitPolicy  [0]      SkipCerts OPTIONAL,
    inhibitPolicyMapping   [1]      SkipCerts OPTIONAL }

SkipCerts ::= INTEGER (0..MAX)

-- CRL distribution points extension OID and syntax

id-ce-cRLDistributionPoints OBJECT IDENTIFIER ::=  {id-ce 31}

CRLDistributionPoints ::= SEQUENCE SIZE (1..MAX) OF DistributionPoint

DistributionPoint ::= SEQUENCE {
    distributionPoint      [0]      DistributionPointName OPTIONAL,
    reasons                [1]      ReasonFlags OPTIONAL,
    cRLIssuer              [2]      GeneralNames OPTIONAL }

DistributionPointName ::= CHOICE {
    fullName               [0]      GeneralNames,
    nameRelativeToCRLIssuer [1]      RelativeDistinguishedName }
```

```
ReasonFlags ::= BIT STRING {
    unused                (0),
    keyCompromise         (1),
    cACompromise          (2),
    affiliationChanged     (3),
    superseded            (4),
    cessationOfOperation  (5),
    certificateHold        (6),
    privilegeWithdrawn     (7),
    aACompromise          (8) }

-- extended key usage extension OID and syntax

id-ce-extKeyUsage OBJECT IDENTIFIER ::= {id-ce 37}

ExtKeyUsageSyntax ::= SEQUENCE SIZE (1..MAX) OF KeyPurposeId

KeyPurposeId ::= OBJECT IDENTIFIER

-- permit unspecified key uses

anyExtendedKeyUsage OBJECT IDENTIFIER ::= { id-ce-extKeyUsage 0 }

-- extended key purpose OIDs

id-kp-serverAuth          OBJECT IDENTIFIER ::= { id-kp 1 }
id-kp-clientAuth          OBJECT IDENTIFIER ::= { id-kp 2 }
id-kp-codeSigning         OBJECT IDENTIFIER ::= { id-kp 3 }
id-kp-emailProtection     OBJECT IDENTIFIER ::= { id-kp 4 }
id-kp-timeStamping        OBJECT IDENTIFIER ::= { id-kp 8 }
id-kp-OCSPSigning         OBJECT IDENTIFIER ::= { id-kp 9 }

-- inhibit any policy OID and syntax

id-ce-inhibitAnyPolicy OBJECT IDENTIFIER ::= { id-ce 54 }

InhibitAnyPolicy ::= SkipCerts

-- freshest (delta)CRL extension OID and syntax

id-ce-freshestCRL OBJECT IDENTIFIER ::= { id-ce 46 }

FreshestCRL ::= CRLDistributionPoints
```

```
-- authority info access

id-pe-authorityInfoAccess OBJECT IDENTIFIER ::= { id-pe 1 }

AuthorityInfoAccessSyntax ::=
    SEQUENCE SIZE (1..MAX) OF AccessDescription

AccessDescription ::= SEQUENCE {
    accessMethod      OBJECT IDENTIFIER,
    accessLocation    GeneralName  }

-- subject info access

id-pe-subjectInfoAccess OBJECT IDENTIFIER ::= { id-pe 11 }

SubjectInfoAccessSyntax ::=
    SEQUENCE SIZE (1..MAX) OF AccessDescription

-- CRL number extension OID and syntax

id-ce-cRLNumber OBJECT IDENTIFIER ::= { id-ce 20 }

CRLNumber ::= INTEGER (0..MAX)

-- issuing distribution point extension OID and syntax

id-ce-issuingDistributionPoint OBJECT IDENTIFIER ::= { id-ce 28 }

IssuingDistributionPoint ::= SEQUENCE {
    distributionPoint      [0] DistributionPointName OPTIONAL,
    onlyContainsUserCerts  [1] BOOLEAN DEFAULT FALSE,
    onlyContainsCACerts    [2] BOOLEAN DEFAULT FALSE,
    onlySomeReasons        [3] ReasonFlags OPTIONAL,
    indirectCRL            [4] BOOLEAN DEFAULT FALSE,
    onlyContainsAttributeCerts [5] BOOLEAN DEFAULT FALSE }
-- at most one of onlyContainsUserCerts, onlyContainsCACerts,
-- and onlyContainsAttributeCerts may be set to TRUE.

id-ce-deltaCRLIndicator OBJECT IDENTIFIER ::= { id-ce 27 }

BaseCRLNumber ::= CRLNumber
```

```

-- reason code extension OID and syntax

id-ce-cRLReasons OBJECT IDENTIFIER ::= { id-ce 21 }

CRLReason ::= ENUMERATED {
    unspecified          (0),
    keyCompromise        (1),
    cACompromise         (2),
    affiliationChanged    (3),
    superseded           (4),
    cessationOfOperation (5),
    certificateHold       (6),
    removeFromCRL        (8),
    privilegeWithdrawn    (9),
    aACompromise         (10) }

-- certificate issuer CRL entry extension OID and syntax

id-ce-certificateIssuer OBJECT IDENTIFIER ::= { id-ce 29 }

CertificateIssuer ::= GeneralNames

-- hold instruction extension OID and syntax

id-ce-holdInstructionCode OBJECT IDENTIFIER ::= { id-ce 23 }

HoldInstructionCode ::= OBJECT IDENTIFIER

-- ANSI x9 arc holdinstruction arc

holdInstruction OBJECT IDENTIFIER ::=
    {joint-iso-itu-t(2) member-body(2) us(840) x9cm(10040) 2}

-- ANSI X9 holdinstructions

id-holdinstruction-none OBJECT IDENTIFIER ::=
    {holdInstruction 1} -- deprecated

id-holdinstruction-callissuer OBJECT IDENTIFIER ::= {holdInstruction 2}

id-holdinstruction-reject OBJECT IDENTIFIER ::= {holdInstruction 3}
    
```

```
-- invalidity date CRL entry extension OID and syntax

id-ce-invalidityDate OBJECT IDENTIFIER ::= { id-ce 24 }

InvalidityDate ::= GeneralizedTime

END
```

Appendix B. ASN.1 Notes

CAs MUST force the serialNumber to be a non-negative integer, that is, the sign bit in the DER encoding of the INTEGER value MUST be zero. This can be done by adding a leading (leftmost) `00'H octet if necessary. This removes a potential ambiguity in mapping between a string of octets and an integer value.

As noted in Section 4.1.2.2, serial numbers can be expected to contain long integers. Certificate users MUST be able to handle serialNumber values up to 20 octets in length. Conforming CAs MUST NOT use serialNumber values longer than 20 octets.

As noted in Section 5.2.3, CRL numbers can be expected to contain long integers. CRL validators MUST be able to handle cRLNumber values up to 20 octets in length. Conforming CRL issuers MUST NOT use cRLNumber values longer than 20 octets.

The construct "SEQUENCE SIZE (1..MAX) OF" appears in several ASN.1 constructs. A valid ASN.1 sequence will have zero or more entries. The SIZE (1..MAX) construct constrains the sequence to have at least one entry. MAX indicates that the upper bound is unspecified. Implementations are free to choose an upper bound that suits their environment.

The character string type PrintableString supports a very basic Latin character set: the lowercase letters 'a' through 'z', uppercase letters 'A' through 'Z', the digits '0' through '9', eleven special characters ' = () + , - . / : ? and space.

Implementers should note that the at sign ('@') and underscore ('_') characters are not supported by the ASN.1 type PrintableString. These characters often appear in Internet addresses. Such addresses MUST be encoded using an ASN.1 type that supports them. They are usually encoded as IA5String in either the emailAddress attribute within a distinguished name or the rfc822Name field of GeneralName. Conforming implementations MUST NOT encode strings that include either the at sign or underscore character as PrintableString.

The character string type TeletexString is a superset of PrintableString. TeletexString supports a fairly standard (ASCII-like) Latin character set: Latin characters with non-spacing accents and Japanese characters.

Named bit lists are BIT STRINGs where the values have been assigned names. This specification makes use of named bit lists in the definitions for the key usage, CRL distribution points, and freshest CRL certificate extensions, as well as the freshest CRL and issuing distribution point CRL extensions. When DER encoding a named bit list, trailing zeros MUST be omitted. That is, the encoded value ends with the last named bit that is set to one.

The character string type UniversalString supports any of the characters allowed by [ISO10646]. ISO 10646 is the Universal multiple-octet coded Character Set (UCS).

The character string type UTF8String was introduced in the 1997 version of ASN.1, and UTF8String was added to the list of choices for DirectoryString in the 2001 version of [X.520]. UTF8String is a universal type and has been assigned tag number 12. The content of UTF8String was defined by RFC 2044 and updated in RFC 2279, which was updated in [RFC3629].

In anticipation of these changes, and in conformance with IETF Best Practices codified in [RFC2277], IETF Policy on Character Sets and Languages, this document includes UTF8String as a choice in DirectoryString and in the userNotice certificate policy qualifier.

For many of the attribute types defined in [X.520], the AttributeValue uses the DirectoryString type. Of the attributes specified in Appendix A, the name, surname, givenName, initials, generationQualifier, commonName, localityName, stateOrProvinceName, organizationName, organizationalUnitName, title, and pseudonym attributes all use the DirectoryString type. X.520 uses a parameterized type definition [X.683] of DirectoryString to specify the syntax for each of these attributes. The parameter is used to indicate the maximum string length allowed for the attribute. In Appendix A, in order to avoid the use of parameterized type definitions, the DirectoryString type is written in its expanded form for the definition of each of these attribute types. So, the ASN.1 in Appendix A describes the syntax for each of these attributes as being a CHOICE of TeletexString, PrintableString, UniversalString, UTF8String, and BMPString, with the appropriate constraints on the string length applied to each of the types in the CHOICE, rather than using the ASN.1 type DirectoryString to describe the syntax.

Implementers should note that the DER encoding of the SET OF values requires ordering of the encodings of the values. In particular, this issue arises with respect to distinguished names.

Implementers should note that the DER encoding of SET or SEQUENCE components whose value is the DEFAULT omit the component from the encoded certificate or CRL. For example, a BasicConstraints extension whose cA value is FALSE would omit the cA boolean from the encoded certificate.

Object Identifiers (OIDs) are used throughout this specification to identify certificate policies, public key and signature algorithms, certificate extensions, etc. There is no maximum size for OIDs. This specification mandates support for OIDs that have arc elements with values that are less than 2^{28} , that is, they MUST be between 0 and 268,435,455, inclusive. This allows each arc element to be represented within a single 32-bit word. Implementations MUST also support OIDs where the length of the dotted decimal (see Section 1.4 of [RFC4512]) string representation can be up to 100 bytes (inclusive). Implementations MUST be able to handle OIDs with up to 20 elements (inclusive). CAs SHOULD NOT issue certificates that contain OIDs that exceed these requirements. Likewise, CRL issuers SHOULD NOT issue CRLs that contain OIDs that exceed these requirements.

The content-specific rules for encoding GeneralName field values in the nameConstraints extension differ from rules that apply in other extensions. In all other certificate, CRL, and CRL entry extensions specified in this document the encoding rules conform to the rules for the underlying type. For example, values in the uniformResourceIdentifier field must contain a valid URI as specified in [RFC3986]. The content-specific rules for encoding values in the nameConstraints extension are specified in Section 4.2.1.10, and these rules may not conform to the rules for the underlying type. For example, when the uniformResourceIdentifier field appears in a nameConstraints extension, it must hold a DNS name (e.g., "host.example.com" or ".example.com") rather than a URI.

Implementors are warned that the X.500 standards community has developed a series of extensibility rules. These rules determine when an ASN.1 definition can be changed without assigning a new Object Identifier (OID). For example, at least two extension definitions included in [RFC2459], the predecessor to this profile document, have different ASN.1 definitions in this specification, but the same OID is used. If unknown elements appear within an extension, and the extension is not marked as critical, those unknown elements ought to be ignored, as follows:

- (a) ignore all unknown bit name assignments within a bit string;
- (b) ignore all unknown named numbers in an ENUMERATED type or INTEGER type that is being used in the enumerated style, provided the number occurs as an optional element of a SET or SEQUENCE; and
- (c) ignore all unknown elements in SETs, at the end of SEQUENCES, or in CHOICES where the CHOICE is itself an optional element of a SET or SEQUENCE.

If an extension containing unexpected values is marked as critical, the implementation MUST reject the certificate or CRL containing the unrecognized extension.

Appendix C. Examples

This appendix contains four examples: three certificates and a CRL. The first two certificates and the CRL comprise a minimal certification path.

Appendix C.1 contains an annotated hex dump of a "self-signed" certificate issued by a CA whose distinguished name is `cn=Example CA,dc=example,dc=com`. The certificate contains an RSA public key, and is signed by the corresponding RSA private key.

Appendix C.2 contains an annotated hex dump of an end entity certificate. The end entity certificate contains an RSA public key, and is signed by the private key corresponding to the "self-signed" certificate in Appendix C.1.

Appendix C.3 contains an annotated hex dump of an end entity certificate that contains a DSA public key with parameters, and is signed with DSA and SHA-1. This certificate is not part of the minimal certification path.

Appendix C.4 contains an annotated hex dump of a CRL. The CRL is issued by the CA whose distinguished name is `cn=Example CA,dc=example,dc=com` and the list of revoked certificates includes the end entity certificate presented in Appendix C.2.

The certificates were processed using Peter Gutmann's `dumpasn1` utility to generate the output. The source for the `dumpasn1` utility is available at <http://www.cs.auckland.ac.nz/~pgut001/dumpasn1.c>. The binaries for the certificates and CRLs are available at http://csrc.nist.gov/groups/ST/crypto_apps_infra/documents/pkixtools.

In places in this appendix where a distinguished name is specified using a string representation, the strings are formatted using the rules specified in [RFC4514].

C.1. RSA Self-Signed Certificate

This appendix contains an annotated hex dump of a 578 byte version 3 certificate. The certificate contains the following information:

- (a) the serial number is 17;
- (b) the certificate is signed with RSA and the SHA-1 hash algorithm;
- (c) the issuer's distinguished name is
cn=Example CA,dc=example,dc=com;
- (d) the subject's distinguished name is
cn=Example CA,dc=example,dc=com;
- (e) the certificate was issued on April 30, 2004 and expired on April 30, 2005;
- (f) the certificate contains a 1024-bit RSA public key;
- (g) the certificate contains a subject key identifier extension generated using method (1) of Section 4.2.1.2; and
- (h) the certificate is a CA certificate (as indicated through the basic constraints extension).

```
0 574: SEQUENCE {
4 423:   SEQUENCE {
8   3:     [0] {
10  1:       INTEGER 2
      :     }
13  1:       INTEGER 17
16 13:     SEQUENCE {
18  9:       OBJECT IDENTIFIER
      :       sha1withRSAEncryption (1 2 840 113549 1 1 5)
29  0:       NULL
      :     }
31 67:   SEQUENCE {
33 19:     SET {
35 17:       SEQUENCE {
37 10:         OBJECT IDENTIFIER
      :         domainComponent (0 9 2342 19200300 100 1 25)
49  3:         IA5String 'com'
      :       }
      :     }
54 23:   SET {
56 21:     SEQUENCE {
58 10:       OBJECT IDENTIFIER
      :       domainComponent (0 9 2342 19200300 100 1 25)
70  7:       IA5String 'example'
      :     }
      :   }
```

```

      :      }
79   19:      SET {
81   17:          SEQUENCE {
83    3:              OBJECT IDENTIFIER commonName (2 5 4 3)
88   10:              PrintableString 'Example CA'
      :          }
      :      }
      :  }
100  30: SEQUENCE {
102  13:     UTCTime 30/04/2004 14:25:34 GMT
117  13:     UTCTime 30/04/2005 14:25:34 GMT
      :  }
132  67: SEQUENCE {
134  19:     SET {
136  17:         SEQUENCE {
138  10:             OBJECT IDENTIFIER
      :             domainComponent (0 9 2342 19200300 100 1 25)
150   3:             IA5String 'com'
      :         }
      :     }
155  23: SET {
157  21:     SEQUENCE {
159  10:         OBJECT IDENTIFIER
      :         domainComponent (0 9 2342 19200300 100 1 25)
171   7:         IA5String 'example'
      :     }
      : }
180  19: SET {
182  17:     SEQUENCE {
184   3:         OBJECT IDENTIFIER commonName (2 5 4 3)
189  10:         PrintableString 'Example CA'
      :     }
      : }
      : }
201 159: SEQUENCE {
204  13:     SEQUENCE {
206   9:         OBJECT IDENTIFIER
      :         rsaEncryption (1 2 840 113549 1 1 1)
217   0:         NULL
      :     }
219 141: BIT STRING, encapsulates {
223 137:     SEQUENCE {
226 129:         INTEGER
      :         00 C2 D7 97 6D 28 70 AA 5B CF 23 2E 80 70 39 EE
      :         DB 6F D5 2D D5 6A 4F 7A 34 2D F9 22 72 47 70 1D
      :         EF 80 E9 CA 30 8C 00 C4 9A 6E 5B 45 B4 6E A5 E6
      :         6C 94 0D FA 91 E9 40 FC 25 9D C7 B7 68 19 56 8F
      :         11 70 6A D7 F1 C9 11 4F 3A 7E 3F 99 8D 6E 76 A5

```

```

      :      74 5F 5E A4 55 53 E5 C7 68 36 53 C7 1D 3B 12 A6
      :      85 FE BD 6E A1 CA DF 35 50 AC 08 D7 B9 B4 7E 5C
      :      FE E2 A3 2C D1 23 84 AA 98 C0 9B 66 18 9A 68 47
      :      E9
358  3:      INTEGER 65537
      :      }
      :    }
      :  }
363  66:  [3] {
365  64:    SEQUENCE {
367  29:    SEQUENCE {
369   3:      OBJECT IDENTIFIER subjectKeyIdentifier (2 5 29 14)
374  22:      OCTET STRING, encapsulates {
376  20:        OCTET STRING
      :          08 68 AF 85 33 C8 39 4A 7A F8 82 93 8E 70 6A 4A
      :          20 84 2C 32
      :        }
      :      }
398  14:    SEQUENCE {
400   3:      OBJECT IDENTIFIER keyUsage (2 5 29 15)
405   1:      BOOLEAN TRUE
408   4:      OCTET STRING, encapsulates {
410   2:        BIT STRING 1 unused bits
      :          '0000011'B
      :        }
      :      }
414  15:    SEQUENCE {
416   3:      OBJECT IDENTIFIER basicConstraints (2 5 29 19)
421   1:      BOOLEAN TRUE
424   5:      OCTET STRING, encapsulates {
426   3:        SEQUENCE {
428   1:          BOOLEAN TRUE
      :        }
      :      }
      :    }
      :  }
      : }
      : }
431  13: SEQUENCE {
433   9:   OBJECT IDENTIFIER
      :     sha1withRSAEncryption (1 2 840 113549 1 1 5)
444   0:   NULL
      :   }
446 129: BIT STRING
      :   6C F8 02 74 A6 61 E2 64 04 A6 54 0C 6C 72 13 AD
      :   3C 47 FB F6 65 13 A9 85 90 33 EA 76 A3 26 D9 FC
      :   D1 0E 15 5F 28 B7 EF 93 BF 3C F3 E2 3E 7C B9 52
      :   FC 16 6E 29 AA E1 F4 7A 6F D5 7F EF B3 95 CA F3
```

```
:      66 88 83 4E A1 35 45 84 CB BC 9B B8 C8 AD C5 5E
:      46 D9 0B 0E 8D 80 E1 33 2B DC BE 2B 92 7E 4A 43
:      A9 6A EF 8A 63 61 B3 6E 47 38 BE E8 0D A3 67 5D
:      F3 FA 91 81 3C 92 BB C5 5F 25 25 EB 7C E7 D8 A1
:      }
```

C.2. End Entity Certificate Using RSA

This appendix contains an annotated hex dump of a 629-byte version 3 certificate. The certificate contains the following information:

- (a) the serial number is 18;
- (b) the certificate is signed with RSA and the SHA-1 hash algorithm;
- (c) the issuer's distinguished name is
cn=Example CA,dc=example,dc=com;
- (d) the subject's distinguished name is
cn=End Entity,dc=example,dc=com;
- (e) the certificate was valid from September 15, 2004 through March 15, 2005;
- (f) the certificate contains a 1024-bit RSA public key;
- (g) the certificate is an end entity certificate, as the basic constraints extension is not present;
- (h) the certificate contains an authority key identifier extension matching the subject key identifier of the certificate in appendix C.1; and
- (i) the certificate includes one alternative name -- an electronic mail address (rfc822Name) of "end.entity@example.com".

```
0 625: SEQUENCE {
4 474:   SEQUENCE {
8   3:     [0] {
10  1:       INTEGER 2
:         }
13  1:       INTEGER 18
16 13:     SEQUENCE {
18  9:       OBJECT IDENTIFIER
:           shalwithRSAEncryption (1 2 840 113549 1 1 5)
29  0:       NULL
:         }
31 67:     SEQUENCE {
33 19:       SET {
35 17:         SEQUENCE {
37 10:         OBJECT IDENTIFIER
:           domainComponent (0 9 2342 19200300 100 1 25)
49  3:         IA5String 'com'
:       }
:     }
54 23:     SET {
```

```
56 21: SEQUENCE {
58 10:   OBJECT IDENTIFIER
      :   domainComponent (0 9 2342 19200300 100 1 25)
70 7:   IA5String 'example'
      :   }
      :   }
79 19: SET {
81 17:   SEQUENCE {
83 3:   OBJECT IDENTIFIER commonName (2 5 4 3)
88 10:   PrintableString 'Example CA'
      :   }
      :   }
      :   }
100 30: SEQUENCE {
102 13:   UTCTime 15/09/2004 11:48:21 GMT
117 13:   UTCTime 15/03/2005 11:48:21 GMT
      :   }
132 67: SEQUENCE {
134 19:   SET {
136 17:     SEQUENCE {
138 10:       OBJECT IDENTIFIER
          :       domainComponent (0 9 2342 19200300 100 1 25)
150 3:       IA5String 'com'
          :       }
          :       }
155 23:   SET {
157 21:     SEQUENCE {
159 10:       OBJECT IDENTIFIER
          :       domainComponent (0 9 2342 19200300 100 1 25)
171 7:       IA5String 'example'
          :       }
          :       }
180 19:   SET {
182 17:     SEQUENCE {
184 3:       OBJECT IDENTIFIER commonName (2 5 4 3)
189 10:       PrintableString 'End Entity'
          :       }
          :       }
          :       }
201 159: SEQUENCE {
204 13:   SEQUENCE {
206 9:   OBJECT IDENTIFIER
      :   rsaEncryption (1 2 840 113549 1 1 1)
217 0:   NULL
      :   }
219 141: BIT STRING, encapsulates {
223 137:   SEQUENCE {
226 129:   INTEGER
```

```

:      00 E1 6A E4 03 30 97 02 3C F4 10 F3 B5 1E 4D 7F
:      14 7B F6 F5 D0 78 E9 A4 8A F0 A3 75 EC ED B6 56
:      96 7F 88 99 85 9A F2 3E 68 77 87 EB 9E D1 9F C0
:      B4 17 DC AB 89 23 A4 1D 7E 16 23 4C 4F A8 4D F5
:      31 B8 7C AA E3 1A 49 09 F4 4B 26 DB 27 67 30 82
:      12 01 4A E9 1A B6 C1 0C 53 8B 6C FC 2F 7A 43 EC
:      33 36 7E 32 B2 7B D5 AA CF 01 14 C6 12 EC 13 F2
:      2D 14 7A 8B 21 58 14 13 4C 46 A3 9A F2 16 95 FF
:      23
358 3:      INTEGER 65537
:      }
:    }
:  }
363 117: [3] {
365 115:   SEQUENCE {
367 33:     SEQUENCE {
369 3:       OBJECT IDENTIFIER subjectAltName (2 5 29 17)
374 26:       OCTET STRING, encapsulates {
376 24:         SEQUENCE {
378 22:           [1] 'end.entity@example.com'
:         }
:       }
:     }
402 29:   SEQUENCE {
404 3:     OBJECT IDENTIFIER subjectKeyIdentifier (2 5 29 14)
409 22:     OCTET STRING, encapsulates {
411 20:       OCTET STRING
:         17 7B 92 30 FF 44 D6 66 E1 90 10 22 6C 16 4F C0
:         8E 41 DD 6D
:       }
:     }
433 31:   SEQUENCE {
435 3:     OBJECT IDENTIFIER
:       authorityKeyIdentifier (2 5 29 35)
440 24:     OCTET STRING, encapsulates {
442 22:       SEQUENCE {
444 20:         [0]
:           08 68 AF 85 33 C8 39 4A 7A F8 82 93 8E 70 6A
:           4A 20 84 2C 32
:         }
:       }
:     }
466 14:   SEQUENCE {
468 3:     OBJECT IDENTIFIER keyUsage (2 5 29 15)
473 1:     BOOLEAN TRUE
476 4:     OCTET STRING, encapsulates {
478 2:       BIT STRING 6 unused bits
:       '11'B

```

```

:      }
:    }
:  }
: }
482 13: SEQUENCE {
484 9:   OBJECT IDENTIFIER
:     sha1withRSAEncryption (1 2 840 113549 1 1 5)
495 0:   NULL
: }
497 129: BIT STRING
:      00 20 28 34 5B 68 32 01 BB 0A 36 0E AD 71 C5 95
:      1A E1 04 CF AE AD C7 62 14 A4 1B 36 31 C0 E2 0C
:      3D D9 1E C0 00 DC 10 A0 BA 85 6F 41 CB 62 7A B7
:      4C 63 81 26 5E D2 80 45 5E 33 E7 70 45 3B 39 3B
:      26 4A 9C 3B F2 26 36 69 08 79 BB FB 96 43 77 4B
:      61 8B A1 AB 91 64 E0 F3 37 61 3C 1A A3 A4 C9 8A
:      B2 BF 73 D4 4D E4 58 E4 62 EA BC 20 74 92 86 0E
:      CE 84 60 76 E9 73 BB C7 85 D3 91 45 EA 62 5D CD
:    }
```

C.3. End Entity Certificate Using DSA

This appendix contains an annotated hex dump of a 914-byte version 3 certificate. The certificate contains the following information:

- (a) the serial number is 256;
- (b) the certificate is signed with DSA and the SHA-1 hash algorithm;
- (c) the issuer's distinguished name is cn=Example DSA CA,dc=example,dc=com;
- (d) the subject's distinguished name is cn=DSA End Entity,dc=example,dc=com;
- (e) the certificate was issued on May 2, 2004 and expired on May 2, 2005;
- (f) the certificate contains a 1024-bit DSA public key with parameters;
- (g) the certificate is an end entity certificate (not a CA certificate);
- (h) the certificate includes a subject alternative name of "<http://www.example.com/users/DSAendentity.html>" and an issuer alternative name of "<http://www.example.com>" -- both are URLs;

- (i) the certificate includes an authority key identifier extension and a certificate policies extension specifying the policy OID 2.16.840.1.101.3.2.1.48.9; and
- (j) the certificate includes a critical key usage extension specifying that the public key is intended for verification of digital signatures.

```

0  910: SEQUENCE {
4  846:   SEQUENCE {
8    3:    [0] {
10   1:    INTEGER 2
      :    }
13   2:    INTEGER 256
17   9:    SEQUENCE {
19   7:    OBJECT IDENTIFIER dsaWithShal (1 2 840 10040 4 3)
      :    }
28  71:    SEQUENCE {
30  19:    SET {
32  17:    SEQUENCE {
34  10:    OBJECT IDENTIFIER
      :    domainComponent (0 9 2342 19200300 100 1 25)
46   3:    IA5String 'com'
      :    }
      :    }
51  23:    SET {
53  21:    SEQUENCE {
55  10:    OBJECT IDENTIFIER
      :    domainComponent (0 9 2342 19200300 100 1 25)
67   7:    IA5String 'example'
      :    }
      :    }
76  23:    SET {
78  21:    SEQUENCE {
80   3:    OBJECT IDENTIFIER commonName (2 5 4 3)
85  14:    PrintableString 'Example DSA CA'
      :    }
      :    }
      :    }
101 30:    SEQUENCE {
103 13:    UTCTime 02/05/2004 16:47:38 GMT
118 13:    UTCTime 02/05/2005 16:47:38 GMT
      :    }
133 71:    SEQUENCE {
135 19:    SET {
137 17:    SEQUENCE {
139 10:    OBJECT IDENTIFIER
      :    domainComponent (0 9 2342 19200300 100 1 25)

```



```
151      3:      IA5String 'com'
      :      }
      :      }
156     23:     SET {
158     21:     SEQUENCE {
160     10:     OBJECT IDENTIFIER
      :         domainComponent (0 9 2342 19200300 100 1 25)
172      7:      IA5String 'example'
      :      }
      :      }
181     23:     SET {
183     21:     SEQUENCE {
185      3:      OBJECT IDENTIFIER commonName (2 5 4 3)
190     14:      PrintableString 'DSA End Entity'
      :      }
      :      }
      :      }
206    439:    SEQUENCE {
210    300:    SEQUENCE {
214      7:      OBJECT IDENTIFIER dsa (1 2 840 10040 4 1)
223     287:    SEQUENCE {
227     129:    INTEGER
      :         00 B6 8B 0F 94 2B 9A CE A5 25 C6 F2 ED FC FB 95
      :         32 AC 01 12 33 B9 E0 1C AD 90 9B BC 48 54 9E F3
      :         94 77 3C 2C 71 35 55 E6 FE 4F 22 CB D5 D8 3E 89
      :         93 33 4D FC BD 4F 41 64 3E A2 98 70 EC 31 B4 50
      :         DE EB F1 98 28 0A C9 3E 44 B3 FD 22 97 96 83 D0
      :         18 A3 E3 BD 35 5B FF EE A3 21 72 6A 7B 96 DA B9
      :         3F 1E 5A 90 AF 24 D6 20 F0 0D 21 A7 D4 02 B9 1A
      :         FC AC 21 FB 9E 94 9E 4B 42 45 9E 6A B2 48 63 FE
      :         43
359     21:    INTEGER
      :         00 B2 0D B0 B1 01 DF 0C 66 24 FC 13 92 BA 55 F7
      :         7D 57 74 81 E5
382     129:    INTEGER
      :         00 9A BF 46 B1 F5 3F 44 3D C9 A5 65 FB 91 C0 8E
      :         47 F1 0A C3 01 47 C2 44 42 36 A9 92 81 DE 57 C5
      :         E0 68 86 58 00 7B 1F F9 9B 77 A1 C5 10 A5 80 91
      :         78 51 51 3C F6 FC FC CC 46 C6 81 78 92 84 3D F4
      :         93 3D 0C 38 7E 1A 5B 99 4E AB 14 64 F6 0C 21 22
      :         4E 28 08 9C 92 B9 66 9F 40 E8 95 F6 D5 31 2A EF
      :         39 A2 62 C7 B2 6D 9E 58 C4 3A A8 11 81 84 6D AF
      :         F8 B4 19 B4 C2 11 AE D0 22 3B AA 20 7F EE 1E 57
      :         18
      :      }
      :    }
514    132:    BIT STRING, encapsulates {
518    128:    INTEGER
```

```

:      30 B6 75 F7 7C 20 31 AE 38 BB 7E 0D 2B AB A0 9C
:      4B DF 20 D5 24 13 3C CD 98 E5 5F 6C B7 C1 BA 4A
:      BA A9 95 80 53 F0 0D 72 DC 33 37 F4 01 0B F5 04
:      1F 9D 2E 1F 62 D8 84 3A 9B 25 09 5A 2D C8 46 8E
:      2B D4 F5 0D 3B C7 2D C6 6C B9 98 C1 25 3A 44 4E
:      8E CA 95 61 35 7C CE 15 31 5C 23 13 1E A2 05 D1
:      7A 24 1C CB D3 72 09 90 FF 9B 9D 28 C0 A1 0A EC
:      46 9F 0D B8 D0 DC D0 18 A6 2B 5E F9 8F B5 95 BE
:      }
:    }
649 202: [3] {
652 199:   SEQUENCE {
655 57:     SEQUENCE {
657 3:       OBJECT IDENTIFIER subjectAltName (2 5 29 17)
662 50:       OCTET STRING, encapsulates {
664 48:         SEQUENCE {
666 46:           [6]
:             'http://www.example.com/users/DSAendentity.'
:             'html'
:           }
:         }
:       }
714 33:     SEQUENCE {
716 3:       OBJECT IDENTIFIER issuerAltName (2 5 29 18)
721 26:       OCTET STRING, encapsulates {
723 24:         SEQUENCE {
725 22:           [6] 'http://www.example.com'
:         }
:       }
:     }
749 29:   SEQUENCE {
751 3:     OBJECT IDENTIFIER subjectKeyIdentifier (2 5 29 14)
756 22:     OCTET STRING, encapsulates {
758 20:       OCTET STRING
:         DD 25 66 96 43 AB 78 11 43 44 FE 95 16 F9 D9 B6
:         B7 02 66 8D
:       }
:     }
780 31:   SEQUENCE {
782 3:     OBJECT IDENTIFIER
:       authorityKeyIdentifier (2 5 29 35)
787 24:     OCTET STRING, encapsulates {
789 22:       SEQUENCE {
791 20:         [0]
:           86 CA A5 22 81 62 EF AD 0A 89 BC AD 72 41 2C
:           29 49 F4 86 56
:         }
:       }

```

```

      :      }
813  23:      SEQUENCE {
815    3:      OBJECT IDENTIFIER certificatePolicies (2 5 29 32)
820  16:      OCTET STRING, encapsulates {
822  14:          SEQUENCE {
824  12:              SEQUENCE {
826  10:                  OBJECT IDENTIFIER '2 16 840 1 101 3 2 1 48 9'
      :                  }
      :              }
      :          }
      :      }
838  14:      SEQUENCE {
840    3:      OBJECT IDENTIFIER keyUsage (2 5 29 15)
845    1:      BOOLEAN TRUE
848    4:      OCTET STRING, encapsulates {
850    2:          BIT STRING 7 unused bits
      :          '1'B (bit 0)
      :      }
      :  }
      : }
      : }
854    9: SEQUENCE {
856    7:     OBJECT IDENTIFIER dsaWithSha1 (1 2 840 10040 4 3)
      :     }
865   47: BIT STRING, encapsulates {
868   44:     SEQUENCE {
870   20:         INTEGER
      :         65 57 07 34 DD DC CA CC 5E F4 02 F4 56 42 2C 5E
      :         E1 B3 3B 80
892   20:         INTEGER
      :         60 F4 31 17 CA F4 CF FF EE F4 08 A7 D9 B2 61 BE
      :         B1 C3 DA BF
      :     }
      : }
      : }
```

C.4. Certificate Revocation List

This appendix contains an annotated hex dump of a version 2 CRL with two extensions (cRLNumber and authorityKeyIdentifier). The CRL was issued by cn=Example CA,dc=example,dc=com on February 5, 2005; the next scheduled issuance was February 6, 2005. The CRL includes one revoked certificate: serial number 18, which was revoked on November 19, 2004 due to keyCompromise. The CRL itself is number 12, and it was signed with RSA and SHA-1.

```

0 352: SEQUENCE {
4 202:   SEQUENCE {
7 1:     INTEGER 1
10 13:   SEQUENCE {
12 9:     OBJECT IDENTIFIER
:         sha1withRSAEncryption (1 2 840 113549 1 1 5)
23 0:     NULL
:       }
25 67:   SEQUENCE {
27 19:     SET {
29 17:       SEQUENCE {
31 10:        OBJECT IDENTIFIER
:          domainComponent (0 9 2342 19200300 100 1 25)
43 3:        IA5String 'com'
:      }
:    }
48 23:   SET {
50 21:     SEQUENCE {
52 10:      OBJECT IDENTIFIER
:        domainComponent (0 9 2342 19200300 100 1 25)
64 7:      IA5String 'example'
:    }
:  }
73 19:   SET {
75 17:     SEQUENCE {
77 3:      OBJECT IDENTIFIER commonName (2 5 4 3)
82 10:      PrintableString 'Example CA'
:    }
:  }
: }
94 13:   UTCTime 05/02/2005 12:00:00 GMT
109 13:   UTCTime 06/02/2005 12:00:00 GMT
124 34:   SEQUENCE {
126 32:     SEQUENCE {
128 1:      INTEGER 18
131 13:     UTCTime 19/11/2004 15:57:03 GMT
146 12:     SEQUENCE {
148 10:      SEQUENCE {
150 3:        OBJECT IDENTIFIER cRLReason (2 5 29 21)
155 3:        OCTET STRING, encapsulates {
157 1:          ENUMERATED 1
:        }
:      }
:    }
:  }
: }
160 47:   [0] {
162 45:     SEQUENCE {

```

```
164 31: SEQUENCE {
166 3:   OBJECT IDENTIFIER
      :   authorityKeyIdentifier (2 5 29 35)
171 24:   OCTET STRING, encapsulates {
173 22:     SEQUENCE {
175 20:       [0]
      :       08 68 AF 85 33 C8 39 4A 7A F8 82 93 8E 70 6A
      :       4A 20 84 2C 32
      :     }
      :   }
      : }
197 10: SEQUENCE {
199 3:   OBJECT IDENTIFIER cRLNumber (2 5 29 20)
204 3:   OCTET STRING, encapsulates {
206 1:     INTEGER 12
      :   }
      : }
      : }
      : }
209 13: SEQUENCE {
211 9:   OBJECT IDENTIFIER
      :   sha1withRSAEncryption (1 2 840 113549 1 1 5)
222 0:   NULL
      : }
224 129: BIT STRING
      : 22 DC 18 7D F7 08 CE CC 75 D0 D0 6A 9B AD 10 F4
      : 76 23 B4 81 6E B5 6D BE 0E FB 15 14 6C C8 17 6D
      : 1F EE 90 17 A2 6F 60 E4 BD AA 8C 55 DE 8E 84 6F
      : 92 F8 9F 10 12 27 AF 4A D4 2F 85 E2 36 44 7D AA
      : A3 4C 25 38 15 FF 00 FD 3E 7E EE 3D 26 12 EB D8
      : E7 2B 62 E2 2B C3 46 80 EF 78 82 D1 15 C6 D0 9C
      : 72 6A CB CE 7A ED 67 99 8B 6E 70 81 7D 43 42 74
      : C1 A6 AF C1 55 17 A2 33 4C D6 06 98 2B A4 FC 2E
      : }
```

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